

NOT A POTATO

THE EIGHTEEN-YEAR ALMANAC

ANDRIES J. GREYLING

Chapter 1 — The Mark in the Grass

The frost was the first thing that read as deliberate.

Naila saw it before she saw the earthwork — a long band of white grass lying lower and brighter than the grass around it, a seam where the cold had pooled in a hollow that ran arrow-straight across the fairway and then turned, at an angle she could feel in her teeth before she could name it, and ran straight again. Somebody had mowed the lawn. The frost had found the geometry anyway.

“That’s it,” she said.

Frik, who had been arguing with the parking situation, stopped. “That’s what? That’s a golf course.”

“That’s the wall.”

He squinted along the line of her arm. Behind them the rental’s engine ticked as it cooled. Ahead, past a low chain and a sign that said the things signs say — PLEASE RESPECT — the land rose into a swell so gentle you’d walk over it carrying coffee and never know you’d crossed anything at all. A man in a windbreaker was wheeling a bag-cart up the rise. He went up one side of the ancient embankment and down the other without a hitch in his stride, the way you’d cross a speed bump.

“Okay,” Frik said slowly. “Okay. He just— he walked over a two-thousand-year-old astronomical instrument to go hit a small ball.”

“He doesn’t know.”

“*Nobody* knows. That’s insane. That’s the most insane thing I’ve—”

He got out his phone, held it up, put it down again. “You can’t even photograph it. There’s nothing to photograph. It’s *grass*.”

That was the part Naila kept turning over. From the air — she’d seen the air, everybody had seen the air, the survey plates and the drone shots that turned the whole thing into a diagram — the Octagon was a thunderclap: eight walls, five hundred and fifty feet each, fifty acres of folded ground holding a near-perfect figure, hinged by a neck of causeway to a circle eleven hundred feet across. From above it was a proof. You could not argue with it. You could barely believe a person had stood on the ground that made it.

And here was the ground. From the ground it was almost nothing. A man with a five-iron crossing a rumple in the lawn.

The Lens came up beside her, camera down at the hip, not raised. He’d ridden the last hour in silence with the window cracked, and he had that stillness on him that meant he was already working, already deciding what he would not take. He looked along the frost-seam the way Naila had, and then he looked at the golfer, and something moved in his jaw.

“Don’t,” she said, quietly, before she knew she was going to.

“I wasn’t.”

“You were going to.”

“I was deciding whether to,” he said, which was honest, and lowered the strap an inch more, and that was that. The golfer reached the green and was gone behind the swell. The frost held its line. The highway, somewhere off past the far walls, made its long indifferent sound, the sound of the present tense going about its business over the top of the past.

Dr. Tessa Redwing was waiting by the gate, and she had been waiting in a way that made clear she had not been waiting *for* them, exactly. She’d been here. They had arrived into where she already was.

She was perhaps sixty, in a field coat the colour of the wet ground, with a thermos and a clipboard and reading glasses pushed up into grey hair, and she watched the four of them come up the path the way you watch weather you’ve seen before — not unkindly, but with

a working knowledge of what it tends to do. When Frik stuck out his hand and started talking she let him finish, which took a while, and then she shook it once.

"You're the film people," she said.

"We're—" Frik started.

"You're the film people." Not a correction. A placement. "The Connection said you'd come. There's a research window open right now, so you'll want to keep off the marked grass and out of the way of the golf, what's left of it. The lease is winding down. It is not wound down. People still live their Saturdays out here." She nodded at the man's distant green. "He paid his dues. He gets his round."

"It's a *golf course*," Frik said, with the particular outrage of a man who has just learned a thing and needs everyone to feel it as freshly as he does. "On *this*."

"For about a hundred years," Tessa said. "Mowed it. Watered it. Put in some sand traps." A beat. "Kept the developers off it, too, which is more than the City of Newark managed for the other nine-tenths of it." She let that sit. "So I'd hold the outrage for a minute and decide where you want to spend it."

Naila liked her immediately, with the clean fast certainty she felt about very few people and almost no rooms.

"Three thousand acres," Tessa went on, because Frik had gone quiet and somebody had to fill the air, and she'd clearly decided it would be her, on her terms. "That's what was here. Circles, squares, octagons, ellipses, miles of parallel walls running off across what's now subdivisions and the bypass and a Kroger. People drive over the parallel walls four times a day. Less than ten percent left. This." She turned and took in the swell of it, fifty acres of folded earth that from where they stood looked like a slightly interesting public park. "And the Great Circle, a couple miles that way. That's the survivors."

"Who built it?" The Lens, low.

"The people the archaeologists call Hopewell." A small pause on the name, a pause Naila clocked and filed and did not touch. "Which isn't what they called themselves, because we don't know what they called themselves, because they didn't write. The name's off a

farm. A man named Hopewell owned a field with mounds in it in the eighteen-nineties, and the name stuck to a whole world of people across four hundred years because a farmer was standing nearby when the archaeologists showed up.” Dry as the frost. “So when you say *Hopewell* in your film, you might let your viewers know they’re hearing the name of a dead man’s pasture, not a nation.”

Frik wrote that down. Naila watched him write it and watched Tessa watch him write it, and understood that the doctor had handed them something true and small to see whether they’d take it greedily or carefully, and that this had been a test, and that they had — barely — passed it by Frik having a pen.

Naila walked the near wall while they talked permissions.

She did it the way she did everything, which was with her whole attention and none of her social cover, walking the top of the embankment with her eyes on her feet and then on the turn ahead, feeling the grade come up under her boots — gentle, gentle, three feet, maybe four, of raised ground, nothing, a child could run up it — and then the turn.

The turn was the thing.

She stopped at the corner where one wall ended and the next began, where the figure changed its mind by a precise and intended amount, and she stood in the cold morning with her arms out a little for balance and felt the angle the way you feel a chord resolve. It was *exact*. Not exact-looking. Exact in the body, exact the way a step is exact when the stair is built true and your foot lands on nothing because there’s nothing there to land on, a perfect absence. The wall didn’t wander to its corner and round it off the way a thing piled up by guesswork wanders. It came in straight and it turned through an angle somebody had *decided*, and it left straight, and the deciding was still in the ground after two thousand years of frost and golf and rain.

Basket by basket. That was the part her mind kept snagging on, the way a coat snags a nail. No machines. No metal. No draft animals — she’d read it on the plane and not believed it and now stood on the proof of it. People had carried this. People had filled baskets with earth from a borrow pit and carried the baskets up the rising

wall and tipped them out and gone back for more, and they had done it to a *plan*, to a number, to a tolerance, for so long that the people who started the wall were dead before the wall was a wall, and the people who finished it had never met the people who'd surveyed it, and somehow the angle was still exact.

You did not get this by accident. You did not get this by feel. You got this by a number, kept.

She crouched and put her palm flat on the frost. It was just cold grass. It was just dirt. It was the most precise dirt she had ever stood on.

"She does that," she heard Frik say, somewhere behind her, fond and apologetic at once, the thing he said about her in new rooms so the new room would file it correctly. *She does that*. Translating her into a register the room could hold.

She didn't mind. She kept her hand on the cold ground and let the angle be exact in her body and didn't say any of it out loud, because there wasn't anybody yet she wanted to say it to who would hear the number under the words.

When she stood up, Tessa was watching her. Not Frik. Her.

The doctor didn't say *she does that*. The doctor didn't say anything. She just watched Naila come down off the wall with an expression Naila couldn't read, which was rare, which was almost unheard of — and then she looked away first, back to her clipboard, and the moment closed like water.

There was a fifth person, and Naila had not seen him arrive.

He was just suddenly there at the gate, weathered, in a canvas coat gone soft with age, holding a paper cup of gas-station coffee and looking at the earthwork with the particular flat attention of a man reading a thing rather than admiring it. He didn't introduce himself to the crew and the crew didn't introduce themselves to him, and Tessa gave him a single nod that had a whole history folded into it, the nod you give someone you've worked a hard site with before.

"Jakobus," Frik said, under his breath, to Naila. "That's the fixer. That's the—"

“I know who it is.”

The man — Jakobus — didn’t walk the wall. He looked at the gap between the far walls, the gateway out at the east end where the two embankments stopped and left an opening to the horizon, and then he looked back along the line of that gap toward the centre of the figure, sighting it the way you’d sight down a rifle or a rule. He did it once. He nodded to himself, the small private nod of a man who has confirmed that a made thing was made *for* something, and that the something points east. Then he drank his coffee and said nothing at all, and the saying-nothing was so complete it was almost a kind of manners.

He was the road, not the lead. Naila had been told that and not understood it. She understood it now: he came in, he read the thing as a thing-for-a-purpose, and then he stood back and let the gateway belong to whoever it belonged to. It belonged to Tessa. He knew that better than any of them, which was why he’d shut his mouth.

“You want the line you’ll put under the picture,” Tessa said.

They were back at the gate. The light had come up gold and the frost was going off the grass in patches, the geometry softening back into lawn as the sun took the cold seam away. In an hour you would not be able to see it at all. In an hour it would be a park again, and the man with the five-iron, and the highway.

“You want the sentence,” Tessa said again, and there was the faintest edge to it, because she knew how the film people worked, knew they needed the line, and she’d decided to give them one rather than let them go off and invent a worse one. “Here it is. It’s the Ohio History Connection’s, not mine, so quote it straight. The surveying out here is twice as precise as Stonehenge.”

Frik laughed — the bark of a man whose decoy has just been kicked out from under him. “*Twice*— Stonehenge is the most famous—”

“Stonehenge is famous,” Tessa agreed. “Stonehenge gets the post-cards. Stonehenge has a gift shop the size of a barn.” She gathered her thermos and her clipboard. “And the people who measured *this* one, in this field, with cord and baskets and no writing and no metal and no wheel, were working to a tolerance that makes the famous one

look approximate.” She looked at the frost-line one more time before it went. “And they were tracking something a great deal harder than the Sun.”

She let that hang. She did not finish it. She picked up her things and she did not finish it, and Naila felt the unfinished part go down into her like the cold going into the ground.

“Harder how?” Frik said.

But Tessa was already walking, and the answer, whatever it was, was hers, and she had decided — for now, for this morning, on her ground — to keep it.

Behind them, somewhere past the far gateway, the Moon was down and the Sun was up and the highway kept making its long ordinary noise over the top of the most precise dirt in North America, and nobody on it knew.

Chapter 2 — The Golf Course Problem

The gate was a chain-link gate with a combination lock, and beyond it a man in a quarter-zip was practising his putting.

Frik watched him through the diamond mesh and felt the specific itch he got when the world refused to make sense in the order he wanted. There was a two-thousand-year-old lunar observatory on the other side of this fence — fifty acres of it, eight earthen walls each longer than a football field, raised basket by basket by people who tracked the Moon to half a degree — and a guy was rolling a Titleist across it toward a little white cup.

“He’s not even good,” Frik said.

“He doesn’t have to be good,” Naila said. “He has a key.”

She was standing very still, the way she got, hands loose at her sides, reading the embankment that ran off to their left and curved away into the trees. To Frik it was a long grassy hump with a fairway mowed over the top of it and a sand bunker scooped out of one flank. To Naila it was — he didn’t actually know what it was to Naila. Something with edges. She’d gone quiet four minutes ago and hadn’t come back yet.

The LENS had the camera up but pointed at the ground, at the worn place in front of the gate where a thousand golf shoes had ground the grass to dirt. She did that. She found the human wear before she found the monument.

“Okay,” Frik said, to nobody, “so. Official version. This is the Mound-builders Country Club. Established nineteen-ten. They have, what—”

“A lease,” said the woman who had come up behind them, and Frik turned around.

She was maybe fifty, in a field jacket and good boots, with a lanyard from the Ohio History Connection and the air of somebody who had said most of what she was about to say a great many times to a great many people who did not listen the first time. She did not offer her hand right away. She looked at the camera first.

“It’s not running,” the LENS said, and lowered it the rest of the way.

The woman nodded once, like that was a question that had needed answering, and only then said, “Tessa Redwing.” She didn’t add the credentials. Frik would learn them later — archaeologist, Tribal Historic Preservation Officer, Hopewell-descendant-affiliated, the kind of CV that meant she was both the scientist and the thing the scientists were supposed to be careful around. Right now she was just a person watching strangers stand at a fence.

“Frik,” said Frik. “That’s Naila, she’s — she’s doing a thing. That’s—” He gestured at the LENS, who supplied nothing. “We’re the Why Files crew. We talked to the office. There were emails.”

“There were a lot of emails,” Tessa said.

The man in the quarter-zip had noticed them. He scooped his ball, tucked the putter under his arm, and came across the green toward the gate at the unhurried pace of someone who had decided, in advance, that he would be polite and that being polite would cost him nothing.

“You must be the documentary people,” he said through the chain-link, and smiled, and it was a good smile. “Daniel Voss. Meridian Heritage Foundation.” He had a card. He fed it through one of the diamonds in the mesh, and Frik took it because what else do you do. Heavy stock. Embossed. *Securing the record for everyone*. “I’d let you in, but it’s not really mine to let you into. Funny old situation here.”

“Funny,” Frik agreed.

Here is the funny old situation, and Frik got it in pieces over the next hour, standing at a fence in the cold while a stranger explained a

lawsuit to him.

The Octagon Earthworks belongs to the Ohio History Connection. It has for a long time. But in 1933 the Connection leased the grounds to a golf club, and the club built a course right over the top of the monument — fairways draped across the embankments, greens nestled in the gateways, a clubhouse, a sprinkler system, the whole apparatus of leisure laid down on top of the single most precise piece of pre-contact astronomical engineering in North America. For ninety years you mostly could not walk it. You could look at it four days a year. If you wanted to stand inside the figure the rest of the time you needed to either play golf or know somebody.

In 2018 the Connection moved to take it back. It went to court. It went, eventually, all the way up, and the Connection won the right to end the lease early through eminent domain, and the price was to be set, and the golfers were leaving, and the gate would open — the actual physical gate, this one, with the combination lock — to the public, soon, properly, for the first time in living memory.

That was the good story. That was the story you put in the press release. Frik loved it. It had everything: the long patient wrong, the slow correction, the monument coming home.

“So it’s basically settled,” he said.

“Basically,” Tessa said, in a way that took the word apart.

Because here was the part that wasn’t in the press release. The lease was ending but it had not *ended*. There was a window — a stretch of weeks before the handover finalised, before full public access, before the consultation protocols with the descendant nations were locked down — during which the grounds were neither golf course nor open monument. They were in between. And in between was a soft, undefended place, and into it had walked Daniel Voss with his heavy card stock, holding a research-access agreement that the Connection had signed back when the Foundation looked like a friendly grant and not a problem.

“Meridian wants to do a new survey,” Tessa said. “High-resolution. They want to re-walk the alignments with current ephemeris data and publish a — they called it a *definitive reading*.” She let the word sit there. “They have a window. The window is legal. The window is

closing, but right now it's open, and it's theirs."

"And ours isn't," the LENS said.

"Yours doesn't exist yet," Tessa said, not unkindly. "You're media. You can stand at the fence and film the fence."

"Here's what I don't get," Frik said.

He had walked the line of the embankment as far as the fence let him, which was about a hundred and forty feet, counting under his breath, before the chain-link cut him off at the bunker. Voss had drifted away to take a call. Naila had finally surfaced and come to stand beside him, and he could tell she'd been doing arithmetic the whole time she was gone, because she had the look.

"It's a country club," Frik went on. "Okay. People play golf on a sacred calendar, which is — that's a whole sentence, I'm going to need to sit with that. But here's my actual problem." He pointed at the wall. "Standard skeptic move, right? You see big mounds, you say *fort*. Defensive earthworks. People built it to hide behind. Reasonable. Walls are walls."

"It's not a fort," Naila said.

"I know it's not a fort, but I want to say *why* it's not a fort, because if I can't say why then I'm just being a fan." He looked at her. "Give me the why."

She didn't answer with words. She crouched, found a stick, and drew it in the dirt of the worn patch by the gate — a circle, and then inside the circle, hugging the inner edge, a second line.

"The Great Circle's a couple miles that way," she said. "Three hundred and sixty-six metres across. Eight-foot walls." She tapped the inner line. "The ditch is here. Inside the wall."

Frik looked at it. He looked at it again.

If you are building a fort, you dig your ditch *outside* your wall, because the ditch is the thing the enemy has to cross before they reach the thing they have to climb. The ditch and the wall do their work together, against an outside. That is the entire logic of a fort; it is almost the definition of one.

These people dug the ditch on the inside. They built a wall to hold something *in*, or to hold the world *out* in a way that had nothing to do with arrows. They made the one shape that announces, at the level of basic engineering, *this is not about defence*. You could stand on the wall and the ditch was at your back, useless against any attacker, perfect against — what? The point was that there was no attacker. The point was that the shape was the point.

“Okay,” Frik said quietly. “Yeah. The fort thing is dead.” He felt the small clean click of a wrong idea going out. “That’s — they did that on purpose. To tell you.”

“To tell *someone*,” Naila said. “Maybe not you.”

The thing about Naila is that she didn’t negotiate, so Frik did the negotiating, which mostly meant he talked to Tessa while Naila stood at the fence and read the monument she wasn’t allowed to enter.

“We don’t want to *take* anything,” Frik said. “We’re not Meridian. We don’t want a definitive reading. We just want to walk the OCD — the Observatory Circle Diameter, the unit, the thousand-and-fifty-four-foot thing — we want to measure it ourselves so we can show people it’s real and not a story. That’s it. That’s the whole ask. Let us walk it with a tape and a phone and we’ll film our own feet doing the boring part.”

Tessa heard him out. That was a thing she did that he was beginning to notice — she actually waited until you were finished, all the way finished, before she said anything, which was so rare in a person that it kept tricking him into saying more than he meant to.

“You want me to give you access,” she said, “to a place I can’t give Voss access to leaving from, faster than I want him in. So that I’ll have two outside crews on a site that’s mid-handover, where the descendant nations haven’t finished saying how they want it treated.” She wasn’t angry. She was just laying it flat so they could both look at it. “You understand that the people who built this kept the record for the better part of a hundred years before the shape of it even existed to be built. Patience is sort of the whole subject here.”

“Yeah,” Frik said. “I do get that. That’s why I want to do it right.”

"Then the right version," Tessa said, "is that this isn't your record to read first. It's not Voss's either. He just thinks faster is the same as righter." She glanced past Frik at Naila, who had gone still again at the fence, head tilted, tracking some line out across the buried geometry that nobody had drawn for her, that she was finding anyway. "Your friend already sees it."

"She sees everything," Frik said. "It's not always great for her."

"It's not a wound," Tessa said. "Don't make it one." And there was something in how she said it — not a correction exactly, more like she'd handed him a thing she'd been holding and now it was his to not drop.

It was the LENS who saw the truck.

She'd wandered off down the road to film the clubhouse, the way she liked to get the ordinary edges of an extraordinary thing — the parking lot, the cart path, the *No Trespassing* sign zip-tied to a post that was, itself, standing on a sightline two thousand years old. She came back at a walk that wasn't quite a run.

"Survey van," she said. "Out by the far gate. The one that faces the trees." She didn't editorialise; she never did. "Two people setting up a tripod. There's a — it's a total station, the laser thing. And Voss is walking out to meet them."

Frik turned. Across the fenced fifty acres, past the green where the man in the quarter-zip had been practising, past the buried walls and the silent gateways, he could see it now: a small bright figure crossing the grass toward the eastern gate, the one that pointed at where the Moon would clear the horizon at its furthest north in a cycle that took eighteen years and seven months to come around, that no single watcher had ever lived slowly enough to see whole.

Voss reached the tripod. He put a hand on the shoulder of one of the surveyors. He looked, even at this distance, pleased.

Tessa watched it too, and her face did not change, which was its own kind of telling.

"His window opens tomorrow," she said. "Six in the morning. They'll walk it at first light." She looked at Frik, and then at the LENS, and

then, longest, at Naila, who had finally turned from the fence and was looking back at all of them with the particular expression of someone who had just finished a sum and didn't like the answer.

"So," Tessa said. "Now you know what you're racing."

The fish, if there had been a fish, would have said: *ninety years it sat under a golf course and nobody hurried. Now everybody's in a hurry. Funny what a deadline does to a calendar.*

But there wasn't a fish. There was just the cold gate, and the lock, and the long grass laid over the arithmetic, and a stranger walking out to read it first.

Chapter 3 — The Straight Line

The dew was still on it when they came through the gap in the wall, and the first thing Naila did was take her shoes off.

Frik watched her do it without comment, which was a kindness she noticed and filed. He had his own ritual — phone out, thumb flicking, narrating to nobody: “Observatory Circle, one thousand fifty-four feet across, twenty acres, joined to the Octagon by a causeway, and we are *standing inside it* and you cannot tell. You literally cannot tell, it’s so big the curve looks straight.” He turned a slow circle. “It looks like a park. It looks like somebody mowed a really nice park.”

It did look like a park. That was the thing nobody put in the brochures, Naila thought — that the largest geometric earthwork on the planet, twice as precise as Stonehenge by the Ohio History Connection’s own measuring, presents to the standing human eye as a long green bowl with a grass wall around it about as high as a man, and a sky on top, and the sound of Route 79 coming over the rim like surf. You had to leave the ground to see it. You had to be a bird, or a satellite, or dead, or carrying it in your head the whole time you built it, which was the part she kept coming back to and couldn’t put down.

She walked the wall barefoot to feel the grade. The grass was cold and the soil under it gave, just slightly, the way deep-built soil does, soil that was carried here. Not bulldozed. Carried. Basket by basket, a handful of pounds at a time, by people with no metal, no wheel that mattered, no animal to pull a load — and no draft of the finished thing to work from, because the finished thing was bigger than any one of them would ever stand far enough back to see.

“Okay, here’s my problem,” Frik said, falling in beside her, scrambling up the inside face. “*Fort*. People used to say fort. There’s a wall, it’s a fort, the Indians were defending it, very *Last of the Mohicans*. And I want to be fair to fort, because fort is at least a guess that respects the labor —”

“The ditch is inside,” Naila said.

“*The ditch is inside.*” He said it like a man laying down a winning card. “On the Great Circle, the one across town. Eight-foot walls, beautiful ditch, and it’s on the *inside* of the wall. You don’t dig your moat where the enemy’s already standing. That’s not a fort, that’s — I don’t know what that is, but it’s not a fort. Fort is a potato.”

“Fort is a potato,” Naila agreed. It was their word for it. The instinct to make the astonishing thing small enough to hold in one hand. *It’s a fort. It’s a coincidence. It’s a fluke.* The world reaching for the nearest dull box to put the wonder in so it could stop being responsible for it.

The LENS was thirty feet off, down on one knee in the wet grass, not shooting. The camera hung. He was looking through the gap toward the causeway where the circle handed off to the octagon, and his face had the still, evaluating quality it got when he was deciding whether a thing was his to take. Naila had learned to read that stillness. It meant *not yet*. It meant *whose is this*.

Dr. Tessa Redwing stood near the centre of the bowl with her hands in her coat pockets, watching the three of them arrange themselves across her grandmothers’ work, and saying nothing, which Naila was beginning to understand was itself a kind of saying.

They had brought a wheel. Not a fancy one — a surveyor’s measuring wheel from a hardware store, the kind that clicks off feet, and a hundred-metre tape, and Frik’s GPS, and Naila’s eye, which was the only instrument she fully trusted. She’d asked Tessa first. *May we measure it.* Tessa had looked at her for a moment that went on a beat too long and then said, “People have been measuring it for a hundred and forty years. Go ahead.” Which was permission and was also something else, something with a hook in it that Naila hadn’t worked out yet.

So they measured. Frik took the wheel and walked a chord. Naila stood at the centre — or what her body told her was the centre, the place where the bowl was equally far from her on every side, a thing she could feel before she could prove — and let the shape come up through her feet.

This was the part she could never explain to people who asked what it was like. They wanted a metaphor and she didn't have one, because it wasn't *like* anything; it was just that the relations were *there*, the way a chord is there in a struck string, and she heard them. The circle wasn't a circle to her. It was a circle and a square at once, the square sitting inside the curve and touching it, and the octagon out past the causeway built off the same square, all of it one figure thinking one thought.

She crouched and sketched it in the notebook. Circle. Square inscribed, corners kissing the curve. Then she drew the diameter of the circle and wrote *OCD* next to it, the way the papers did — Observatory Circle Diameter, the ruler the whole site was cut from. Then she paced off the distance to where the causeway started and counted under her breath and wrote 6 and circled it.

"Six," she said.

"Six what?" Frik dropped down next to her, breathing.

"The circles. The Observatory and the one out past the octagon. They're six of these apart." She tapped the diameter. "The thing's built in units of itself. The diameter is the brick. You don't measure it in feet. You measure it in *circles*."

Frik looked at the sketch, then up at the impossible flatness of the green, then back at the sketch. "That's — okay, that's the thing I came here to be annoyed by and I am instead going to be normal about it." He wasn't going to be normal about it. "They had a *standard length*. A custom foot. An — a Hopewell metre."

"Mhm."

"And they used it to lay out a fifty-acre octagon next door with eight five-hundred-and-fifty-foot walls, by *hand*, with *cord*, and the walls came out straight." He was up again, pacing, hands going. "How do you make a straight five-hundred-foot wall with no instrument? You

can't sight it, the ground rolls. You'd peg a cord, but cord stretches, cord sags, cord's wet today —"

"You'd hand it down," Tessa said.

She had come closer without Naila hearing her, which on grass that gave underfoot was its own small marvel. She wasn't looking at the notebook. She was looking out across the bowl, at the long green nothing that was the most measured nothing in North America.

"You make the cord, you set the length, you teach somebody the length." Tessa's voice was even, conversational, the voice of someone who had given this answer many times to many people who heard half of it. "Then that person teaches it. The cord rots. The length doesn't, if you keep handing it on. You're not asking how one clever man built a straight wall. You're asking how a *people* kept a measurement true across the time it takes to build something nobody alive at the start would see done."

Naila stopped sketching.

"You keep talking about it like it's geometry," Tessa said. Not unkindly. There was even something close to affection in it, the patience of a teacher with a bright student who has the wrong end of the thing. "You came in here and took your shoes off — I saw that, thank you — and you went straight to the centre and felt for the middle. Like it's a puzzle. Like it's a beautiful machine and the question is how the gears fit."

"It is a beautiful machine," Naila said, because it was, and she wasn't going to pretend it wasn't to be polite.

"It is." Tessa nodded slowly. "And a song is just air pressure. You can measure a song with the right instrument and never once hear it." She let that sit. Out past the rim a truck downshifted on 79 and the sound poured over the wall and pooled in the bowl with them. "My grandmother's grandmothers carried this dirt. The unit you're calling OCD — the *length* — that got carried too. Hand to hand. It's not a number that happens to repeat. It's a thing somebody was *responsible* for. Somebody held the cord and somebody checked it and somebody said *no, again, it's long* and they did it again. For longer than any of them lived."

Frik had gone quiet, which almost never happened, and Naila could

see him wanting to make the joke that would let him off the hook of the largeness of it, and not making it, which she respected more than anything he'd done all day.

"It's not just geometry," Naila said. Trying the shape of it.

"It's a *record*," Tessa said. "The geometry is how you can read it. But you don't write a record for a hundred years to do trigonometry." She looked at Naila then, full on, and there was the hook again, gentle and absolute. "You crack the math fast. You're good — I can see you're good. But the math being crackable is the least interesting true thing about this place. The interesting true thing is *who keeps a record longer than a life*. And that part isn't a puzzle you get to solve. That part's a people you have to ask."

The LENS lowered to a sitting position in the wet grass without a sound and let the camera lie in his lap. He had heard all of it. He didn't lift the lens. Naila understood, distantly, that this was the most useful thing he could do, and that it cost him to do it, and that the cost was the point of having him there.

She went back to the diameter. She wanted, suddenly and badly, to know the number. Not to own it. To stand next to it. To know what the cord measured when they finally laid it down satisfied.

Frik read out the wheel. They walked it twice. They argued the centre by a foot and a half and split the difference. Naila held the tape with cold fingers and Frik called the GPS and the two methods fought to a draw the way real measurement does, never clean, always a smudge of disagreement around the truth.

She wrote the number. She did the arithmetic against the published figure, the one in Hively and Horn, the one in the survey done before anyone here was born.

"Frik."

"Yeah."

"Read me the official diameter again. The good one. The careful one."

He read it. One thousand fifty-four feet, near enough, the figure that had stood through decades of being checked.

Naila looked at what they'd got, two amateurs with a hardware-store wheel and a tape and a freezing-handed argument about where the middle was, on a circle so large the curve read as a straight line to the standing eye.

It closed to within four feet.

Four feet. On a circle a fifth of a mile across, basket-loaded by hand, by people with cord and patience and no way to ever stand back far enough to check, the round of it held true to within four feet — and the gap between *their* clumsy morning's work and the careful survey was smaller than that. The error in the measuring was bigger than the error in the thing.

She sat down where she was. The grass soaked through and she didn't care.

The fish would have something to say about this, she thought — would lean in and go *four feet, on a fifth of a mile, with a basket, and you sweated through your good shirt to measure it worse than they built it, congratulations, genius* — and the fish would be right, and that was the joke and also not the joke at all.

Tessa was watching her sit down in the wet. Something in the older woman's face had eased, very slightly, as if a thing she'd been braced for had not happened.

"You felt for the centre with your feet," Tessa said again, quietly. "And then you measured it, and it beat you. Good. Hold onto that. The not-knowing how. People rush past the not-knowing how. They want it to be a clever trick or a lost secret. It's neither." She turned to look out toward the causeway, toward the octagon they hadn't walked yet, the eight straight walls and the gateway where, on the right dawn once in a generation, the Moon was said to come up dead through the middle. "It's just people. Being more patient than you can stand. For longer than you'd believe."

Naila looked at her notebook, at the circle holding the square holding the circle, all of it cut from one length somebody had carried in their hands and refused to let drift.

Four feet.

Fifty-three miles south, she knew, there was another one. The only other octagon they ever built. And the cord that laid out that one, the books said, was this same cord, this same length, this same impossible kept-true number — and she did not yet understand, sitting cold and stunned in the most measured grass on the continent, how you could hand a thing that exact across fifty-three miles and four hundred years and have it still close to four feet, or who had stood at the far end holding the other end of it.

“We should walk to High Bank,” she said.

Tessa didn’t answer for a moment. When she did, it wasn’t yes and it wasn’t no.

“We’ll see,” she said, “what you’ve earned by then.”

Chapter 4 — The Potato and the Fort

The Great Circle didn't announce itself. That was the first thing wrong with it, if you came looking for a fort.

Frik came over the lip of the embankment expecting — he didn't know what. A rampart, maybe. Something that said *keep out* in the universal grammar of dirt thrown up by frightened men. Instead the wall rose gentle and green under his boots, mowed like a municipal park, and on the far side it sloped down into a ditch, and past the ditch the ground opened into a bowl of grass big enough to lose a cathedral in. Eight feet of wall. A thousand-odd feet across. Geese on it. A man in a windbreaker walking a beagle along the inner crest as though it were the perimeter path of any suburb in Ohio, which it was, because it was.

"Okay," said Frik, to no one. "Okay, hold on."

He had a notebook. He had, specifically, a page headed FORT in capital letters with a box drawn around it, because he liked to argue with the boring answer before he let himself like the interesting one. Naila said this was the most ADHD thing about him, the need to wrestle the dull theory to the ground first, and she said it with affection, which was how she said most things to him, in a flat voice that took him a year to learn was warmth.

The fort theory was a good theory. He wanted to be fair to it. It was, in fact, the *first* theory — eighteenth-, nineteenth-century surveyors riding up on these walls and reaching for the only word their world handed them for big regular earth, which was *defensive*. Of course they did. You see a wall, you think someone's behind it with a bow.

The whole continent of Europe had taught them to read a raised ring of earth as a thing that meant *war*. They weren't fools. They were doing exactly what Frik was doing right now: bringing the ruler they already owned to a thing that didn't come with one.

"So," he said, louder, to Naila, who was twenty feet off with her shoes already in her hand because she said she couldn't feel a grade through rubber. "Pretend you've never heard the real story. You ride up here in 1840. What do you write down?"

"Fort." She didn't look up. She was watching her own feet on the slope, the way she did, reading the angle through the soles. "Earth-work enclosure. Defensive. Probably."

"Right! Right. Big wall, you hide your village inside it, you shoot the bad guys off the top." He paced along the crest, arms going. "It's clean. It's the kind of answer that lets you go home and have dinner. I *like* it. It doesn't need anything weird. No calendars, no genius, no eighteen-point-six years of anything. Just scared people and a shovel."

And it'd be a fine answer, something in him said, in a voice that sounded suspiciously like the fish, *if you'd quit walking and look at where you put your feet*.

He stopped. He looked at where he'd put his feet.

He was on top of the wall. To his left, the inside — the great green bowl. To his right, the ditch.

The ditch was on the inside.

The thing about a defensive ditch is that a child understands it. You dig the trench, you pile the dirt up *behind* the trench facing your enemy, so the man charging you has to drop down into a hole and then climb a wall while you stand on top and discourage him. Moat, then keep. Trench, then parapet. Ditch *outside*. It is not a subtle piece of engineering. It is arguably the least subtle thing humans have ever agreed on. Every hillfort in Britain, every motte-and-bailey, every Iron Age rampart from here to Anatolia: ditch out, bank in. You could lose the blueprints for the whole species and rederive it from one frightened afternoon.

Here the ditch was *inside* the wall.

Frik stood very still and let it land, because it was the kind of fact that didn't land if you read it off a page — and he *had* read it off a page, he'd read it in three books, *the ditch is interior, indicating ceremonial rather than defensive function*, and it had gone in one ear and out the other as the dry kind of sentence experts wrote to fill space. It was a completely different animal standing on the wall with the trench at your right hand, knowing in your body that an attacker coming over this would *land on the high ground*. You'd have dug your own enemy a parapet. You'd have built the fort backwards. You'd have, with enormous patient labour, basket by basket, hand by hand, constructed the single worst defensive structure it is possible to imagine, a thing that doesn't merely fail to defend but actively assists the assault.

"Naila." His voice had gone strange. "Naila, the ditch is —"

"On the wrong side. Yeah." She'd known the whole time. She'd let him get there. "It's been on the wrong side for two thousand years."

"That's —" He laughed, the bark he did when a thing he'd been condescending to turned around and looked at him. "That's not a mistake. You don't make that mistake *once*, never mind around the whole —" He turned, following the green ring of it with his arm, all the way round, the geese, the beagle, the windbreaker man now a far comma against the far wall. "It goes the whole way round. The wrong side goes all the way round."

Took you long enough, said the fish, who was not there, who was a habit Frik had instead of a god. *That FORT page looking a little embarrassed, champ?*

He flipped to the FORT page and drew a line through it. Then, because he was honest, and because the whole job was being honest about your own reach, he wrote underneath: *me, ten minutes ago, very confident*. He'd put it in the chapter. He always put himself in. The day he stopped puncturing his own overconfidence was the day he became one of the men on the internet who knew everything.

Tessa Redwing watched all of this from down in the bowl, near the centre, where she'd set up nothing and was doing nothing, which Frik had come to understand was its own kind of doing. She had

a thermos. She let them flail. She was a culture-bearer with a doctorate and the patience of a person who has watched a great many enthusiastic strangers discover, with the joy of a toddler finding its own foot, a thing her family had carried as ordinary knowledge for longer than enthusiasm had been a word. She didn't correct them. She didn't help. She watched the way you watch someone reading a letter that was addressed to you.

The LENS was further off, on the south arc, camera down for once. Friek clocked him clocking Tessa — the photographer's instinct to frame her against the wall, the visible decision *not* to, the lens swinging instead to the empty geese-dotted grass. The man had rules about whose face belonged to whom. Friek didn't fully understand them but he'd learned to trust the way the camera hung loose at the end of his arm when it meant *not mine*.

Friek came down off the wall to where Tessa stood, sliding the last few feet on the wet grass like a kid, recovering badly.

"It's not a fort," he announced, as though breaking news.

"No," she agreed.

"The ditch is on the inside."

"Mm."

"I drew a *box* around the word." He showed her the crossed-out page. He wanted, he realised, for her to be amused, and was annoyed at himself for wanting it, because that was a kind of asking-to-be-let-in, and it wasn't his to ask. She looked at the page. She was amused, a little, around the eyes, which was a gift and he knew it.

"People have been crossing out that word for a hundred and fifty years," she said. "And then drawing it again. There's a thing that happens." She screwed the cap back on her thermos. "They give up *fort*, because the ditch embarrasses them. And then they need a new word that does the same job, that lets them stop. So they reach for the next one."

"Coincidence," said Naila, from up on the wall. She'd been listening. She was always listening; she just didn't always look like it.

"Coincidence," Tessa agreed, and there was something under the word, the faintest weariness, a person who had heard this one too.

Frik had a coincidence page too. He flipped to it now, because Naila had come down and was already at work in the dirt with a length of mason's line and two tent pegs, doing the thing she did where the whole social world fell away and there was only the geometry, and he knew the look, and the look meant the chapter was about to get harder.

The coincidence theory went like this, and it was *also* a good theory, and Frik tried to give it the dinner-and-go-home respect it had earned: people see patterns. People are pattern machines. You build a big regular shape, you've got eight walls and a bunch of gateways, and you've got a sky full of things rising and setting in every direction, and *of course* if you go looking you'll find some gateway that lines up with some moonrise, because the sky is crowded and the gateways are many and human beings will draw a line between any two dots you hand them. Astrology runs on this. So does the face on Mars. You can align Stonehenge to the bus stop if you let yourself. The honest skeptic's whole job is to refuse the line your hungry eye wants to draw.

"This one's harder to kill," Frik admitted, crouching by Naila. "Because it's *true* about people. We do that. I do that. I aligned my breakfast to my mood this morning."

"Walk it with me," said Naila. She wasn't really asking. She'd run the line from a peg to the gateway gap in the wall and she was sighting along it with one eye shut, and her voice had the steadiness it got when the numbers were holding her up. "The coincidence story works if there are lots of targets and lots of lines. Lots of arrows, lots of haystacks, one's bound to hit. Right?"

"Right."

"So the killer isn't *is there an alignment*. The killer is *how many things does this one alignment have to hit at once, and how tight*." She moved the line a hand's width and the gateway gap went from framed to not-framed, just like that, a door that was a door only from exactly here. "Hold this."

He held it. Through the gap, the far wall, the morning sky going pale over Newark, the hum of the highway coming over the grass

like weather.

“Hively and Horn,” Naila said, “came here in ’eighty-two to *kill* the alignment. Physicist and a philosopher. They wanted it to be coincidence. They expected coincidence.” A peg in, a peg out. “And what they got was eight. Not one lucky moonrise. All eight standstill points. Both ends of the swing, north and south, rise and set, the whole envelope the Moon makes when it walks — built into the gateways. To half a degree.”

“Half a degree’s the width of —”

“The Moon. About. Yeah.” She sat back on her heels in the wet. “You can have a coincidence hit one target. You start asking it to hit eight, all from the same figure, all to inside the Moon’s own width, and the haystack runs out of needles. It stops being *the sky is crowded so something had to fit*. It becomes *somebody aimed*.”

The fish had gone quiet in Frik’s head, which was how he knew it was real. The fish heckled the reach. The fish had nothing to say to a number that held.

“And here’s where I want to overreach,” Frik said, because he could feel it, the pull, the thing in him that wanted to take the lovely hard fact and yank it past where it would go. “Here’s where I want to say *therefore* something. Therefore advanced. Therefore taught. Therefore —” He shut his mouth. He’d seen the documentaries. He knew the shape of the slide, the one that ended with a man in a leather jacket saying *now, mainstream archaeologists won’t tell you*. “Therefore nothing. The fact’s the fact. Eight points, half a degree. The fact doesn’t owe me a *therefore*.”

Tessa, who had drifted close, said nothing. But she was looking at him differently now, the way you look at a stranger who’s just declined an easy thing, and Frik understood that the not-reaching was the entrance fee, and that there were further rooms, and that they were hers to open.

He climbed back up on the wall before they left. He wanted the ditch again, the wrong-side ditch, the embarrassing trench that the whole century of fort-talk had stepped over.

Down in the bowl Naila was coiling her line, and Tessa was standing where she'd stood the whole time, near the centre, on the spot that was the spot for some reason none of them had been told yet, and the LENS had finally raised the camera and was photographing the grass, only the grass, the empty patient grass with the geese on it and not one human face.

The fort theory was clean. Frik wanted to be fair to it to the end. It was clean and it was wrong, and it was wrong for the simplest reason in the world, a reason you could feel through your boots: the dirt was on the wrong side to keep anyone out. Whatever this was, it was not built by people afraid of an enemy at the wall. It was built by people who'd decided, with two thousand years of labour, that the important direction was *in*. Toward the centre. Toward whatever Tessa was standing near and not naming.

He looked at the crossed-out word in his book.

Then he turned the page, because the next theory was already lining up — *local fluke, one clever village, a one-off* — and he could see, from the way Naila kept glancing southwest toward Chillicothe, toward the second octagon fifty-three miles down the road with the same ruler in its bones, that the page was going to get a line through it too.

The geese lifted off the inner ditch all at once, the way they do, for no reason a person can see.

Chapter 5 — Eight Gates, One Moon

The frost held the shape of the grass until the sun got to it, and then it let go all at once, the whole eastern slope of the wall going from white to wet in the time it took Naila to walk its length. Fifty-five hundred feet of earthen wall, give or take. She had paced it twice now. Her boots were soaked through and she did not care.

The Octagon did not look like fifty acres from inside it. It looked like a fairway with delusions of grandeur — flat green, a maintenance shed off to one side, eight long banks of earth standing maybe five feet proud of the turf, with gaps between them where the walls didn't quite meet. From the ground you couldn't see the figure. That was the whole trick of the place and the whole insult of it. You were standing inside one of the largest pieces of applied geometry on the planet and your eyes told you it was a park with some humps in it.

Frik had found the eighth gate and was standing in it like a man who'd located the bathroom. "Okay," he called. "Okay. So this is a gap in a wall. Yes? This is just — there's a gap here. People walked through here. It's a door."

"It's a door," Naila agreed.

"It's a door to *what*."

"To the next bit of grass."

"See, that's what I'm saying." He came toward her across the open interior, arms out, a man in a graphic tee under three jackets. "Because the official line — and I want to say I have read the official line, I did the reading, I want a sticker — the official line is that these eight

gaps point at the Moon. Specifically. To within half a degree. And I look at this, Naila, and I see a gap. A gap in some dirt. I do not see a precision instrument. I see where the lawnmower turns around."

This was Frik working. He did his thinking out loud and at people, and the thing he was doing right now was the most honourable thing in his whole repertoire: he was trying as hard as he possibly could to make the wonder small, so that if it survived him, it had earned the right to stay big.

The Lens was forty feet off, low to the wall, not shooting yet. Waiting for the light to mean something. He had a way of holding the camera against his chest like it was warm.

"Tell me the survey," Naila said.

"Hively and Horn," Frik said immediately. "Eighty-two. Earlham College — and here's the part I *like*, here's the part that should worry you if you're a debunker, which I am, professionally — they came here to take this apart. Physicist and a philosopher. They thought the Moon stuff was wishful. People had been muttering about lunar alignments for years and these two showed up to do it properly and *kill it*." He spread his hands. "And then they couldn't. They found all of it. Every standstill point the Moon has, going up and going down, north end and south end — eight of them — and the gates and the wall axes line up to those points to about half a degree. They went to bury it and they ended up confirming it."

"That's the cleanest kind of result there is," Naila said.

"That's the kind of result that makes me itch."

She walked back toward the centre of the figure, where the surveyors' line ran — the long sightline, the one that mattered. From here, if you faced the right way, you looked down the throat of the whole construction: across the open fifty acres, between two of the wall segments, and out through the far gateway where the bank stepped aside to let the world in. A clean tube of air a third of a mile long.

She crouched. Sighted along the turf with one eye, the way you'd sight down a rifle, or a length of pipe.

The gate framed a piece of sky over the eastern trees. Just sky. Pale, going gold at the bottom. But it was a *chosen* piece of sky. Somebody

had built fifty acres of earth to make a frame for that exact slot of horizon, and the slot wasn't aimed at where the sun came up, and it wasn't aimed at anything you'd care about on an ordinary night.

It was aimed at where the Moon came up on the most extreme night of an eighteen-year-long argument the Moon was having with itself.

"Here's the thing that gets me," Frik said, dropping into a crouch beside her, following her line. "The *sun* is easy. I'll give them the sun for free. Sun goes up, sun comes back, longest day, shortest day, you can sort the sun in one year flat. A kid could do the sun. But these gates aren't sun gates. Right?"

"They're not sun gates."

"They went *harder*. They went for the worst possible target."

He wasn't wrong, and it was the part she'd been turning over since they got here. The Moon doesn't keep the sun's tidy schedule. The Moon's rising point on the horizon swings north and south every month, fast, a wide nervous arc — but the *width* of that arc breathes, slowly, getting wider and then narrower and then wider again, and one full breath of it takes eighteen years and change. At the widest, the Moon rises further north and sets further south than it ever otherwise will. That's a standstill. That's the extreme. That's the night this gate was cut for.

And you cannot see it.

Not in one night. Not in one year. You watch the Moon come up over those trees for a whole year and you've seen the monthly swing a dozen times and you've learned exactly nothing about the eighteen-year breath, because the breath is too slow and too wide and your one year is a single held note inside a song you cannot hear the shape of.

Naila felt the size of that and went very still inside it.

"You've gone quiet," Frik said. "You've got the face."

"The gates are right," she said. "That's not in dispute. Hively and Horn settled the gates."

"Then what."

"The gates being *right* is the small problem."

Behind them, boots on wet grass. Tessa Redwing came up the slope with two travel mugs and the particular unhurried walk of someone who had watched a great many visitors stand exactly here and say exactly this. She handed one mug to Naila and kept the other. She did not offer Frik one, and he had the grace to understand he hadn't earned it yet.

"You found the line," Tessa said.

"It's hard to miss once you crouch."

"Most people never crouch." She turned and looked down the sight-line herself, out through the far gate, and something in her shoulders changed — not reverence exactly, more the set of someone checking that a thing they're responsible for is still where they left it. "Hively and Horn," she said, without being prompted. "You're at the eighty-two survey."

"Frik did the reading," Naila said. "He wants a sticker."

"They did good work," Tessa said. "I'll say that plainly. They came to disprove it and they had the integrity to publish what they actually found instead of what they came for. That's rarer than it should be." She sipped. "And."

"And?" Frik leaned in.

"And it's a survey. It measures the dirt. It's true and it's careful and it tells you the walls point where they point." She let a beat go by, watching the light come up the slope toward them. "It doesn't tell you what it cost to know where to point them."

The Lens lifted the camera now. Naila heard the shutter once, soft, aimed not at any of them but down the long line of the gate.

"Walk it with me," Tessa said, and started across the fifty acres toward the far gateway, and they went.

It took a while. That was part of it — Naila understood that walking it slowly was itself an argument Tessa was making with her feet. You did not skim this. The wall on their left ran beside them the whole way, low and grass-furred and absolutely straight, straight in a way that earth doesn't want to be, straight the way only a decision can be. Basket-loaded. That was the word that wouldn't leave her. No carts, no animals to pull them, no metal, no plans drawn on anything that

survived. A wall this long and this true was a number of baskets she didn't want to calculate because the number was a kind of devotion and she wasn't ready for it before coffee.

"Twice as precise as Stonehenge," Frik offered, to the air. "That's your lot's line. The History Connection."

"It's a fair line," Tessa said. "Stonehenge does the sun. Stone, and the sun, and the sun comes round every year to check the work." She didn't slow down. "We did the Moon. And the Moon doesn't come round every year to check anything."

They reached the far gate. Standing in it, the interior of the Octagon opened behind them, the whole figure they still couldn't see, and ahead the land fell away toward the trees and the slot of sky the gate framed.

"All right," Tessa said. She set her mug down on the top of the wall. "You want the thing the survey doesn't carry. Here it is, and I'll go slow because everybody nods at this part and almost nobody lets it land." She looked at Naila when she said it. Chose her, the way you choose the person in the room who'll feel the weight. "The cycle these gates are built for is eighteen years and seven months long, near enough. Eighteen point six. To find where that gate goes — to *know* that the Moon, at its widest, comes up *there* and not a thumb's width to the left — you have to watch the Moon come up over that horizon for the better part of two decades. And you have to keep a record while you do it. Mark the spots. Hold the marks. Because no single night tells you, and no single year tells you, and your own eyes will lie to you if you trust the swing you saw last month."

Frik had gone uncharacteristically still.

"Two decades," Tessa said. "And then think about who started it. The person who set the first marker on that horizon — who decided this was worth watching, and started keeping count — was not the person who finished. Couldn't have been. You start that record young and you're not laying out a fifty-acre octagon at the end of it. Somebody else finished. Somebody who got the marks handed to them. And before *them*, somebody had to have already done it once, just to know the shape was worth chasing — to know there even *was* a cycle, before you'd build a city of dirt to hold it." She picked her mug back up. "So the survey tells you the answer is right. It doesn't tell you

the answer was kept. By people who knew, every one of them, that they would die before the figure was complete, and kept it anyway. Handed the count forward. Trusting the next set of hands to not drop the count.”

The sun cleared the trees then, and the frost finished going, and the slot of sky in the gate filled with ordinary morning. No Moon. Wrong time of the month, wrong year of the cycle, nothing to see. Just the place where a thing was true once a generation and the people who’d built a frame around it so that when the rare night came the whole town could stand here and watch the Moon walk up the centre of the gate like it had been invited.

“The Moon’s not even here today,” Frik said quietly. It wasn’t a heckle. It was the realest thing he’d said all morning.

“It’s almost never here,” Tessa said. “That’s the whole point you keep wanting to skip past. They built fifty acres for a night that comes around once every eighteen years. They built it so exact that when it *does* come, the Moon clears that gate within half a degree of dead centre. And not one of the people who started the record got to stand here and see if they’d got it right.”

Naila looked down the line, back through the figure she couldn’t see, and the thing she’d been circling all morning finally sat down inside her with its full weight, and it was heavier than she’d let herself expect.

The gates being right was the small problem. The survey had closed that one. Hively and Horn had walked in to kill the wonder and instead they’d nailed it to the ground at half a degree, and the half-degree was real and checkable and not going anywhere.

The big problem was the one the half-degree *implied* and could not explain. To put a gate where these gates are, somebody had to already know a number that no living person can find inside their own life. The cycle is invisible to a single pair of eyes. It is invisible to a single lifetime. You cannot watch it close. You can only inherit the watching, and add to it, and hand it on still unfinished, and trust.

She didn’t say any of that. It wasn’t hers to say. She looked at Tessa, and Tessa looked back at her with the patience of someone who already knew the shape of the question forming, who had carried it

longer than any survey, and was deciding, slowly, how much of it she was going to lend the strangers on the wall.

“So who kept it?” Naila asked. The smallest possible version of the question. “Across all those lifetimes. Who held the count.”

Tessa picked up her mug, and looked out the gate, and did not answer right away.

The Lens lowered the camera.

“That,” Tessa said, “is the right question. And it isn’t the one you came up the hill thinking you’d ask.” She started back down the wall toward the centre, and over her shoulder, almost kind: “Walk it again. We’ll get to who. We won’t get to all of it today.”

Frik watched her go. “She didn’t tell us who kept it.”

“No,” Naila said.

“They had to use the *same ruler* at the place down south. Fifty-three miles. The same — Frik, the *same unit*. You don’t share a ruler across fifty-three miles and two hundred years by accident. Somebody taught somebody. Somebody kept the *length* the same way they kept the count—”

“I know.”

“So who—”

“She told us we don’t get to who today.” Naila looked down the long straight wall, basket by basket, toward the centre where the line ran out through the gate to the slot of sky that held nothing this morning and would hold the whole Moon, dead centre, on a night none of them would get to choose.

The cycle was invisible to a single human lifetime.

Somebody had seen it anyway.

That was the door, and Tessa was the one holding it, and she’d just shown them exactly how far it would open and no further.

Chapter 6 — The Almanac Nobody Saw

The frost held the field flat and white at seven in the morning, and the Lens shot it the way you shoot a thing you don't fully understand yet — wide, then wider, then a single frame down at his own boots to remember the cold was real.

Naila was already out on the wall.

She walked the crest of the Octagon's south embankment with her arms held a little away from her sides, the way she did when her body was busy doing the arithmetic her mouth wasn't. Two thousand years of grass had softened the wall into something a golfer would call a swale, but from up here you could feel it was a made line. The Lens framed her against it. The fairway markers were still in the ground a hundred yards off — little flags, a sand bunker raked into a crescent, a wooden bench with a brass plaque thanking somebody for forty years of membership. He left them in the shot. They belonged. The whole strange grief of the place was in the fact that they belonged.

"Okay," Frik said, jogging up the slope with a tape measure and a print-out flapping. "So talk me through it like I'm a golf cart. Somebody stands here—"

"Not here." Naila didn't turn. "There. The gateway."

She pointed across fifty acres to the far gap in the earthwork, where the wall broke and the causeway ran out toward Observatory Circle. The Lens raised the long lens and found it: a notch of pale sky framed in two grass shoulders, a thousand feet away and dead ahead, and centred — he could feel it before Frik's tape confirmed anything —

centred the way a doorway is centred when somebody loved the room.

“You stand at the centre,” Naila said. “You watch the Moon come up through that gap. And on one morning, once, the Moon rises as far north as it will ever rise. It clears the gateway.” She breathed. “Within half a degree of the exact middle. That’s the OHC line, twice as precise as Stonehenge, and it’s not marketing, Frik, I checked the survey, it’s *true*—”

“I believe you.” Frik was crouched, sighting along the tape. “I’m not the potato. I just want to know how anybody *found* the morning.”

That was the question. The Lens lowered the camera and let it hang. He had spent a career standing where the question was hardest and pressing the shutter instead of asking it, and the habit had cost him things he was still counting. So when Frik asked the right question, he listened with his whole body.

“Found the morning,” Frik said again, to himself this time. He stood up. “Right. So I’m a Hopewell sky-watcher. Hundred BCE. I want to know where the Moon rises. I watch for a year.” He turned in place, mapping it on the air. “The Moon’s a liar, though. It doesn’t rise in one spot like the Sun roughly does. It swings — way north, way south — every *month*. Twenty-eight days, the whole arc, north to south and back. So in one year I see the swing — what, a dozen times?”

“Twelve, thirteen full swings,” Naila said.

“And every month it’s a *different* width of swing.” Frik’s hands stopped. “Oh. Oh, that’s mean.”

“There it is,” said Naila, and for a moment she almost smiled, which on Naila was a structural event.

Tessa Redwing came up the causeway on foot, no cart, in a wax jacket with a travel mug and the unhurried gait of a person who had walked this ground in every season and was not going to be rushed across it by anybody’s morning. The Lens had asked, the first day, whether he could photograph her on the works. *You can photograph the works*, she’d said. *I’ll tell you when I’m in them*. He’d kept the frame loose around her ever since. It was her edge to set.

"He's got it," she said, nodding at Frik, who was vibrating slightly. "Mostly."

"It's mean," Frik told her, delighted. "The swing changes width every month."

"And the *peak* of the width changes too." Tessa took a slow sip. "Some years the Moon's furthest-north rise is further north than other years. Creeps out, year on year, like a tide coming in over a tide. Reaches the maximum—" she lifted her chin at the far gateway "—and then for years it pulls back in. The whole breath of it, in and out."

"How long's the breath?" said Frik, already wincing because he could feel the shape of the answer coming.

"Eighteen years," Tessa said. "And about seven months."

The Lens watched Naila go very still on the wall.

"Eighteen point six," she said quietly. "That's the standstill cycle. That's what Hively and Horn found in the walls. All eight points — the far-north moonrise, the far-south, the sets, all of it, in the gateways." She looked down the field. "I knew the number. I've known the number for years. I didn't—" She stopped. Naila did not say *I didn't realise*; she let the gap stand, and the gap said it. "Eighteen years is most of a life out here. To find the *edge* of the breath, you'd have to watch one rise a month, mark it, and not lose the marks, until the marks themselves outlived the watching."

"You don't see it," Frik said. The delight had drained out and left something steadier. "That's the thing. You can't see it. You could stand on this exact wall every clear night for *ten years* and the pattern wouldn't close. There's no morning where it *clicks*. You'd just have a pile of marks that don't mean anything yet."

Hecklefish, riding in the Lens's jacket pocket the way he did when the morning was good, surfaced enough to be heard. "*So the genius move is, you die before the answer. Love that for them. Real strong life-goals energy.*"

"He's not wrong," said Tessa, who had taken to the fish faster than anyone the Lens had ever introduced him to. "That's exactly the part people can't hold. They want a genius. One person, one flash, one

aha. There's no *aha* here. There can't be." She turned the mug in her hands. "There's a girl who watches the eastern horizon and cuts a notch in something — a post, a tally, we don't know what kept the count, it doesn't survive. There's the man she taught to read the notches after her eyes went. There's *his* apprentice, who never met the girl, adding to a record she started, for a reason none of them will live to see proved. That's the instrument. Not the walls. The walls are just where they wrote the answer down once they finally had it."

The Lens lifted the camera then. Not at Tessa. At the gateway — the empty notch of sky a thousand feet off, with frost on the grass shoulders and a highway humming somewhere behind the south wall, and a sand bunker raked smooth in the foreground like a small white scar. He held the frame for a long time. *The almanac nobody saw*, he thought, and did not say it, because saying it would have been captioning a thing that was perfectly able to mean itself.

The Foundation rep arrived with the sun properly up, which the Lens privately thought was the most honest thing about him.

His name was Aaron Voss and he was good at his job, which was the problem with him. He had a soft voice and a real reverence for the place and a laptop bag that probably cost more than the crew's van, and he came across the causeway with two of his own people and an ephemeris program already open on a tablet he held the way other men hold a winning hand they're trying not to grin about.

"Dr. Redwing." Genuinely warm. "Naila. You're seeing it too." He turned the tablet so the screen caught the light, a black field of plotted dots, every standstill point modelled to a decimal Hively and Horn could only have dreamed of in '82. "We've run the alignments forward and back against modern lunar theory. The agreement is — honestly, it's better than the published figures. We're tightening the half-degree to a quarter. I'd love to get the crew on record for the rollout. The story tells itself: ancient observatory, modern math, the precision *confirmed*." He said *confirmed* the way you'd lay down a trump card.

"It was confirmed in 1982," Naila said.

"It was *suggested* in 1982." Gentle correction. "We're closing it."

The Lens kept the camera down and watched the field instead. He had photographed men like Voss in a dozen countries — the ones who arrive after the hard thing is over with a clean shirt and a press release, and the awful thing was that Voss meant it, he wasn't lying, the dots were probably real. That was what made the room dangerous. Not malice. Sincerity with a publication date attached.

"You can model it to a quarter degree," Naila said slowly. The Lens recognised the tone; it was the one she used when a pattern was wrong in a way she hadn't located yet. "Sure. You've got four hundred years of telescopes behind you and a satellite that—" She shook her head. "That's not what they had. That's not the achievement. You're confirming our own arithmetic with our own arithmetic. They *built* it before any of the arithmetic existed."

"Which is exactly the wonder we'd be selling," Voss said, and to his credit he heard the word leave his mouth and tried to walk it back. "Sharing. The wonder we'd be sharing."

Hecklefish again, muffled, vicious, fond. "*He said the quiet part with a footnote.*"

Frik, who had been quiet a suspiciously long time, finally spoke. "Can I ask the boring question." It wasn't really a question. "Your model. It tells us the Moon *did* clear the gateway on the right mornings. Cool. Stipulated. Does it tell us how the people who built the gateway *knew that*, before they had your model?"

Voss opened his mouth.

"Because that's the only interesting fact on this whole field," Frik went on, "and your tablet can't touch it. You're proving they were right. We already knew they were right. The question is who *kept the record long enough to find out they were right* — and that data didn't survive, did it. There's no spreadsheet from 80 BCE. There's just the answer carved into the dirt and zero working shown." He spread his hands. "You can't publish the working. Nobody can. The working was a hundred and fifty years of people's lives and it's gone."

The Lens watched Voss take that, and to the man's credit again, he took it like a professional — a small recalculation behind the eyes. "Then the alignment's all we *can* responsibly present," he said. "The measurable part."

“No,” said Tessa.

She didn’t raise her voice. She just put the word down in the middle of the field where everyone could see it, and the Lens felt the morning rearrange around it.

“You can present a measurement,” she said. “You can’t present a record that isn’t yours to read aloud.” She looked at the gateway, not at Voss. “You keep saying *the data*. There’s no data here, Mr. Voss. There’s a count somebody kept for longer than they were alive, on faith that the keeping mattered, and they handed it down a line of people whose descendants are the ones the Park Service is consulting this month — not the ones with the better ephemeris. The alignment is a *fact*. What it’s *for* is a relationship. You can publish the fact. The relationship isn’t on your tablet, and it isn’t mine to sell you either.”

It was, the Lens thought, the cleanest thing he’d heard said on a field in years, and he did not photograph her saying it, because she had not told him she was in the works.

Voss was quiet. One of his people shifted. Far off, on the road behind the south wall, a truck changed gear and the sound came over the grass like weather.

“I’d still like the rollout,” Voss said at last, but something had gone out of it, the trump card lowered.

“I’d like to walk High Bank with the crew before anybody rolls out anything,” Tessa said, and it was not a no, and it was not a yes, and it was entirely hers to leave hanging.

They came off the wall as the frost let go and the green turned ordinary and the place stopped being a calendar and went back to being a field with old earth in it.

The Lens hung back, the way he’d trained himself to, and got the shot he actually wanted: not the gateway, not Voss, not even Naila. He got Frik standing at the centre point with the tape measure dangling forgotten in one hand, looking down the thousand feet of grass to the far gap, doing the thing in his head that nobody could do on the ground — running the impossible relay forward, watcher to watcher

to watcher, none of them living to the end of it.

"It can't be one person," Frik said, mostly to himself. "It's the one thing that's actually impossible here. Not the math. The *math's* fine. You just need the lifetime." He turned. "You need *more* than the lifetime. You need somebody to start counting a thing they'll never see finish, and somebody else to believe the count enough to keep it going after the first one's gone, and a *third* one to trust two dead people's marks enough to build it into the ground at half a degree."

"You need a relay," Naila said.

"You need a relay," Frik agreed.

Hecklefish, from the pocket, very quietly for once. *"...and somebody had to hand the next runner the baton without ever seeing the finish line. Okay. Okay, that one got me."*

Tessa was already walking back down the causeway toward the cars, not waiting, because it was her ground and she didn't have to. Over her shoulder, not turning: "Fifty-three miles south, they built another one. Same gateway. Same Moon." A beat. "Same ruler."

She let that lie in the frost behind her, and kept walking, and did not explain.

Chapter 7 — The Shared Ruler

The Ohio History Connection's reading room smelled of cold coffee and toner, and Naila had been happy in it for four hours.

She had pulled her chair too close to the long table — close enough that the edge pressed a line into her forearm — because the survey sheets did not lie down flat and she needed to read them the way they wanted to be read, all at once, from above. The 1848 Squier and Davis plate. The 1982 Hively and Horn overlays. A modern theodolite trace that someone at OHC had laid over both, a ghost of dotted blue where the old ink wobbled. Three centuries of people drawing the same thing, getting better cameras for it, and the thing under all the cameras not changing one inch.

"You've gone still," Frik said. "You only go still for two reasons and we're indoors, so it's the good one."

"Bring me High Bank." She didn't look up. "Whatever they have. The Chillicothe folder."

Frik did the thing where he rolled across the floor in the wheeled chair rather than stand, because standing was a whole separate negotiation with his attention, and came back with a grey clamshell box labelled in a curator's careful hand: HIGH BANK WORKS — SURVEYS — DUPLICATES — DO NOT REMOVE. The LENS, by the window, lowered the camera he hadn't been taking pictures with and watched the light instead. He had learned to tell when a room was about to mean something. He was usually a beat ahead of the rest of them on that.

“High Bank’s the other one,” Frik said, half to himself, the way he checked his own facts out loud so somebody could catch him. “The only other octagon. Fifty-three miles south. Built by the same — well. The same people, probably. Generations apart, maybe. We don’t actually know the order.”

“We don’t,” said Dr. Redwing.

She had come in quietly with her own folder and a paper cup and had not announced herself, which Naila appreciated more than she could have said. Tessa Redwing did not perform being in a room. She set her cup down on a coaster she’d brought from somewhere, because that was how you treated a table that held a record, and she looked at the spread of plates without touching them.

“Go on,” she said. “You were about to find something. I’d rather watch you find it than tell you.”

Here is the official story, and it is not in dispute, and it ought to make the hair stand up on your arms before anybody says one strange word.

Two octagonal earthworks exist in the world that were built by the Hopewell. Two. There is the one at Newark, the fifty-acre figure with its eight five-hundred-and-fifty-foot walls and its gateways open to the Moon. And there is the one outside Chillicothe, on the high bank of the Scioto River, which gives it its plain and lovely name. They were raised by people who carried the soil up basket by basket — no metal, no draft animals, no writing to leave a note for the grandchild who would lay the next course. They built ceremonial geometry on a continental scale, and they built it twice, and the second time they built it fifty-three miles away.

Ray Hively and Robert Horn — a physicist and a philosopher, who came to Newark in 1982 hoping to put the astronomy claim quietly to bed — found that High Bank, too, holds the lunar alignments. Same hard target. Same eight standstill points. The mainstream archaeology does not flinch from this. It is in the literature. It is, as far as it goes, settled.

That is the official story. It is true. Hold it a second longer before the floor tilts.

Naila laid the Newark plate and the High Bank plate side by side and turned the southern one ninety degrees, then back, then thirty more, hunting the angle where her eye stopped fighting it.

The octagons did not match. That was the first thing, and Frik said it before she could.

“They’re not the same shape. The Newark one’s — squashed. High Bank’s rounder. Different orientation entirely.” He tapped, not touching, a half-inch above the paper. “So that kills it, right? If somebody had a master template they’d have stamped them out identical.”

“That’s the right question,” Tessa said. “Keep asking it.”

Naila was not listening to either of them now. She had stopped listening the way you stop hearing a fridge.

Because the octagons were different, yes. The site of Newark sloped one way and the site at High Bank sloped another, and the builders had turned each figure to take the horizon it actually had, the real eastern hills where the real Moon would really rise. That was not sloppiness. That was the opposite of sloppiness. You only turn the figure to fit the sky if the sky is the point.

But there was a number under both of them that did not turn.

“Give me a ruler,” she said. “A real one. Metal.”

Frik produced an architect’s scale from the bottom of his bag, because of course he had one, and Naila set it against the Observatory Circle on the Newark plate and read the diameter off the survey’s stated scale, and wrote it down, and then she did not measure it again at High Bank. She didn’t need to. She had read the figure off the High Bank survey’s own notation an hour ago without knowing it would matter, and it had sat in her head the way a wrong note sits in a tune you can’t stop humming.

She wrote two numbers on the back of a sheet of scrap and turned it to face the table.

The circle that anchors the Newark octagon — the Observatory Circle, a thousand and fifty-four feet across, true to within a stride of a man’s

pace over a fifth of a mile. And the circle that anchors High Bank, fifty-three miles south, on a different river, on different ground, turned to a different horizon.

The same length. Not close. The same.

“That’s the OCD,” Frik said, very quietly, the comic-book voice gone right out of him. “The Observatory Circle Diameter. They use it as a unit. The circles at Newark sit six of those apart — the spacing’s the diameter, over and over, like somebody pacing it out. And it’s the same length down at —” He stopped. “Naila.”

“I see it.”

“No, but — Naila.” He had his hand flat on the table now, not pointing at anything, just needing the table to be solid. “That’s a ruler. That’s an actual ruler. Somebody made a ruler and carried it fifty-three miles.”

“Or carried the number,” she said. “You don’t have to carry the stick. You have to carry the knowing of how long the stick is.”

The LENS lifted his camera at that and then, deliberately, did not press the shutter. He set it down on the windowsill. Some things you photograph. Some things you stand inside of.

Now the easy story comes back to make itself small, because that is what easy stories do, and it deserves its hearing.

It’s a coincidence. Two circles, two builders, the same width by chance — earthworks come in a range of sizes and you’d expect some overlap. Fair. Except a coincidence is one circle the same as another circle. This is a *unit*. It’s the spacing at Newark, six times over. It’s the diameter that ties the square to the circle to the octagon in a single squared-circle figure you could draft today and not improve. It’s that whole grammar, carried south, and used again, by people who then turned their octagon to a completely different angle because they were too good at this to copy lazily. You don’t get a grammar by accident. You get a width by accident. Not a grammar.

It’s a fluke of the survey. Different surveyors, different errors, the numbers only *look* matched. Also fair, and the OHC has chased exactly this — which is why there are theodolite traces ghosting the old

ink, why the modern numbers are tighter than Squier and Davis ever managed. The match held. Tighten the measurement and it gets *better*, not worse, which is the wrong direction for an artefact of sloppy nineteenth-century chains.

It's a lone genius. One brilliant Hopewell surveyor who walked between the sites. Now we're getting warmer, and now we're also getting colder, because the alignment built into both figures is the eighteen-point-six-year standstill — and no genius, however brilliant, lives long enough to find that cycle alone. You cannot reach the number by being clever for one lifetime. You reach it by handing the watch forward. So the lone genius doesn't shrink the wonder. He multiplies it. He needs a school behind him, and grandparents in front of him, and the same length agreed by all of them before he was born.

Each time you try to make it smaller it gets larger in your hand. That is the tell that you are holding the real thing.

"How," Naila said. It came out flat, almost a complaint, and she heard it and didn't soften it because Tessa, of all people, would not need it softened. "How do you keep a length right across fifty-three miles and across — how long? A century? Two? Without writing it down. Without a metal standard you can take to a museum. How do you make sure the stick down there is the same as the stick up here when the person who cut the first stick is four generations dead?"

Tessa Redwing did not answer quickly. She had a way of letting a question finish being asked even after the words stopped.

"You're thinking of it as a stick," she said. "A thing that wears, that gets lost, that has to be copied and copies drift." She turned the Newark plate a few degrees, the way Naila had, settling it. "What if the length isn't in the stick. What if it's in the circle."

Naila waited.

"You can lose a measuring rod," Tessa said. "You can't lose a thousand-and-fifty-four-foot circle laid into the ground at a tolerance you can pace and check. The circle *is* the standard. It's not a record of the unit. It's the unit, kept full-size, walkable, out under the sky where the next generation comes of age inside it and learns

the length in their own feet before anyone hands them a single word about it.” She looked at Naila, and there was something careful in it, something that was deciding how much to give. “We didn’t keep the number in an archive. We kept it in a place you stand in. And then we made a second place, far enough away that you couldn’t see one from the other, and we made it the same size, so the people there would carry the same length in their bodies without ever walking north.”

The room was quiet. The highway was a long way off and you could hear it under everything, the way you always can in Newark, a sea-sound behind the walls.

“That’s not a copy,” Frik said. “That’s — you’d have to teach it. Person to person. Somebody who knew the north circle in their feet walks south and helps lay the new one and checks it against — against what’s *in* them.”

“Now you’re closer to it,” Tessa said. And she did not say what *it* was. She let *it* sit on the table between the plates, unowned, and she did not invite them to pick it up.

Naila stood, finally, because her arm had gone dead against the table edge and because she could not sit any longer with two numbers that were the same. She walked to the window where the LENS had set his camera down, and the late light was doing the thing to the parking lot it does to everything, making the ordinary look like it had been waiting to be seen.

Two octagons in the world. The whole world. People built ten thousand things in earth across that continent and across those centuries, and exactly twice they built this figure, this eight-walled gate-keeper of the Moon — and both times they reached for the same circle, the same thousand and fifty-four feet, the same squared-circle grammar tying it all together, and both times they turned the gates to catch a Moon they could only have learned about by keeping a record longer than any of them lived.

She had come into the room wanting the satisfaction of the click. The thing solved. The pattern closed in her own two hands. And the pattern was closing — she could feel it close — and the closing did not

belong to her. It belonged to the people who had kept a length in their feet for two hundred years so that a circle in the north and a circle in the south would agree without a word passing between them, and to the woman across the table who knew, and was choosing, slowly, how much of the knowing to set down where outsiders could pick it up.

Frik had gone back to the plates, was holding the architect's scale against the southern circle one more time, just to feel the number arrive again.

"Two," he said. "There's only two octagons. And they're the same size to —" He checked the scale, checked it again. "To inside what you could walk off and not notice the error. Fifty-three miles apart. Naila, somebody agreed on this. Somebody agreed."

Tessa picked up her paper cup. She had not touched the plates the whole time, Naila noticed. Not once.

"Somebody agreed," she said. "Yes."

She did not say who.

She did not say how the agreement was kept, beyond the part she'd given them — the circle as the ruler, the body as the archive — and Naila understood, watching her gather her folder, that the part she'd given them was the part that was the crew's to know. The rest of it, whatever it was, the *why two and not ten*, the *who taught the first feet the length*, went out of the room in the THPO's folder, under her arm, off the table, home.

The same ruler. Twice. Fifty-three miles down a river that didn't run between the sites, laid into ground that didn't share a horizon, by people who had never seen the finished pattern of either.

Somewhere under the floorboards of the official story, the fish surfaced and said, to nobody, *you'd think a person who could carry a number that far would've written it down somewhere we could read it*, and went back under, and did not stay to argue the point.

Chapter 8 — Chasing the Standstill

The OHC conference room smelled of burnt coffee and toner, and Frik had covered the long table with paper.

Not in the metaphorical sense. Actual paper — a roll of brown banner stock he'd liberated from a closet, taped down at both ends with the desperate authority of a man who has had three coffees and intends to be vindicated. On it he'd drawn the Octagon. Eight walls, the gaps between them, the causeway running off to the Observatory Circle like a tail. He'd drawn it freehand first and it had come out looking like a stop sign that had eaten a doughnut, so he'd started over with Naila's protractor, and the second one was better, and the third one was actually good, and now he was drawing lines.

That was the part Naila wouldn't let him do casually. Lines from the centre of each gateway, outward, into the empty white where the horizon would be.

"You're at fifty-two point one," she said, not looking up from her laptop. The screen showed a sky simulator, a black field of stars wheeling over a flat green plane, a date readout ticking through the centuries. "Azimuth. Off true north."

"It looks like fifty-three."

"It's fifty-two point one. Use the protractor like a person."

Frik used the protractor like a person.

Across the table, the LENS had a camera in his lap and wasn't using it, which for the LENS was a kind of resting state, the way other people

had a screensaver. He was watching the door. Dr. Redwing had said she'd come by once the morning's call with the Foundation's lawyers was done, and the way she'd said *the Foundation's lawyers* had put a particular flatness in the room that nobody had wanted to disturb.

"Okay," Frik said. "Okay. Here's my problem."

"You always have a problem," the LENS said. "It's your contribution."

"My contribution is *load-bearing*." Frik pointed his pencil at the brown paper. "Here's the deal. I believe the math. I do. Hively and Horn, Earham, 1982, came to call it a coincidence and left calling it the most precise lunar instrument north of the Rio Grande, fine, I have read it, I have *underlined* it. But."

"There it is."

"But the Moon doesn't rise in *one* spot. That's the whole thing people don't get when they hear 'aligned to the Moon.' The Sun, sure, you can build a doorway to the solstice sunrise, the Sun behaves, the Sun is a good employee. The Moon is — the Moon is *all over the sky*." He flapped a hand at the ceiling, indicating the Moon's general lack of professionalism. "Month to month it swings way north, way south, north, south. Eighteen-ish degrees of horizon, back and forth, every twenty-seven days. So if you build *eight* walls and *eight* gateways pointing at *eight* directions, and the Moon spends all month sweeping back and forth across that whole arc anyway —"

"Then you'll catch *something*," Naila said. "Eventually. By accident."

"Then you'll catch something eventually by accident. *Thank* you." Frik sat back, vindicated, briefly. "That's my hole. That's the one I want to dig. Eight gates is a lot of gates. You throw eight darts at a wall the Moon's going to walk past, statistically you hit a moonrise. It doesn't *prove* anybody planned it. It proves you drew enough lines."

He meant it. That was the thing he'd want a reader to know about him, if readers thought about him at all: he wasn't *performing* the doubt. He'd lain awake on the air mattress in the rental with the doubt sitting on his chest like the cat he didn't have. He wanted the wonder, badly. He'd wanted it since he was nine and reading about this stuff under the covers. Which was exactly why he didn't trust it. The thing you want is the thing that lies to you.

“So model it,” Naila said. “You don’t get to feel clever about a hole you haven’t measured.”

This was, Frik would later admit, the most loving thing anyone said to him that week.

They modelled it. Naila ran the simulator, Frik ran the paper, and they did the cheap version of what two physicists had done with theodolites and forty years less computing. They picked the eight gateways. They took the azimuths off the published survey because Frik refused to trust his own protractor for the final numbers and Naila agreed, on the grounds that he was correct to distrust himself. They told the software: here is Newark, Ohio. Here is the flat eastern horizon, give or take the trees the Hopewell would have cleared. Run the Moon. Run it for the whole cycle.

“Watch the spread,” Naila said.

On screen the Moon climbed the eastern edge night after simulated night, and its rising point crawled north, then stalled, then crawled south, then stalled, a slow pendulum that took a month to swing once. Frik watched the rise-points pile up. They didn’t fill the arc evenly. They bunched at the ends — the Moon lingered at the extremes of its swing the way a swing lingers at the top before it comes back. The bunching had a name. He could feel the name in his mouth and refused to say it yet.

“It loiters,” he said instead.

“It loiters,” Naila agreed.

And the loitering points moved. Not within a month — across years. The northernmost rise the Moon reached this year wasn’t the northernmost it would reach next year. The whole envelope of the swing breathed wider, then narrower, over — Frik checked the readout, watched the date counter spin — over about eighteen and a half years. At its widest, the Moon at its most northerly rose further north than the Sun ever managed even at midsummer. At its narrowest, nine-odd years later, the swing pulled in close.

“Okay,” Frik said quietly. “Okay, that’s the thing. That’s the eighteen-six.” He didn’t sound triumphant anymore. “The far edges. The maximum north the swing ever reaches, once every eighteen and a half years. That’s the *major standstill*.”

“That’s the major standstill.”

He looked at his eight lines on the brown paper. He’d been ready to argue eight darts, a wall, dumb luck. But the simulator wasn’t showing him a wall the Moon swept past. It was showing him that the Moon only paused at the very edges, and the very edges only reached their furthest *once in nearly two decades*, and that furthest point — the rise-point at the major standstill — was a *single, specific azimuth*.

Not an arc. A spot.

“Where’s it land,” he said.

Naila was already overlaying it. She brought up his gateway azimuths, the ones from the survey, and laid the simulated standstill moonrise on top.

The door opened.

Dr. Redwing came in with a travel mug and the particular composure of a person who has spent ninety minutes on the phone declining, politely, to be hurried. She looked at the brown paper, at the protractor, at Frik’s face, and something in her eased a fraction — not warmth exactly, but the recognition of work being done in the open rather than behind her back.

“You’re at the alignments,” she said.

“He’s trying to break them,” the LENS said.

“Good.” She set the mug down and didn’t sit. “People should. The two who proved it tried to break it first. That’s why it held.” She nodded at the screen. “What’s your objection?”

Frik told her. The eight gates, the wide arc, the darts. He told it straight, didn’t soften it for the room, because he’d have wanted someone to tell him straight. *Eight directions is a lot of directions. The Moon covers most of that ground anyway. Catching a moonrise in one of eight gateways isn’t a miracle, it’s arithmetic.*

Redwing listened all the way through. She didn’t interrupt, which Frik had learned to respect in the small number of people who could do it. When he finished she picked up his pencil — asked first, with her eyebrows — and put the tip on the screen, on the loitering points at the northern extreme.

“You’re right that the Moon covers the arc,” she said. “You’re wrong about which part you have to hit.” She tapped the extreme. “It doesn’t pause in the middle. It pauses *here*. And the place it pauses moves a little every year for eighteen and a half years, and then it’s reached as far as it’ll ever go, and *that* —” the pencil rested on one point, “ — is the standstill. You don’t get to throw eight darts at the whole arc. You only get to throw at the edge. And the edge is narrow. And you only see where the true edge is once in a generation.”

“So how would you even know where to build the gate,” Frik said. He wasn’t arguing now. He was asking. There’s a difference and the room could hear it. “If the furthest north only happens once every eighteen years — you’d build the gate, and then wait nine years to see the Moon *not* get there, wait another nine, and only then —”

“Only then could you check your work,” Redwing said. “Yes.” She let that sit. “You’d have to have watched the one before you build it. Which means somebody watched the one before *that*. Which means it was written down — kept — in the only way they had to keep it, which was in people. Handed forward. By the time you laid the basket that fixed that gateway, the person who’d seen the *previous* standstill from this spot was likely dead. You built it on their word. And whoever checks it after you is going to need *your* word, because you’ll be gone before it swings back.” She looked at the eight lines, fond and unhurried. “That’s not eight darts, Frik. That’s a relay nobody alive ever ran the whole length of.”

The fish — Frik thought of him as the fish, the heckling skeptical thing that lived in his own head and looked, for reasons he’d stopped examining, like a cranky aquarium occupant — the fish opened its mouth to say *sure, but show me the number*. And Frik, for once, agreed with the fish. He wanted the number.

“Show me where it lands,” he said. “Versus the gate.”

Naila turned the laptop.

She’d done it cleanly. The Octagon’s gateways as surveyed; the simulated major-standstill moonrise dropped on top; and the angle between them rendered out to a decimal because Naila would no more round it for drama than she’d lie. The crucial gateway — the one the moonrise was supposed to clear as it cleared the figure end to end, the longest sightline across the whole fifty acres — sat under

the modelled rise-point.

The gap between the centre of the gate and the rise-point was half a degree. A hair over, a hair under, depending on how you treated the horizon trees and the wall heights, all of which the published survey treated with more care than Frik's banner paper ever could. But it didn't move out of half a degree. Frik nudged the parameters. He bumped the assumed observer height. He shifted the horizon. He gave it every chance to fall apart, because that was the job, because a hole you don't honestly test isn't a hole, it's a wish.

It stayed inside half a degree.

Half a degree. Frik knew what half a degree was, in the sky, with no telescope, with cord and stakes and baskets and the naked eye. Half a degree was about the width of the Moon itself. They'd built a gateway, across the better part of a quarter-mile, by hand, out of dirt carried in baskets by people without writing or metal or the wheel — and lined it up on a point in the sky that only shows its true edge once every eighteen and a half years, and they'd hit it to within the Moon's own width.

"That's not a fluke," Frik said. It came out smaller than he'd planned.

"No," Redwing said.

"You can't get to half a degree by accident. Not on the one azimuth that only —" He stopped. He looked at the eight gates again and this time saw the argument from the other side: not *eight chances to get lucky*, but *eight points that all have to be right at once*, the whole figure a single instrument, every wall a line in a sentence, and the sentence was the sky. "Eight gates isn't more chances. It's eight things that all had to land." He laughed once, not happily, the laugh of a man whose favourite objection just got up and walked off the table. "I came in here to throw darts. You're telling me they didn't have a dartboard. They had a *blueprint*, and the blueprint was a clock that takes eighteen years to tick, and they couldn't read the whole clock face in a lifetime, so they — they passed the clock down. Until somebody got to *finish* it."

"Somebody got to finish it," Redwing said. "And nobody got to see both ends."

She picked up her mug. The Foundation's lawyers had given her a

deadline, the room knew, and the deadline had not been generous, and she had walked in here anyway to watch a man with banner paper try and fail to make the wonder smaller, because she'd seen people try to make it smaller before and she would rather watch it done in good faith.

"You should be there for it," she said, at the door. "Not on a screen. The standstill moonrise. It comes back —" she gave the date, the next major standstill the Octagon would catch, and it was close, close enough that the Foundation wanted to own the cameras for it, which was a whole other fight, which was the fight, really. "Stand at the right end. Watch it clear the far gate. Then you'll stop arguing with the half-degree." She almost smiled. "Or you'll argue better. Either way I'd take you."

The door shut.

Frik looked at his brown paper, at the eight lines he'd drawn to break a thing and had only managed to draw more carefully than before. On the laptop the modelled Moon sat in the gateway, half a degree off centre, exactly as patient as it had been for two thousand years.

"Half a degree," he said to no one.

"Half a degree," Naila agreed, and saved the file.

The LENS, who had not lifted the camera once, lifted it now — and then, watching the two of them lean over the same number from opposite sides of the same table, lowered it again without a frame.

Chapter 9 — Permission and Custody

The conference room at the Ohio History Connection smelled of carpet glue and yesterday's coffee, and through the window the LENS could see the parking lot, the strip of grass, and beyond it the first faint swell of an earthen wall that most of the people in this building drove past without seeing. He had set the camera down on the table, lens cap on. A small discipline. There were rooms where you shot and rooms where you watched, and he had learned — slowly, expensively — to tell them apart.

This was a watching room.

Eight people around a table built for twelve. A woman named Diane from OHC with a binder she kept squaring against the table edge, the way you straighten a thing you cannot control. Two more staff, younger, taking notes nobody would read. Tessa Redwing at the head of the table — not because anyone had seated her there, but because she had walked in last and that was the chair that was open, and she had taken it without apology, set down a thermos, and folded her hands.

And across from her, the Foundation. A man named Marcus Hale, who wore the kind of soft grey jacket that costs money to look like it costs nothing, and who had a way of leaning in when he spoke, as if every sentence were a confidence.

"We're not asking for anything that isn't already in the public record," Hale said. "The Hively-Horn work is forty years old. Published. Cited a thousand times. What we're proposing is synthesis. A modern ephemeris run against the gateways, lidar over the whole figure, a

clean popular account that finally tells the world what this place *is*.” He spread his hands. “We’d credit OHC. We’d credit everyone.”

Synthesis, the LENS thought. He had heard the word in other rooms, in other countries, usually from people standing on something that belonged to someone else.

Naila was beside him, very still, which was its own kind of loud. She had a notebook open and she had not written anything in it for four minutes. Her eyes were on the laminated site plan in the middle of the table — the Octagon and the Observatory Circle, joined by their little neck of causeway, the whole figure rendered in clean printer’s lines as though it had come out of a drawer rather than out of the dirt. The LENS watched her watch it. He knew that stillness. It meant she had found the part of the diagram that was wrong, or the part that was right in a way nobody else had noticed, and she was deciding whether it was hers to say.

“Twice as precise as Stonehenge,” Frik said, to no one, reading off the wall where OHC had hung the phrase on a banner in their own brand colours. “That’s your line, isn’t it? OHC’s.” He tipped his chair back. “It’s a good line. Bit unfair to Stonehenge, but a good line.”

Diane almost smiled. Hale did not.

The official story sat in the room with them, and it was true, and the LENS gave it its due the way you give a dead man his due — quietly, with both hands. A hundred BCE to four hundred CE. Three thousand acres of geometry raised basket by basket by people with no metal, no draft animals, no writing, no plan they could photograph from above and check. Eight walls of five hundred and fifty feet. A circle a thousand feet across true to within a few feet all the way round. And the gateways set so the Moon, at the far end of an eighteen-and-a-half-year swing, would rise and clear the far gate within half a degree of the figure’s exact centre.

Half a degree. The width of the Moon itself, more or less, held over two thousand years and a golf course.

It was not woo. That was the part the LENS kept coming back to. There was nothing here to debunk. The miracle was on the plaque. The miracle was *settled*.

“Here’s our difficulty,” Diane said, and squared the binder again. “The

lease transition isn't finished. We have public-access obligations now that we didn't have eighteen months ago. We have a tribal consultation protocol that —" she glanced at Tessa, the smallest tilt of the head, asking permission to describe a thing in front of the person it belonged to — "that we are bound to honour. And we have the academic community, who would like very much to be first."

"Everyone would like to be first," Tessa said.

It was not sharp. That was what the LENS noticed, lifting the cap off the camera now after all, slow, because some sentences want a face on them. It was not sharp at all. She said it the way you'd point out the weather.

Hale leaned in. "Dr. Redwing, with respect, the science here doesn't change hands. Hively and Horn didn't ask anyone's permission to measure the alignment, and the alignment is what it is. The Moon does what the Moon does. We're not interpreting a *belief*. We're documenting a *fact*."

The LENS had the shot before Hale finished — Tessa's hands, unmoved, and the thermos sweating a small ring onto the table, and behind her through the window the green line of the wall. He did not press the shutter. He had a rule, lately, about whose moment a thing was, and this was not his to take yet.

"Walk it with me," Tessa said.

—

The frost was off the grass by the time they got out there, but the cold had stayed in the ground, and the LENS could feel it come up through his boots as they crossed onto the Octagon. Hale had come too, in shoes that were wrong for it. Diane stayed by the cars, on her phone, doing the thing OHC was always doing now, which was standing in the gap between a law and a promise.

Up close the walls were not impressive. That was the trap of the place, and the LENS loved it for that. They were low, grassed, soft-shouldered, the height of a man's chest where they survived at all. You could step over them. Kids had sledded them. For a hundred years men in spiked shoes had walked them looking for golf balls, and the walls had let them, the way the truly large things let the small things crawl on them.

You could only see it from the air. That was the whole cruelty and the whole point. The people who built it never saw it. Not one of them. They moved baskets of earth for a shape they could only hold in their heads, passed down by mouth and by hand, basket to basket, and the shape did not close in a lifetime, and the *cycle* it tracked did not close in a lifetime, and they did it anyway.

Naila stopped at the causeway, where the Observatory Circle let into the neck.

“Six,” she said.

“Six what,” said Frik.

“The circles.” She wasn’t looking at any of them. She was looking at a thing that wasn’t there, the way she did. “The diameter of that one.” She pointed at the Observatory Circle, a thousand feet of grass curving away gentle as a held breath. “It’s the unit. You can lay the whole site out with it. The two circles sit six of them apart. The Octagon’s set off it. It’s a ruler. They had a ruler and they kept using it.” She turned, finally, and the LENS caught it — the small open astonishment she never bothered to hide, because hiding things cost her more than it cost other people. “It’s not a drawing. It’s *measured*. Somebody measured this.”

“Somebody measured it,” Tessa agreed.

“And then somebody measured High Bank with the same one.” Naila’s voice did the thing it did when the pattern got too big for the room. “Fifty-three miles. The same length. The only other octagon they ever built, and the same —” She stopped. “Sorry. Sorry, I’m —”

“You’re not wrong,” Tessa said.

Hale had got his phone out. Not to make a call. To make a note. The LENS watched his thumbs move and felt something go cold and familiar in his own chest, the thing he’d felt in marketplaces where men photographed grief and called it coverage. *He’s writing her down*, the LENS thought. *He’s writing the moment down so he can have it later.*

“The shared unit,” Hale said, looking up, bright now, “is exactly the kind of thing a synthesis brings out. People don’t know about High Bank. They should. The network angle is — frankly it’s the headline.

A continent-scale standard of measurement, two thousand years before —”

“It’s not a headline,” Tessa said.

The fish would have had something here. The LENS could almost hear it, that wet little heckle from off-camera somewhere — *oh, it’s not a headline, it’s a five-part docuseries with a merch tier* — but the fish, when it mattered, knew to shut its mouth, and this mattered.

“It’s not a network,” Tessa went on, mild, walking them slowly along the inside of a wall so they had to follow or be rude. “It’s not a *system*. You keep reaching for the word that makes it a thing you can hold. A network, a standard, an instrument. I understand why. It’s how you were taught to make something real — you give it a noun and you publish the noun.” She stopped at a gateway, one of the gaps in the wall where, on a particular morning at the far end of eighteen and a half years, the Moon would come up and thread the eye of it. “It’s a *record*. Somebody watched, and kept what they saw, and gave it to somebody who wasn’t born yet. And that one watched, and kept it, and gave it on. For longer than any of them lived. That’s not a system. A system you could buy. That —” she nodded down the long sightline, out past the grass, out past the highway you could just hear, a thin tide of trucks — “that you can only inherit.”

Frik, who could not help himself, who the LENS loved precisely because he could not help himself, said: “There’s the soft one, though. The size ratio. The Great Circle to the Observatory — somebody’s done the maths, it comes out about one-point-one-three, and the Moon’s own size changes by about —” he scrunched his face — “one-point-one-two-something between when it’s close and when it’s far. Apogee, perigee. Point-three percent off. People say they built the *Moon getting bigger and smaller* into the size of the rings.”

“People say that,” Tessa said.

“Did they?”

She looked at him with something that was almost fondness. “It’s a pretty number.”

“That’s not a yes.”

“No,” she said. “It isn’t.”

And she let it sit there, the pretty number, unconfirmed and undenied, and Frik grinned at it the way you grin at a card someone won't turn over, and the LENS understood that this was the discipline of the place — not the alignment, which was a fact, but the refusal to close the thing that didn't need closing. The willingness to stand in the gap and say *maybe*, and mean it as a gift and not a failure.

—

Back in the watching room the light had gone flat and afternoon.

Hale made his ask cleanly, the LENS would give him that. No villain in it. A research window. The lidar flight, the ephemeris run, a publication embargo of six months so that the Foundation's account came out first, ahead of the popular flood. In exchange, money OHC needed, named credit, a permanent display.

"And the descendant nations are consulted throughout," Hale added, to Tessa, the leaning confidence again. "Of course."

"Consulted," Tessa said.

"At every stage."

The LENS watched Diane look at her binder and not open it. He watched the two young staff stop writing. He watched Tessa unscrew her thermos, pour a half-inch of something into the cap, and not drink it — just hold it, warming her hands, taking the room's clock and slowing it to her own.

"You can have the flight," she said at last. "The lidar. It's over public ground now, and the OHC can permit it, and there's nothing up there your machine can take that the morning sun doesn't already give to anyone who stands here long enough."

Hale started to speak.

"The dawn is ours," Tessa said. "The major standstill morning. When it comes around — and you know when it comes around, you've got the same ephemeris I have — that morning belongs to the people whose grandfathers' grandfathers kept it coming. You can be in the parking lot. You cannot be in the gate."

"That's —" Hale's confidence wobbled, found its feet. "That's a significant restriction on—"

“Yes,” Tessa said.

Diane, very quietly, opened the binder. “We can structure the permit that way,” she said, and the LENS heard the relief in it, the relief of a person handed back the authority she’d been holding without owning. “Survey access, with a closure on the consultation dates. I’d need to draft it.”

“Draft it,” Tessa said.

So the Foundation got its window. That was the shape of the turn, and the LENS named it to himself coldly, the way you name a thing so it can’t surprise you later: Hale would fly his lidar, and run his ephemeris, and write his synthesis, and very possibly publish it first, and call it documentation of a fact. He had the law’s narrow gap and the money to fill it.

But not the morning.

In the parking lot, packing the camera, Frik said, “We’re going to have to work around him.”

“Yeah,” Naila said. She was looking back at the wall, at the gateway, doing the geometry again in the air with one finger, the angle of it, the half a degree.

“He gets the picture from above,” Frik said. “The whole figure. The thing nobody who built it ever saw.”

“He gets the picture,” Naila agreed. “Tessa gets the morning.” She lowered her hand. “I think she knows which one’s the record.”

The LENS got the shot then — not of the wall, and not of Hale’s grey jacket disappearing into a rented car, but of Tessa’s thermos, abandoned on the conference table through the window, the cap still full, the half-inch of coffee gone cold in it, waiting for a sunrise eighteen and a half years in the keeping that she had just, very calmly, refused to sell.

Then he lowered the camera, because that one was hers.

Chapter 10 — The Great Circle's Secret

The frost hadn't burned off the infield yet, and Naila could see her own track across it — one clean line of bruised grass, toe-heel, toe-heel, the way you walk when you're counting and don't trust your legs to keep the count for you.

The Great Circle sat on the south edge of Newark like a bowl somebody had pressed into the suburbs and then forgotten. From the parking lot it read as a low green berm, the kind a city builds to hide a retention pond. You walked up to it expecting nothing. Then you came through the gateway and the ground fell away on the inside — *the ditch is on the inside* — and the wall stood up over your head, and the interior opened to twelve hundred feet of flat grass ringed by an embankment you could not see over, and the highway behind you went quiet the way a room goes quiet when somebody important stops talking.

"Okay," said Frik, turning a slow circle with his arms out, like a man checking the size of a parking spot. "Okay. That's bigger than it has any right to be."

"Three hundred and sixty-six metres," Naila said. She didn't mean to say it out loud. The number had been sitting on her tongue since the car.

Dr. Tessa Redwing came up the inner slope behind them, unhurried, hands in the pockets of a canvas coat that had seen a lot of mornings like this one. She didn't look at the Circle. She looked at the three of them looking at it, which Naila had begun to notice was a thing Tessa did — measured you against the place before she let you have

any of it.

"The eight-foot walls are mostly slumped now," Tessa said. "Two thousand years of rain. When it was new it would've stood higher than a man could see over from the floor. You came in here, you were *inside* something. The world was on the other side of that."

"And the ditch—" Frik started.

"On the inside. Yes." A small, dry tilt of the mouth — she'd done this part before, many times, for many Friks. "A fort puts the ditch on the outside, so the attacker has to climb down and back up under your spears. This one's backwards. You'd be handing the high ground to anybody who wanted it. So." She let it sit. "Not a fort."

"Not a fort," Frik agreed, happily, the way he agreed with anything that killed a lazy explanation.

The LENS had drifted left along the embankment, camera down at his hip, not raised. He was reading the light off the frost, the long blue shadow the wall threw across the floor. Naila watched him decide not to take the picture. He did that more than he took them, lately.

She started walking the diameter.

She didn't announce it. She just set off across the floor on the cleanest line she could hold, eyes forward, counting her own pace because she'd measured her own pace in a hotel hallway the night before — eighty-one centimetres, repeatable to within a thumb's width if she didn't think about it — and a person who knows the length of her own stride is a person who carries a ruler everywhere she goes.

The ground was dead level. That was the first thing her feet told her, before any number did. Twelve hundred feet of grass with no draw to it, no tilt, nowhere the water would pool. You don't get level ground for free. Level ground is a decision. Somebody had stood here with cord and water and patience and made the dirt agree to a plane, and then made a circle on the plane so round that when she finally reached the far embankment her line met the wall square, dead-on, no drift left or right that she could feel.

Four hundred and fifty-one paces. She did the arithmetic without wanting to. Three hundred and sixty-five, three hundred and sixty-six

metres. Close enough that the gap was *her*, her stride loosening over the distance, not the Circle.

"They didn't have a metre," she said, when she'd walked back to where Tessa stood. Her voice came out flatter than she meant. It did that when the thing in her chest got too big for the casual register. "The metre's French. Eighteenth century. So three hundred and sixty-six metres is — that's just *our* number landing on *their* circle. They had their own unit."

"They did." Tessa watched her. Waiting to see if she'd find it.

"The OCD." Naila said it like a held breath. "Observatory Circle Diameter. Whatever they actually called it. Ten fifty-four feet — also not theirs, also our number — but the *length* is theirs. It's real. It's a stick they cut once and copied." She looked across the floor. "This circle isn't that one. This is smaller."

"The Great Circle is its own figure," Tessa said. "The Octagon and the Observatory Circle are the famous pair, three miles north, where the golf course was. People come for those. This one—" she turned her hand palm-up at the great quiet bowl, the frost, the joggers they could no longer hear— "this one gets walked through on a lunch break. It's the one they think they understand."

The way she said *think* had a hook in it.

Frik had found the thing he always found, which was the hardest version of the easy question.

"Here's what I don't get," he said, jogging to catch up, gesturing with his whole arm because his hands couldn't keep up with his head. "Round is easy. Round is the *easiest* shape. You put a peg in the dirt, you tie a cord, you walk it. Kid does it with a stick and a string. So — round, fine. Why's round impressive?"

"Try it," Tessa said.

"What?"

"Twelve hundred feet of cord. Find me a length of cord twelve hundred feet long that doesn't stretch when it's wet, doesn't sag in the middle, doesn't shrink when it dries. Walk it through grass that grabs

it. Keep the peg from drifting. Keep the *tension* identical all the way around — because the second your cord stretches an inch on the far side, your circle isn't a circle anymore, it's an egg, and from up high everybody can see it's an egg." She wasn't unkind about it. She was just laying the cord on the ground in front of him, foot by foot. "Now do it again three miles north and make *that* circle the exact same size to within four feet across a thousand. With baskets of dirt and no writing and your grandfather's memory of how long the stick was."

Frik opened his mouth. Closed it. "The stick," he said. "It always comes back to the stick. Somebody's *minding the stick*."

"Somebody's minding the stick," Tessa said.

Somebody better be minding it now, Naila thought, because the Foundation's rep had been at the OHC offices again that morning, the man with the good coat and the gentle voice and the phrase *narrative custody*, and she'd watched Tessa go very still and very polite, which was the stillness of someone who had been measured for a coat she did not intend to wear.

"It's a place you came to be made over," Tessa said.

She'd walked them to the centre, or near it — there was no marker, just a sense of equidistance Naila's feet kept trying to confirm. Tessa stood with the embankment ringing her at an even height all the way round, and for a moment the geometry did the thing the geometry was built to do: it put a person in the exact middle of a horizon they had made themselves.

"This isn't my read to give you in full, and I won't," Tessa said, even-voiced. "Some of it isn't written down on purpose. But the shape of it — you can stand inside the shape of it. You come through the gateway, you leave the world. The wall takes the world away; you can't see the town, can't see the other mounds, can't see anything but sky and the inside of the figure. You're held. Whatever you came in carrying, you set it down in here, and you carry something else out." She looked at the gateway, the one true gap in the ring, where the embankment opened and the path ran out toward the place the others stood. "A door you go through to become. The circle's the wall around the becoming."

Nobody said anything for a while. The LENS, who had been to the worst rooms human beings make for each other, lowered his camera the last inch to his side and just stood in it.

"It's a church," Friik said, quietly, for once not puncturing anything.

"It's older than that word and it's still being used," Tessa said. "So — be careful with *was*."

It was the LENS who set the last beat going, and he did it without a word — just turned, finally, and lifted the camera, and framed the two circles the way you can only frame them in your head, because you can't get both in one shot from the ground. North circle, south circle. Big and bigger.

"They're different sizes," Naila said. She was still standing in the centre. The number had been waiting in her since the parking lot and she'd been refusing it because she knew where it went. "This one and the Observatory one. They're not the same circle. They're a—" her hands made the shape before her mouth had the word—"they're a *pair*. A ratio."

"They are," Tessa said.

"Great Circle, three sixty-six metres. Observatory Circle, the OCD." Naila ran it. She was good at this; it was the one thing the world had always let her be unambiguously good at, the place where her wiring stopped being a thing teachers wrote notes home about and started being a key. "Twelve hundred feet to ten fifty-four. That's —" she didn't reach for paper, she didn't need paper— "one point one three. One point one three one."

The morning held very still.

"And the Moon," Friik said. He'd gone wide-eyed. He'd read the same dossier she had, on the drive, out loud, doing voices. "The Moon's not the same size all month. Closest it gets, perigee — farthest, apogee — the disc looks bigger up close. The ratio of those is—"

"One point one two seven," Naila said.

"That's—" Friik's hands stopped, which never happened. "That's the *same number*. That's a third of a percent off. The ratio of your two

circles is the ratio of the close Moon to the far Moon. They built the *size of the Moon* into the size of the dirt.”

He turned to Tessa, lit up like a kid who'd found the last page of the comic, and Naila felt it in her own chest too, the pull of it, the gorgeous click of a pattern closing — and then she watched Tessa not close it.

“Maybe,” Tessa said.

“Maybe?” Frik said. “Doctor, it's a *third of a percent*.”

“It's a third of a percent,” Tessa agreed. “And a circle has to be *some* size. Pick two circles, any two circles, and you'll get a ratio, and somebody clever can find a number in the sky to match almost any ratio you hand them, if they go looking hard enough.” She let Frik feel the cold of that. “The standstill alignment up at the Octagon — that one I'll stand on. That's measured, that's eight points to half a degree, that's a cycle you can't see in one life. That one's a rock. This—” she nodded at the two circles only the camera could hold together— “this is pretty. This might be true. I'm not going to tell you it's true, because I don't know that it's true, and the day I pretend to know a thing I don't know is the day somebody puts a fence around it and sells tickets.”

She wasn't preaching. She said it the way you'd point out that the ice on a pond is thin in one spot and you've watched a man go through.

“Frik's reaching,” Naila said. She said it gently. She loved him for the reach.

“Frik's reaching,” Frik admitted, grinning, deflating with grace, the way only he could. “Frik reaches. It's a *condition*.”

“It's a good number,” Tessa said, and for the first time something in her face opened a hairline. “I'll give you that. It's a good number. I think about it.” A beat. “I'm not going to hand you what I think about it.”

And there it was — the line she would not cross, drawn clean in the frost. Not the alignment, which was theirs to share and which they shared, measured and cold and undeniable. This other thing, this third-of-a-percent maybe, this whisper that two thousand years ago somebody pressed the changing face of the Moon into the ground

in two sizes of circle three miles apart — that one Tessa Redwing turned in her hand, and considered, and kept.

Naila stood in the exact centre of a horizon that people had made by hand, basket by basket, minding the stick across more years than any one of them had been alive, and she did the thing she was still learning to do, which was nothing. She didn't solve it. She didn't need to be the one who closed it. She stood in the gap where the answer should be and let the draught come through, cold and clean off the wall, and found — to her own surprise — that the not-knowing was not a hole at all.

It was a door. The Circle's last one. And it was not hers to walk through.

"One point one three one," she said again, softly, to nobody, just to feel the number stand there in the open air where she had decided to leave it.

Out past the embankment, faint, the highway started up again.

Chapter 11 — The Ditch Inside

Frik had a theory, and the theory was a fort.

He liked it because it was the boring answer, and the boring answer is where you should always start. A man two thousand years ago looks at a flat place by a river, thinks *I would not like to be murdered here*, and builds a wall. Walls are not mysterious. Walls are the oldest idea humans have after fire and after complaining about the people on the other side of the wall. So Frik came to the Great Circle ready to like the fort, the way you'd warm up to a dog that doesn't bark — relieved, almost. No baskets-across-generations. No invisible lunar cycle making his teeth ache. Just engineering and the human wish to be left alone.

He stood at the gateway in the grey morning and the wall came up over his head, and for a second the fort theory was beautiful.

Eight feet, give or take. A clean green rampart curving away on both sides into the mist, swallowing the noise of Route 79 the way a closed door swallows an argument in the next room. Inside it was hushed. The grass was wet and the frost was going to silver in the low light. You could believe a hundred people had stood here once with the wall at their backs and felt the particular animal comfort of being inside something.

"Okay," Frik said, walking the line of it. "Sell me. Three hundred and sixty-six metres across. Walls higher than a man. You're the chief of the I-Don't-Want-To-Be-Murdered tribe. This is your wall."

"It's a good wall," Naila said. She wasn't looking at the wall. She was

looking at the ground at the base of it, the way she looked at everything — as if the thing had a number written on it she was politely waiting for it to say out loud.

The LENS was a little behind them, camera down at the moment, just walking. He did that on the first lap of a place. Looked with his own eyes before he made them anyone else's.

"It's a *great* wall," Frik went on, warming up. "Look at this thing. You'd need — what — basket after basket, you'd need years, but people build forts, that's the most normal thing in the world to spend years on. I'm not embarrassed by this theory. I want it on record that the fort theory is dignified. There's even a name for the people who think it's a fort."

"There were," Naila said. "For about a century."

"Right! Whole reputable nineteenth-century guys. Squier and Davis came through, drew these as 'works of defence,' beautiful surveys, very serious moustaches." Frik patted the wall like the flank of a horse. "Earthworks. *Works*. Military word. Forts. I rest my—"

"Frik."

"—case, what."

"Walk to the bottom of the wall," Naila said.

He walked to the bottom of the wall.

There's a moment your body knows a thing before your mouth catches up, and Frik felt his feet do something his theory hadn't authorised. The ground didn't meet the wall and stop. It dipped first. Right at the foot of the rampart, on the *inside* — the side where his hundred imaginary defenders were standing — the earth fell away into a long, shallow, grass-filled trough. A ditch. It ran the whole curve of the circle, hugging the inner foot of the wall, going off into the mist in both directions like a moat that had got the geography exactly backwards.

He stood in it. The wall went up on his left. The wide open interior of the circle lay on his right, flat and innocent.

"Huh," Frik said.

“Say it,” Naila said.

“The ditch is on the inside.”

“The ditch is on the inside.”

He climbed up out of it and went and stood on the crest of the wall to make sure he wasn’t being fooled by the morning. He wasn’t. From up here you could read it like a cross-section in a textbook: the high ground was the wall, and the low ground — the ditch, the place an attacker would have to slog through, the place you’d dig to make life hard for whoever was coming — was *behind* the wall. Tucked safe inside, where your own people would be.

“Okay,” Frik said, “okay, okay, let me — let me defend the fort.” He held up both hands. “Maybe they dug the dirt from inside because it was handy. You build the wall out of the nearest mud. You end up with a ditch where you took the mud. It doesn’t *mean* anything, it’s just where the dirt came from. That’s not a moat, that’s a — borrow pit. They borrowed the dirt.”

He liked *borrow pit*. It had the right boring music to it.

“Then dig it outside,” Naila said. She came up onto the wall beside him, and pointed down the curve, where the inner ditch ran on and on, deliberate as a drawn line. “If you’re going to dig out the dirt anyway, and you want a fort, you dig it on the *outside*, where it does double duty — wall *and* trench against the enemy, in one move. Every hillfort in the world does exactly that, for exactly that reason, because dirt is heavy and you only want to move it once.” She let that sit. “They moved it the wrong way. On purpose. Around the entire circle. For no defensive benefit and at extra cost.”

“Maybe they were bad at it,” Frik tried, and heard how that sounded the instant it left him.

“These are the people,” Naila said gently, “who built an octagon four miles from here that tracks the Moon to half a degree.”

“Yeah,” Frik said. “Yeah, scratch that one. They were not bad at it.”

A heron came low over the far wall, saw them, thought better of the whole arrangement, and banked off toward the river. Somewhere out past the rampart a truck changed gears on the highway and the sound came over softened, the way everything from out there came over

softened, which Frik was beginning to notice was *the point* — the wall didn't keep arrows out, it kept the world quiet.

"So it fails," he said. "The fort fails."

"On the dirt," Naila said. "It fails on the dirt. You don't even need the calendar. You just need to notice which side the ditch is on."

They'd worked themselves down off the wall and into the great flat bowl of the interior when Tessa found them — Dr. Redwing, in a field coat with a thermos, coming through the gate with the unhurried walk of someone who had been to this place enough times that it owed her nothing and she owed it everything. Frik had learned in three days not to perform the cleverness at her. She had a way of letting your cleverness finish and then saying the actual thing, and you didn't want to be still talking when she did it.

"You found the ditch," she said. Not a question. She'd watched them from the gate. She'd watched a lot of people find the ditch.

"We found the ditch," Frik admitted. "I had a whole fort. It was a good fort."

"It usually is." She poured something into the thermos cup and didn't offer it, just held it, warming her hands. "People want it to be a fort. I understand why. A fort, you know what it's *for* the second you see it. You can stop thinking. The wall is between you and the thing you're afraid of, and that's the end of the story, and it's a story everybody already owns." She looked up the green curve of the rampart. "This isn't that story."

"Because the ditch is on the inside," Frik said, fishing — wanting her to confirm the win, the way you do.

She didn't confirm the win.

"You're still reading it like a wall," she said. "High thing, low thing, which side keeps you safe. You're thinking about *out*." She turned, slowly, taking in the whole eight-foot ring of it standing around them in the mist, the gateway the only break in it. "Try the other direction. A defensive wall asks: how do I keep them *out*." A small pause, and her voice didn't rise, it dropped. "This one asks: how do I keep something *in*."

Frik felt the back of his neck do the cold thing.

“Keep what in?” he said.

And here is where a worse book would have her answer. Frik wanted her to answer. Naila, beside him, had gone very still — Naila who could read a squared circle like a face, standing there with her hands quiet, *waiting*, because she’d understood something Frik was still catching: that this wasn’t theirs to take.

Tessa drank from the cup. Let the question hang in the wet air with the heron-sound and the muffled truck.

“You stand in the middle of this,” she said at last, and she was looking at the centre of the bowl now, the flat empty heart of the circle where nothing was, where two thousand years of nothing very carefully was, “and the wall takes the world away. The road. The town. The arrow you were so worried about. All of it goes down under the rim. What’s left over the wall is the sky and the horizon line, and that’s *all* that’s left.” She nodded, more to herself than to them. “The ditch and the wall together. They don’t divide the inside from an enemy. They divide it from the *ordinary*. You go down into the ditch and up onto the floor and you’ve crossed out of one kind of place into another kind of place.” She looked at Frik, and there was something close to kindness in it, the kindness of someone handing a tourist back his small idea without breaking it. “You don’t dig a moat to do that. You dig it because the crossing has to be *made*, in the ground, with your hands, so the people coming after you cross it too. The barrier is the meaning. It’s a threshold you built so heavy that it’ll still be a threshold when nobody alive remembers the word for it.”

Nobody said anything for a moment.

“That’s a better fort,” Frik said quietly. It was the most respectful thing he had.

“It’s not a fort at all,” Tessa said. Not sharp. Just closing a door he’d been leaning on. “It never was. The ditch told you that in the first ten minutes. You just hadn’t decided to believe it yet.”

Later — laps of the rampart, the frost gone, the LENS finally up and working the long curve of the wall with the gateway breaking the line

— Frik couldn't let it lie all the way down. It's a disease he has and he knows it.

"Okay but here's the thing that's killing me," he said, half to Naila, half to the air. "The Great Circle here. The Octagon four miles that way with the Moon thing. They're not even the *same shape*. One's a ring, one's eight walls and a calendar. And then there's a whole other octagon fifty-odd miles south at High Bank doing the same Moon trick. Why isn't somebody saying these are three different ideas by three different—"

"The unit," Naila said.

"The unit."

"The Observatory Circle. Its diameter shows up over and over here, and it shows up at High Bank, fifty-three miles away." She said the number flat, the way she said all the numbers, but Frik had learned that the flatter Naila said a number the more it had cost her to keep her voice off it. "Same ruler. Two sites. Generations apart. Whoever laid out the second one used the length of the first." A beat. "You don't carry a ruler fifty-three miles by accident."

The LENS lowered the camera. Frik noticed him do it, and noticed Tessa, a way off down the wall, notice the LENS do it — the photographer choosing, for a second, not to take the thing.

"That's the part that gets me," Frik said. "Not the Moon. Not even the ditch. It's that somebody *taught* somebody the length. Same length, both sites, both Moons. Like a—" He stopped, because the word that wanted to come out was *school*, and the word after that was *who*, and he could feel himself reaching, feel the floor of the known thing dropping into the floor of the wanted thing.

"Don't," Naila said. Lightly. The fish-bite of it.

"I'm not reaching."

"You're about three words from reaching."

"I am *appreciating*," Frik said, with enormous dignity, "the verified fact that two thousand years ago a person here knew exactly how long another person's circle had been, and used it. That's it. That's all I'm doing. I'm a model of restraint." He looked back across the bowl, at the ring of wall standing patient in the warming light, the ditch

running its quiet wrong-way line around the inside of it. “I came here for a fort and I’m leaving with a threshold and a ruler and absolutely no idea who held it. This is a terrible vacation.”

Tessa, far off, didn’t laugh, but the line of her shoulders changed.

She had let them find the ditch. She had let them watch the fort die on the dirt. And the thing she hadn’t told them — the *what’s-kept-in* of it, the rest of the answer past the threshold — she still held, easy as the thermos cup, hers to pour out or carry away.

Frik looked at the gateway. The one gap in the wall. The single place the makers had left for the world to get in, or for whatever-it-was to get *out*, and he found he didn’t want her to say which.

He let it stand open.

Chapter 12 — Lines Across the Land

Fifty-three miles is nothing in a car. Naila kept saying it to herself on the way down — *fifty-three miles, an hour with the radio on* — because the number had been doing something strange in her chest since Newark, and she wanted it to behave.

High Bank Works sits south of Chillicothe, where the Scioto bottomlands run flat and green and the road dead-ends in a way that feels less like arrival than like the land deciding to stop pretending. There's no visitor centre worth the name. There's a fence, a field, a sign that under-describes everything behind it, and the long low swell of walls that you don't see until you're standing on them, the way you never see the Octagon either, the way these things refuse to be a postcard.

Frik got out of the van and turned a full circle with his hands on his head. "Okay. Okay. So this is the *other* one."

"The only other one," Tessa said. She had driven separately, in her own truck, and she said it the way you'd correct a child gently and on purpose. "The Hopewell built earthworks all over this valley. Circles. Squares. They built exactly two octagons." She let that sit. "One up there. One here."

The LENS was already low to the grass with the camera, not shooting yet, just looking, the way he did when he was deciding whether a thing deserved to be looked at by a machine. Naila stood at the foot of the nearest wall and felt the old vertigo — the conviction, physical and absurd, that she was standing inside a figure she could only understand from a height she would never reach.

The official version is clean, and it is true, and the people who study this place will tell you so without flinching. High Bank is a Hopewell ceremonial and astronomical complex, a circle joined to an octagon, c. 100 BCE to 400 CE, basket-loaded by hand over generations by people who had no draft animals, no metal, no writing — and who nonetheless laid a circle so true that surveyors with instruments come to measure it and go home humbled. It encodes the lunar stand-stills, the same hard eight-point cycle the Octagon up north encodes. Hively and Horn read it here too. None of this is contested. None of it needs a single grain of woo to be the strangest thing in three states.

“Show me the unit,” Naila said.

Tessa looked at her for a moment. Not unkindly. There was a thing Tessa did with a pause, a weighing, where you understood you were being decided about, and Naila had stopped finding it uncomfortable somewhere around day three. It was just accuracy. Tessa measured people the way the makers measured ground.

“You brought your own cord,” Tessa said. It wasn’t a question.

Naila had. A hundred metres of surveyor’s tape, a couple of ranging poles Frik had liberated from a hardware store outside Lancaster, and the number written on the back of her hand in marker so she wouldn’t fumble it: **1,054 feet**. The Observatory Circle Diameter. The OCD. The ruler that built Newark.

Here is the potato, and you should hold it honestly before you let it fall apart, because it falls apart better that way.

The easy story says: *coincidence*. Two big round things sixty miles apart, both circle-and-octagon, both built by the same broad culture — of course they’ll look alike. Build enough earthworks over five hundred years and some of them will rhyme. People love a pattern. People will find a face in a tortilla. You can’t get excited every time two ancient circles are roughly the same size, or you’ll spend your whole life excited and broke and writing to magazines.

It’s a good potato. It’s the responsible one. Frik tried it on out loud, walking the south arc with the ranging pole over his shoulder like

a fishing rod, because trying on the boring answer first is the only honest way to earn the interesting one.

“Roughly the same,” he said. “*Roughly*. That’s the word that does the work, right? Roughly is where all the crimes are committed.”

“Then let’s not be rough,” Tessa said.

So they weren’t.

They started at the Observatory Circle here — High Bank’s circle, the one the octagon hangs off — and Naila did the thing she does, which is to stop being a person and become a measurement for a while. She found the entry causeway, paced the chord, set Frik with a pole at one node and walked the tape out across the flat with the wind pulling at it and the highway humming somewhere behind the trees, that distant traffic-hush that follows you everywhere these places survive, the sound of the present leaning on the past and the past not noticing.

The circle came up true. Not roughly. The diameter at High Bank ran out under her tape and her steps and her cross-checks and it sat where it was supposed to sit, where the same diameter sits a county away in Licking, the same 1,054 feet give or take the tolerance of two people and a tape on a windy afternoon — which is to say, give or take *nothing that matters*.

“Talk me through it,” Tessa said. She wasn’t asking because she didn’t know. Naila understood that much. Tessa had walked this ground her whole professional life and a good deal of it before that, with her family’s names in it. She was asking the way a teacher asks — to hear whether the student had the thing in their hands or only in their mouth.

“The unit,” Naila said. “The diameter of the Observatory Circle up at Newark. That length. It’s not just a length they happened to use. It lays out the whole site. The circles up there sit six of these apart, centre to centre. Six OCD. It’s a module.” She paused with the tape sagging between her fingers. “And it’s *here*. The same module. The same ruler. Fifty-three miles south, built — when? Generations apart?”

“We don’t have it to the year,” Tessa said. “We have it to the lifetimes.”

“The lifetimes.” Naila tasted it. “So someone carried the number.”

“Someone carried the number.”

Frik had gone very quiet, which for Frik is a kind of alarm. He stood at his node with the pole planted and looked at the line of the tape running away from him to Naila, this hundred-metre fact lying across an empty Ohio field, and you could see him trying to make it into a coincidence and failing, the way you fail to un-see a word once you’ve read it.

“Hang on,” he said. “Hang on, hang on. A standard unit of length. That’s — that’s not a tool. That’s an *agreement*. You can’t have a shared ruler without people who agreed to share it. Across — how do you even —” He turned to Tessa, and there was the puncture-instinct firing, but inward now, at his own building excitement, the way the best of him does. “Okay. Stop me. Tell me why this is nothing. Tell me everybody used the same arm-span or paced the same way or — give me the boring version, because I want it and I can’t find it.”

“I can’t give you the boring version,” Tessa said, “because there isn’t one. Not that I’d lie to you about.”

The fish — and there is always a fish, somewhere just off camera, heckling the moment so nobody gets to be solemn about it — would point out here that a shared unit of measurement is the single most aggressively *un-mystical* thing a civilisation can have. It’s a standards committee. It’s a weights-and-measures office. It’s the least romantic possible explanation, and it is *enormously* more interesting than aliens, and that’s the joke the whole place is telling at the expense of everyone who came looking for a flying saucer.

They aligned the gates after lunch, with the light going long.

This is the part the chapter has been circling, and it has to be done with a cold eye, because the temperature of the eye is what separates this from a séance.

The octagon’s gateways are not random. At Newark, Hively and Horn established it past serious dispute: the walls and the openings between them frame the rise and set points of the Moon across the full standstill cycle, all eight of them, to within roughly half a degree. The

far gateway clears the major-standstill moonrise within about half a degree of the figure's exact centre. Twice as precise as Stonehenge, the Ohio History Connection says, and they are not a body given to overstatement.

High Bank does it too. Same alignments. Same Moon. Sighted along its own axis, the octagon here opens onto the standstill points like the one up north — a second instrument, tuned to the same impossible-to-watch sky, built by the same ruler.

Naila stood in the gateway Tessa pointed her to and sighted down the axis to the horizon, and the cold thing arrived in her chest where the warm thing about the fifty-three miles had been all day, and the two of them sat together without arguing, which is the only honest way to hold this place.

Because here's the hole, and it is not a hole in the evidence. The evidence is whole. The hole is in the *easy feeling* you wanted to have about it.

The 18.6-year standstill cycle is invisible to one person. You cannot find it in a year. You cannot find it in ten. There is no clever observer, no lone genius on a hill with good eyes and a free evening, who sees this pattern and goes home and builds it. The cycle is longer than the patience of any single watching. You find it one way only: somebody watches the eastern horizon, and marks where the Moon turns back, and keeps the record — and dies — and hands the record to someone who keeps watching, and marks, and keeps it, and hands it on. For the better part of a lifetime, before the *shape* of the thing is even knowable. And then they build the finished shape into the dirt, basket by basket, to half a degree, in two places, with the same ruler.

"You keep saying *handed forward*," Naila said. Her voice came out smaller than she meant. "The record. Handed forward."

"Yes."

"That's not a figure of speech for you."

Tessa stood in the gateway beside her and looked down the axis at the same horizon, and the LENS, off to the side and low, did not raise the camera, because some moments you photograph and some you let belong to the people standing in them, and the LENS, who has

gotten people killed by knowing the difference too late, knows the difference.

“My grandmother handed me things she got from people who got them from people,” Tessa said. “Not measurements. I’m not telling you there’s a secret cord in a box somewhere with the OCD knotted into it. I don’t have that, and if I had it, it wouldn’t be yours.” She said it flatly, no heat, a fence post going into the ground. “But I know what it is to be a link in something you didn’t start and won’t finish. To keep a thing accurately because keeping it accurately is the whole job, and the meaning of it is somewhere up ahead where you won’t get to stand.” She nodded at the gateway, at the gone Moon’s appointed door. “*That*, I understand from the inside. That’s not a mystery to me. That’s a *family*.”

Frik had wandered up the wall and now he came back down it sideways, sliding a little in the grass, and he had the look — the one where he’s about to push a soft thing too hard, and the value of him is that he knows it even as he does it.

“So if there’s a shared ruler,” he said, “and it’s at two sites, and it’s the same to within nothing — doesn’t that mean somebody *taught* it? Like — a person, a teacher, somebody who went from here to there with the number, somebody who —”

“Careful,” Naila said, before Tessa had to.

“No, I know, I know, I’m reaching, I’m watching myself reach —” Frik held both hands up, surrendering to his own arrest. “It just — it wants a single hand, doesn’t it? Your brain wants one guy. One genius. The Sky-Watcher of Ohio, striding the valley with his magic cord.”

“Your brain wants one guy,” Tessa agreed. “Mine doesn’t.”

And there it was — the turn, delivered not as a revelation but as a correction, the way she’d corrected Frik about the octagons in the parking lot, gently and on purpose.

“One genius dies,” Tessa said. “One genius takes the number into the ground with him. You can’t build *this*” — the gateway, the axis, the half a degree, the fifty-three miles, the lifetimes — “out of one genius. One genius is exactly what *can’t* do it. The cycle’s too long for a life.

The whole thing only works if the knowledge is bigger than any one person who held it. If it's *taught*. If there are people whose job is to hold the standard and hand it on and check the new ones against the old, so the ruler stays the same when the hands holding it have all changed." She looked at the two of them, the pattern-reader and the puncturer, standing in a doorway built by people who measured better than the woo-merchants would ever credit them. "You don't want a genius. You want a *school*."

The wind came across the flat and the highway hummed and somewhere up north, sixty road-miles away, the same doorway stood open to the same Moon, cut by the same length, by people who agreed on a thing and taught it and died and were replaced by people who agreed on the same thing, on and on, until the shape closed in the ground and nobody who'd started it was alive to see it close.

Naila wound the tape back in slowly, the marker number sweat-faded on the back of her hand, *1,054*, and she did not say the obvious thing, which was the question the fish would heckle and the question the whole season had been leaning on like the highway leans on the field: *who taught the teachers. Who set the length the first time. Whose ruler is it, before it's anyone's.*

She didn't ask it. Tessa wouldn't have answered, and it wasn't Naila's to answer, and the not-asking felt, for the first time on the whole trip, less like a door closing than like one being held open for somebody else to walk through.

"A school," Frik said quietly, trying it on, and finding it fit, and hating a little how much better it was than the magic cord.

"A school," Tessa said. "And it's still in session."

She got in her truck and drove back to Chillicothe, and left them standing in the gateway with the number on Naila's hand and the Moon's appointment in the dirt, fifty-three miles long, and the one length running the whole way between, set by a hand they hadn't read yet.

Chapter 13 — The Relay

The camera was off. That was the first thing the LENS did when Tessa Redwing started talking, and he was glad of it later, because there are moments a lens cheapens just by being raised, and he'd learned the cost of finding that out too late, in other countries, with other people's grief.

They were in the OHC conference room that smelled of carpet glue and old coffee, the blinds half down against a flat October light. Outside, past the parking lot, the Octagon sat in its long grass with the fairway mowed across its back like a careless hand. You could not see the geometry from here. You could not see it from anywhere on the ground. That was the whole point, and it had taken the LENS three days to feel the weight of it.

Tessa had a topographic survey spread on the table — Hively and Horn's, the 1982 one, the lines of the eight walls drawn clean and the gateways notched in. She wasn't pointing at it. She was looking at the middle distance the way people do when the thing they're describing isn't in the room.

"You want me to tell you how they knew," she said. "About the eighteen-point-six."

Naila nodded once, too fast, then caught herself and went still. The LENS had learned to read that stillness. It was not boredom. It was the opposite — Naila banking the fast thing down so the slow thing could come through.

"Everybody who comes here wants that," Tessa said. There was no edge in it. "How did people without writing track a cycle they couldn't see in a lifetime. And I get the question. It's a good question." She

put one finger on the eastern gateway of the Octagon, the far one, the one the major-standstill Moon clears. “But the question’s built wrong. You’re imagining one genius on a hill with a notebook he doesn’t have.”

“There’s no other way to know the cycle,” Frik said. He said it gently, for Frik. “You can’t see it. I ran the numbers. The Moon’s rise point swings north and south every month, sure, but the *envelope* of that swing — the widest the swing ever gets — that only comes around every eighteen and a half years. You’d watch your whole adult life and see it go wide once. Maybe twice if you started young and the gods were kind. You can’t fit a curve to two points.”

“No,” Tessa agreed. “You can’t.”

She let that sit. The highway hummed somewhere past the walls, the sound of it coming over the earthworks the way water comes over a weir, steady and indifferent.

“So you don’t fit the curve in one life,” she said. “You hand it forward.”

The LENS had photographed a lot of things people called sacred. Most of them were buildings, or objects, or the worn places where many hands had touched the same stone until it shone. This was the first time he’d watched someone describe a sacred thing that had no object at all. The thing was the *handing*. The thing was the relay.

“Think about who you tell,” Tessa said. “You’re old now. You’ve watched the eastern horizon since you were a girl — not every night, you have a life, but the nights that matter, the ones the keepers before you said to watch. The Moon’s risen as far north as you’ve ever seen it. You mark where. Not on paper. On the land — a post, a stone, a notch you cut in a sightline that points at the place on the horizon. And you say to the young one beside you: *it will come back to here. Not in your childhood. When you are as old as I am now. Watch for it. And tell the one who comes after you the same.*”

“And the young one believes you,” Naila said. Quiet.

“The young one believes the keeper before her, who believed the keeper before *him*, who believed —” Tessa turned her hand over, palm up, the gesture going back and back. “It’s not faith. It’s record. The

post is still in the ground. The notch is still cut. When the Moon comes back to the place the old woman marked — and it does, it always does, that's the gift of the thing — the young one isn't taking it on trust anymore. She's *checking the prediction*. The dead woman was right. The horizon proved her. So now the young one trusts the next mark, the one for the southern extreme, the one she won't live to verify either."

Frik had gone very quiet. The LENS watched him do the arithmetic he was famous for puncturing other people with, and find nothing to puncture.

"That's —" Frik started. "How many generations? To get all eight points to half a degree? You'd need the cycle measured clean at least —"

"Generations," Tessa said. "Yes. We don't put a number on it like that, but yes. Long enough that the people who started watching the horizon were dust before anybody knew the *shape* of what they were watching. And the people who finally knew the shape — who could lay it out in the ground, all eight points, the squared circle, the whole instrument — *those* people had never seen a single mark made by the first watchers. They inherited the answer. They never met the question."

"They built a machine," Naila said, "out of trust."

The LENS saw Tessa look at her then — a longer look, a recalibrating look, the look of someone deciding whether the outsider had said a true thing or a clever thing. He'd seen Tessa give the crew the clever-thing look more than once that week, polite and final, a door closing soft.

This was not that look.

"Out of *obligation*," Tessa said. "Trust is what you feel. Obligation is what you do at the horizon on the cold night when you'd rather be inside, because the keeper before you stood there too, and the one after you is watching how you stand. The Moon doesn't care if you're tired. The record doesn't care. You keep it because it was kept *for* you."

There was a man in the corner the LENS had been not-watching for twenty minutes. The Foundation rep — Dell, the badge said, lanyard and a soft grey jacket and a tablet he kept angled so the screen wouldn't catch the window. He'd been quiet, which the LENS had clocked as a tactic rather than respect, and the LENS had photographed enough careful men to know the difference and to be sour about both.

Dell chose now. He always would have chosen now. The story had crested and the room was open and soft, and that was exactly the moment a certain kind of man reached in.

"This is extraordinary, Dr. Redwing," he said, and the warmth in it was real, the LENS would give him that — it wasn't fake warmth, it was *unearned* warmth, which is worse. "This is exactly the framing the public never gets. The astronomy, sure, everybody's read Hively and Horn. But *this* — the relay, the multigenerational ethics of it, the — the human-instrument idea —" he turned the tablet slightly, and the LENS saw a slide deck on it, a title in a clean serif, a logo. "We could have a draft up by spring. Peer-reviewed track, popular companion piece, the works. Get this out before somebody flattens it into 'ancient calendar, very impressive.' You'd be lead author or co-equal, your framing, your —"

"My framing," Tessa said.

"Your framing. Absolutely."

The room had changed temperature. Naila had gone rigid in the way that meant she'd registered a wrongness and was running it down before she'd let herself name it. Frik's jaw worked. The LENS kept his hands flat on the table and his camera in his bag and watched the small muscles at the corner of Tessa Redwing's mouth.

"You've already written the title," Tessa said. Not a question.

Dell didn't quite flinch. "A working title. Placeholder."

"Read it to me."

A pause. "*The Eighteen-Year Almanac: Decoding the Hopewell Lunar Observatory.*"

"Decoding," Tessa said.

"It's a — it's accessible. People respond to —"

"It's a word for a thing that's locked." She wasn't loud. The LENS had wanted, for a second, for her to be loud — for the satisfaction of it, the war-photographer's appetite for the decisive frame. She gave him nothing to photograph and it was better. "A code is a thing somebody hid that somebody else cracks open and takes out. You think there's a thing in the ground out there that's locked and you've got the key and you'd like to publish the contents by spring."

"That's not —"

"That's *exactly* it. And the worst part is you mean well." She said *mean well* the way you'd say a diagnosis. "You came here to save it from being flattened, and you'd flatten it yourself, gently, into a thing that gets decoded. You'd take the relay — the obligation, the cold nights, the dead women who were right — and you'd make it a *finding*. Past tense. Solved. Filed."

The LENS had a rule he'd broken too many times: don't make a person's worst moment your best shot. He kept the camera down. But he was a camera regardless of glass, and what he saw and could not stop seeing was Naila, leaning forward now, *agreeing* with Dell — not his manners, his *want*. The want to finish it.

"But it *is* finished," Naila said, and the LENS heard the trap in her own voice before she did, the trap of the thing she loved. "The geometry closes. The standstill points are all there, all eight, to half a degree. High Bank uses the same unit fifty-three miles south. The pattern *resolves*. It's the most resolved thing I've ever — it's a closed proof, Tessa. It's *done*. Why can't we say it's done?"

It was, the LENS thought, the most honest thing anyone had said in the room. It was also the question the whole book had been circling, and Naila had walked straight into the center of it and put her hand up.

Tessa looked at her for a long moment.

"The math is done," she said. "You're right. Hively and Horn closed the math forty years ago and nobody serious has reopened it and nobody will. That part's a building. It's built. You can measure it and

I'll measure it with you and we'll get the same numbers because the numbers are *true*." She tapped the survey. "But you asked me to tell you how they *knew*. And the answer to that isn't math. The answer is that somebody kept watching after the math could've stopped. The Octagon's *built*, Naila. The instrument works. So why are my people still here?"

Nobody answered.

"Because the relay didn't end when the building was finished. That's the part you keep wanting to decode and there's nothing to decode, there's only —" she searched for it, and the LENS understood that she was choosing, in real time, how much to give, and that the choosing was itself the thing she would not hand over. "There's only the next person who stands at the horizon because the last one did. The record isn't the walls. The walls are the record's *handwriting*. The record is alive and it's being kept right now, by people in this town and people who drove back here from Oklahoma, and it is not a thing you publish. You can publish the handwriting. You can measure the half a degree. That's yours to do and I'll help you do it cleanly."

She put her hand flat over the eastern gateway on the page.

"This isn't."

Dell said, into the quiet, "The Foundation's window is until December. I'm trying to *protect* this. If we don't establish the interpretive frame, somebody else will, somebody with worse intentions, and then it's drones and lasers and 'mystery of the Moon temple' on a streaming service, and you'll *wish* —"

"Then somebody will," Tessa said. "And we'll outlast them too. We've outlasted a golf course." A flicker, almost a smile, gone. "A hundred years they drove little carts over the gateways. Members complained about the *bumps*. And the instrument kept working. The Moon kept clearing the gate. We're patient people, Mr. Dell. We invented patience here. You have a window until December. We have eighteen-point-six years, over and over, until the river takes it. I'll back myself."

Frik laughed — one short bark, surprised out of him, and then he looked like he might apologize, and then he didn't.

The LENS reached into his bag. Not for the camera. For the thing he'd been carrying, the question that was not his to ask and had never once, in five sites, belonged to the crew.

He didn't get to it. The door opened and Jakobus was there, in his road coat, off his own ground as always, smelling of the parking lot and somewhere further than that. He did not come in. He stood in the gap of the door the way he stood in every door, like a man who'd learned the hard way that thresholds belong to the people on the inside of them.

He looked at the survey, at the eight clean walls and the notched gateways and Tessa Redwing's hand laid flat across the place the Moon comes through.

"What's it for," he said. Not to the crew. To her.

And the LENS, who had spent a career learning when to lower the camera, lowered the camera, though it was already down, lowered something behind it, and waited with everyone else in the room for an answer that was hers to give.

She did not give it yet.

"Come back at dawn," Tessa said. "Standstill's close enough this week. I'll show you the handwriting." She folded the survey along its old creases, slow, and the LENS understood that the showing and the saying were two different gifts, and that she had only promised the one.

Outside, past the lot, the long grass moved over the walls, and somewhere under it the geometry held, closed and true and waiting, kept by people who would not be told what it was for by anyone holding a tablet — or a camera.

Chapter 14 — The Night Before

The grass was already going to frost when Naila stopped trying to count the gates.

There were eight. She knew there were eight. She'd walked them in daylight twice and traced them on the OHC survey at the kitchen table in the rental until Frik threatened to laminate her. But standing inside the Octagon at dusk with the walls going blue-grey and the highway sighing somewhere past the treeline, the count kept coming loose in her hands, the way a word does if you say it too many times. *Octagon*. Eight walls, each about five hundred and fifty feet, fifty acres of figure laid into a town's worth of ordinary Ohio, and you could not see the shape of it from the ground. That was the part nobody warned you about. You could stand dead in the middle of the most precise lunar instrument on the continent and see only embankment, mowed lawn, the silhouette of a maintenance shed, and a sky doing its slow business overhead.

You had to get above it. Or you had to wait.

"You're doing the thing," Frik said. He'd come up behind her with two thermos cups and a face that meant he'd already had most of his. "The thing where you go quiet and your hands move like you're knitting the air."

"I'm checking the eastern gateway azimuth against where it'll come up."

"With your *fingers*."

"My fingers are calibrated."

He handed her a cup. Coffee, gone tepid, which was how he liked it, which was wrong. Behind them the LENS was a low shape against the western wall, tripod legs splayed, not shooting yet — just sitting with the camera the way she sat with everything, letting the place decide when it wanted to be looked at. She'd been quieter since Chillicothe. They all had.

The Hopewell built this between roughly two thousand and sixteen hundred years ago. No metal. No draft animals. No wheel that mattered for hauling, no writing to leave a note for the next crew. Basket by basket — that was not a flourish, that was the method, soil carried in loads on human backs and human shoulders and tamped into walls eight, ten, twelve feet of effort at a time, over generations, by people who would not live to see the figure close. Naila kept trying to hold the arithmetic of that and kept dropping it, not because it was hard but because it was *heavy*. You build a wall your grandmother started so your granddaughter can stand where the Moon will be.

“Twice as precise as Stonehenge,” Frik said, reading her, because he always could. He said it in the OHC’s own voice, the brochure voice, half a quote. “And nobody’s heard of it. There’s a *golf course*. There was a golf course on it for a hundred years, Naila. People played the back nine across a calendar.”

“There’s still a residual lease in the paperwork.” She sipped. It really was wrong. “That’s half of what tomorrow’s about.”

Tomorrow was about the Moon.

Specifically it was about the major standstill — the far edge of an eighteen-point-six-year swing the Moon makes against the horizon, the slow widening and narrowing of where it rises, a rhythm so long that no single human watching the eastern sky their whole life would ever catch it whole. You’d see the Moon rise a little further north each year, then a little less, then turn around, and you would die before the wheel came back. The only way to know the pattern was to *write it down* — except they had no writing — so the only way to know the pattern was to hand it forward. Watcher to watcher. A mother’s mark on a horizon her daughter would extend. Hively and Horn came here in 1982 to prove the astronomy was a coincidence and went home having found all eight standstill points built into the dirt to within

half a degree, and the worst thing about that, the thing that kept Naila awake, was that it was *settled*. Peer-reviewed. Boring, in the way the most astonishing facts are boring. Nobody serious disputed it. There was no debunk left to do.

You could not even comfort yourself by solving it. It was already solved. The mystery had only ever been the size of the patience.

“Heads up,” Frik murmured.

Headlights swung across the southern gateway. Two vehicles, then a third, parking along the access road in the careful staggered way people park when nobody wants to be the one blocking anybody in. Doors. The OHC site manager, Reyes, in a fleece that said OHIO HISTORY CONNECTION across the back like a referee. A small cluster Naila knew were the tribal representatives — three of them tonight, in for the consultation and the dawn, one of them on the phone with someone in a time zone where it was earlier and warmer. And, of course, the Foundation.

The Foundation came as one man tonight. Calloway. Clean fleece, clean boots, a tablet under his arm in a rugged case, the careful warmth of someone who has read the room before entering it. He was not a villain. Naila had wanted him to be one and he refused, which made him worse. He shook Reyes’s hand, nodded to the tribal reps with exactly the right weight of deference, and made his way over to the crew with the open palms of a man who has nothing to hide and a great deal to publish.

“Big morning,” he said. “You can feel it, can’t you. The site knows.”

“The site is dirt that doesn’t know anything,” Frik said pleasantly. “The *math* knows. The math’s the only thing that’s ever known.”

Calloway smiled like that was charming. “We’ve got a clean ephemeris model running. I can show you the projected azimuth for clearance over the far gateway — we’ll have it overlaid on the live feed. First time anyone’s done a real-time validated capture.” He tilted the tablet so its glow lit his chin. “It’s going to be the definitive record.”

“There’s already a record,” said the LENS, without turning around. “Eighteen hundred years of it. Built in.”

Calloway took that the way water takes a stone. “Of course. Of course. I only mean a *contemporary* one. For access. For the public good.” He said *public* the way some people say *love*. “If the standstill clears the gateway tomorrow and we’ve got it clean, validated, first — that’s the thing that gets this place the funding it’s owed. That gets the rest of it dug out from under the suburbs. I’m on your side here.”

“Whose side is that,” Naila said. It wasn’t a question, exactly. She was still watching the eastern gateway, where the sky was the last pale thing left.

She felt rather than saw him not have an answer, and then choose a smile, and then move off toward Reyes to talk about cable runs.

Dr. Redwing came in from the dark on foot.

She didn’t park with the others. Naila clocked that and filed it — she came from the direction of the causeway, the long earthen avenue that ran from the Octagon to the Observatory Circle, the thing that joined the figures the way a sentence joins two clauses. She’d walked the joint. She came up beside Naila at the eastern gateway without announcing herself, hands in her jacket pockets, and looked at the same patch of horizon Naila had been measuring with her calibrated fingers.

“You’ve found where it’ll come up,” Tessa said.

“I think so. I keep checking it against the survey.”

“Show me where you think.”

Naila lifted her hand and laid the edge of it along the dark notch of the far gateway, the gap in the opposite wall a third of a mile off, where tomorrow — if the cloud held off, if eighteen point six years of orbital mechanics did what the dirt said it would — the Moon at its furthest north would rise and pour itself straight down the throat of the figure, clearing the gateway within half a degree of the octagon’s exact centre. Within half a degree. Across two thousand years and fifty acres.

Tessa looked along Naila’s hand for a while.

“Close,” she said. “You’re a hair north.” She didn’t move Naila’s hand.

She just let the correction sit there in the cold air, accurate, unhurried. “It’s an easy mistake. Everybody puts it where the geometry wants it. The Moon doesn’t care about your geometry. The geometry was built to catch the *Moon*. You’re reading it backwards.”

Naila felt the words land somewhere structural. She turned the figure over in her head and the north-hair was *right*, it had to be, the alignment wasn’t a prediction the builders made, it was a measurement they recorded — they’d watched first and built after, watched for a lifetime and built for the next one, and you couldn’t reconstruct the watching from the wall any more than you could reconstruct a song from a record sleeve.

“How did you—” Naila stopped. Started again, smaller. “You’ve stood here for it before.”

“My grandmother’s people stood here for it before there was a *here* to argue about.” Tessa said it without any weight on it, which was how Naila knew it had all the weight. “I’ve done the dawn three times. I’ll do it tomorrow. The math is the math — your Mr. Hively did good work, I’ve got no quarrel with the number. The number’s true.” She paused. “It’s just not the answer.”

“To what?”

“To what you’re going to ask me.” There was the faintest thing in her voice that wasn’t quite a smile. “All of you. Tomorrow, when it clears the gateway and everybody’s got it on a screen, somebody’s going to turn to me and ask what it’s *for*. The fixer. The quiet one with the question. I can see it coming like weather.”

Naila didn’t say anything. Out past the western wall she could hear Friek and Calloway in some low diplomatic skirmish about whether the live feed counted as a publication, and the highway, always the highway, and underneath all of it the enormous patient silence of a figure too big to see.

“You started something you can’t finish,” Tessa said. “That’s the part you keep snagging on, isn’t it. I watch you do it. You want to close the pattern. You want to be the one standing here when the eighteen years come round and the wheel clicks shut.” She turned and looked at Naila full-on for the first time, and her face was kind and completely immovable, both at once. “Nobody who started this saw

it. The woman who marked the first standstill point — she was forty years dead before the second one came round to prove her right. She handed her daughter a horizon and a number and said *watch this spot, write what you see, tell yours*. That's the instrument. Not the dirt. The *handing*."

"And you inherited it," Naila said. "The handing."

"I inherited it." A beat. "And he wants to validate it. With an ephemeris. First, clean, definitive." She didn't look toward Calloway. She didn't have to. "You can validate a number. You can't validate a relay you weren't running in."

The cold came up through Naila's boots. Over east the first stars were out and the gateway was a darker dark in the dark, and somewhere fifty-three miles south at High Bank there was a second octagon she and Tessa had walked four days ago, the only other one the Hopewell ever built, laid out with the same unit — that 1,054-foot ruler, the Observatory Circle Diameter, the *same length*, across two sites and the generations between them. Naila had stood in that figure and felt the floor of the world tilt. Not aliens. Worse. Better. A *people*. A school. A shared measure carried over fifty-three miles of nothing by somebody who taught somebody who taught somebody, and the same hand, or the same lesson, kept reaching out and leaving the same ruler in the ground.

"The High Bank unit," Naila said carefully. "It's the same. It's exactly the same."

"It is."

"So somebody carried it. The standard. Somebody kept *that* record too, the length, the ruler — who—"

Tessa let her get all the way to the edge of it. Then she did the thing the experts always did in the papers, the thing the OHC did, the thing that drove Frik up the wall and that Naila was only now beginning to understand wasn't evasion at all.

"That's a good question," Tessa said.

And stopped.

The silence after it was not empty. It had a shape, the way the figure had a shape you couldn't see from inside it. Naila waited for more and there wasn't going to be more, and the not-more was the most honest thing anyone had said all night.

Who keeps a record longer than a life, Naila thought. *And who taught them all the same length*. She had the whole question now, both blades of it, sharp and clean and entirely unanswered, and the woman standing beside her in the frost had the answer or had a piece of it or had at least the *right* to it, and was going to hand Naila the dawn and the geometry and the cold half-degree of validated wonder.

Just not that.

"You'll get the alignment tomorrow," Tessa said, gentle now, turning to go meet her relatives by the southern gate. "Film all of it. It's true, it's astonishing, it's yours to tell." She paused at the edge of the gateway, where Naila's hand had been a hair too far north. "What it's *for* — that part's not on the screen. That part I keep."

She walked off down the causeway toward her people, the way she'd come, by the joint that held the two figures together.

Behind Naila the LENS finally lifted the camera — and then, very deliberately, lowered it again, and let the dark have her.

Frik came back with the wrong-temperature coffee and looked at Naila's face.

"What'd she say?"

"She said it's a good question." Naila kept her eyes on the far gateway, where in a few hours the Moon at its furthest would come up and clear the centre by half a degree, exactly as the dirt had promised for eighteen hundred years to anyone patient enough to still be watching.

"And?"

"And that's all she said."

Frik thought about it. Sipped. Made a face — at the coffee, or the answer, or the two thousand years of people who'd died without the punchline and handed the watch on anyway.

"Huh," he said. "Hate that." He didn't, though. Neither of them did. The not-knowing sat between them in the cold like a third thermos

cup, and over the eastern wall the sky kept its long appointment, and nobody — not the Foundation, not the OHC, not the crew, not even, quite, the woman walking the causeway — held the whole of it in one hand.

They waited for the Moon.

Chapter 15 — Moonrise at the Gateway

The frost came up out of the grass like breath off a sleeping animal, and I had been awake long enough to watch it do it.

Four-forty in the morning, Licking County, Ohio. I had the camera on the monopod and the monopod planted in the turf of what used to be a fairway, and I had not lifted the lens to my eye yet. There is a discipline to that — you can burn an hour through the viewfinder and come away with nothing, because the glass eats the thing it shows you. So I stood with the camera at my hip and let the dark do what dark does before it stops.

The Octagon Earthworks at this hour are not impressive. That is the first thing nobody tells you. Fifty acres of low grassed walls, eight of them, each about as long as two football fields and only as tall as a man's chest, joined to a great pale circle by a neck of earth you would step over without noticing if a sign didn't say so. In the half-dark they read as berms. Drainage features. The kind of thing a city builds and forgets. There was a flag still standing on a green out past the third wall, white and limp, because the golf course had only just stopped being a golf course in the eyes of the law and nobody had pulled the flags.

I have photographed war. I have photographed the places where the worst things people do to each other are still warm in the ground. I know how to wait for light, and I know that the light decides what was true.

To my left, Naila had her arms crossed and was not looking at the sky. She was looking at the far gate — the gap in the northeastern

wall, where the earth opened and the causeway ran out toward the horizon — and she had been looking at it the way she looks at a thing she has already finished calculating and is now only confirming. She does not fidget when she is sure. The stillness is the tell.

Frik fidgeted enough for both of them. He had his hands jammed in his armpits and was rocking heel to toe and muttering numbers under his breath, which is how he keeps awake, the way other men hum.

“Half a degree,” he said, to nobody. “Half a degree off the center of the whole figure. You know what half a degree is at that range? It’s the width of —” He stopped, because he’d done this bit before, and because Hecklefish, who had no business being at a sacred site at four in the morning, said from somewhere near the cooler:

“It’s the width of nothing, that’s what it is. You walked it nine times yesterday. If you tell me about your thumb at arm’s length one more time I’m walking into the ditch.”

“The ditch is sacred.”

“So am I.”

Dr. Tessa Redwing stood apart from all of us, which was correct, because it was hers to stand apart and ours to be guests on the grass. She had a thermos she was not drinking from. She was not looking at the gate the way Naila was, with the hunger of someone wanting to be proven right. She was looking at it the way you look at a door in your own house in the dark — knowing where it is, not needing to check.

A few yards from her, the Foundation’s man — Croft, good coat, good boots, the kind of well-meaning that costs money — had a tablet out and the tablet’s screen turned down low so it wouldn’t kill his night vision. He’d learned that much, at least. He had an ephemeris running. He had a drone in a hard case at his feet that the OHC had told him, twice, in writing, he was not flying today. He had the look of a man who has bet on a horse and is now watching the horse to make sure it knows about the bet.

And he had every right to be standing there. That was the part that itched. The Foundation had paper. A research window, a lease residual, a grant with a logo. They had earned their place on this grass

the way you earn anything with lawyers, and now they wanted to be the first voice the world heard say what was about to happen. First to publish. First to frame it. First to tell you what it means before the people whose ancestors built it could finish their coffee.

The official story is simple and it is true and I want to give it to you straight, the way the morning gave it to us.

Somewhere between two thousand and two thousand one hundred years ago, the people we call the Hopewell — not their name for themselves; the name of the farm where a collector dug them up — stood on this ground and watched the moon. Not for a season. Not for a year. The moon does a slow thing, a thing that takes eighteen years and seven months to come back around, where the points on the horizon where it rises and sets drift out to their widest spread and then drift back. You cannot see that drift in a human glance. You cannot see it in a winter. You can only find it by watching the eastern sky, marking where the moon broke the horizon, and writing it down — except they had no writing — and so handing it forward, mouth to mouth, post to post, one keeper to the next, across more years than a watcher could be sure of living.

And when they had it — when somebody, or some long chain of somebodies, finally had the whole shape of that eighteen-and-a-half-year breath — they built it into the dirt. Eight walls. Five hundred and fifty feet each. Gates set so that all eight of the moon's standstill rise and set points are written into the figure to within half a degree. Basket by basket. No metal. No wheel. No draft animal. Just earth carried in arms and the patience of a people who knew they were building something none of them would see proven, because the proving took longer than they had.

That's the part Croft wanted to own. That's the part Hively and Horn came here in 1982 to *disprove*, two college men, a physicist and a philosopher, sure the alignment was wishful thinking — and went home having found that it was not. That it was real. That it was, in the words the Ohio History Connection itself uses on the brochure in the visitor center, *twice as precise as Stonehenge*.

So. We were going to watch it happen.

The major standstill moon — the widest, the rarest, the one that doesn't come back for the better part of two decades — was going

to rise. And if the dead were as good as the textbook said they were, it was going to rise *through that gate*.

The sky in the northeast went from black to the color of old iron. The flag out on the green caught it first, a gray rag where there had been nothing. Then the far wall came up out of the dark — I could see the gate now, the deliberate gap, the causeway running through it like a tongue. Frik had stopped rocking. Hecklefish had stopped heckling, which I have learned to read as the most serious thing in any room.

“It’s late,” Croft said, very quietly, to his tablet. The strain in it. He wanted the math to be human-fallible. He wanted a few degrees of slop he could write a paper around, a *refinement of the Hively-Horn model*, a reason the Foundation’s name needed to be on the door.

Naila answered without looking at him. “It’s not late.” She wasn’t unkind about it. She just knew. “The atmosphere refracts it up at the horizon. It’ll read about a half-degree high of true. Which —” and here she did the smallest thing, a tilt of her head toward Tessa, an offering — “is inside their tolerance. They knew that too. You’d have to. You can’t watch the horizon for eighteen years and not learn that the air lies.”

“You don’t know they corrected for it,” Croft said.

“No,” Naila agreed. “I don’t.”

That is the thing about her that the Foundation could not metabolize. She would not overreach to win. She held the awe and the *I don’t know* in the same hand and did not seem to find them heavy.

The light kept coming. There is a particular minute before a moonrise where the whole eastern edge of the world seems to lean, where you can feel the thing under the horizon the way you feel someone behind you in a doorway. I lifted the camera. Finally. I put my eye to the glass and I framed the gate — the two earthen shoulders of it, the causeway between, the iron sky above — and I waited, and I did not breathe, because I have learned that the shot is decided in the second you stop wanting it.

The moon came up in the gate.

Not near it. Not gesturing at it. *In* it. The lower limb broke the horizon dead in the gap, between the two walls those people had

piled by hand before the Roman Empire was finished being born, and it climbed, swollen and copper at the rim of the world, straight up the throat of the causeway, and for a long moment the whole fifty acres of grass and frost and forgotten golf flags arranged itself around that one fact like filings around a magnet.

I took the picture. I want to be honest about that — I took it, three frames, because that is the only thing I know how to do with my hands when something is true. But I want to be honest about the rest of it too. The picture is not the thing. The picture will show you a moon in a gap and you will think, *nice, a moon in a gap*. You had to be standing on the frost with the highway hissing faint behind the walls and these particular dead being right after two thousand years to understand what your own body understood, which was: somebody *meant* this. Somebody stood where I am standing, in a body like mine, and aimed it.

Nobody spoke. Croft's tablet glowed up at his face, and his face had nothing to write down. The ephemeris had told him where the moon would be. It had not told him this. There is no column in the ephemeris for the thing that happens in the chest when the calculation and the dirt agree across twenty centuries.

Frik made a sound. Not a word. The sound a kid makes.

"Yeah," said Hecklefish, finally, very low, like even a fish knows when to keep it down. "Okay. Yeah."

I held the moon in the gate as long as it stayed in the gate, which was not long — it slid clear of the northern shoulder and went silver and ordinary, just the moon, a Tuesday's moon — and I lowered the camera, and I made myself look at the people instead of the glass.

Tessa Redwing had not lifted a phone. She had not taken a single picture. She stood with her unopened thermos and watched her ancestors' work do exactly what they built it to do, and her face was not the face of someone who'd had a theory confirmed. It was the face of someone keeping an appointment that had been kept, in both directions, longer than any of us could hold in our heads.

Croft turned toward her. You could see him assembling it — the roll-out, the *we witnessed*, the press contact. He had every legal right and you could watch the rights curdle in his mouth as he understood,

a beat too late, that being *present* at a thing is not the same as it being *yours*.

“Dr. Redwing,” he started. “If we move quickly on the announcement —”

“We’re not announcing anything this morning,” she said. She didn’t raise her voice. She didn’t need the camera and she knew none of us would lift one at her without asking, and that knowing was the whole transaction. “My relatives have been watching this gate longer than your foundation has existed. Some of them drove eleven hours to be here. They’ll decide what’s said about it, and when, and to whom.” She finally drank from the thermos, then, as if the matter was now small enough to drink over. “You can stand on the grass. The grass is public now, thank God. But the meaning isn’t a thing you found lying out here for the taking. It was never lost. It was kept. There’s a difference, and the difference is the entire point.”

Croft did not have a lawyer for that one.

Naila walked out toward the gate as the ordinary day came up, and I went with her, because that is my job, to follow the one who’s reading the thing. She crouched at the lip of the causeway and put her flat hand on the cold turf of the wall the way you’d put your hand on a sleeping ribcage.

“Half a degree,” she said, but it was a different voice now than Frik’s had been all night. The number had stopped being a brag. “Frik. You know what makes me — it’s not that they hit it. People hit things. It’s that the first one of them who started writing it down knew they’d be dead before the answer came. They started a sentence they’d never get to finish saying. And they trusted that somebody downstream would keep the count honest.”

Frik, beside her, did not puncture it. For once he had nothing to deflate, because nobody had reached past the true thing into the convenient one. The half-degree was real. The eighteen years were real. The dead watcher who’d never see the wall was real.

“And the same ruler at High Bank,” Frik said, quiet. “Fifty-three miles. Same length. Same moon.”

“Same people,” Naila said. “Or —” and she stopped herself, and looked back at Tessa, who was now walking toward the relatives gath-

ering at the far edge of the field, and who had not handed any of us the last word and was not going to. “Or something I don’t get to finish either.”

The sun came up behind us and threw all eight walls’ shadows long and clean across the frost, and for one minute you could see the whole figure the way the makers never could — they only had the ground; we had the angle of an early sun doing the work a drone wasn’t allowed to. Eight walls. One circle. A door pointed at a moon that wouldn’t come back this wide for eighteen more years.

I did not lift the camera again. The moment had an owner, and it wasn’t me, and the best picture I took all morning was the one I left on the grass.

Chapter 16 — The Foundation's Offer

The conference room at the Ohio History Connection smelled of carpet glue and burnt coffee, and Frik counted the ceiling tiles twice before the meeting started because his knee wouldn't stop bouncing and someone had to put the energy somewhere. Forty-eight tiles. One with a brown ring where the roof had wept, decades ago, into a building meant to keep the past dry.

He liked this room. It was the most honest room he'd been in all week. No earthwork pretended to be a fairway here. No fairway pretended to be flat. It was just a folding table, a whiteboard nobody had erased, and a window that looked out onto a parking lot, beyond which — three miles east, invisible from this chair — eight walls of piled dirt tracked the Moon to within half a degree.

The Foundation's man was named Webb. He had a good handshake and a worse one ready in reserve. He'd brought a laptop and a leather folio and a bottle of the kind of water that costs money, and he'd set all three down like he was laying out a hand of cards he already knew would win.

"I want to start by saying," Webb said, "how much we admire the work being done here. Genuinely. The reclamation. The consultation process. It's a model."

Here it comes, Frik thought. The praise was the wind-up.

Tessa Redwing sat across from him with her hands folded and her coffee untouched. She had the stillness of someone who'd watched a lot of people wind up. Naila was beside her, not folded at all — Naila

had her fieldbook open and was drawing the Observatory Circle for the ninth or tenth time that week, the same 1,054 feet rendered again in pencil because her hands thought better when they were busy. The Lens stood at the back by the weeping ceiling tile, camera down at his hip, not raised. He'd been at the back of rooms for thirty years. He knew which rooms wanted the camera up and which ones it would ruin.

This one it would ruin.

"What the Foundation can offer," Webb went on, opening the folio, "is reach. We're talking documentary partners. A peer-reviewed special issue with our imprint behind it. A traveling exhibition — Newark, Columbus, then national. The Octagon is twice as precise as Stonehenge and almost nobody on earth knows it exists. We can fix that. We can make this site the story it deserves to be."

He slid a single page across the table toward Tessa. Frik couldn't read it upside down, but he could read the shape of it — a logo, a budget line, the white space architects use to make a number look gentle.

"Co-authorship," Webb said. "Full credit to the descendant nations. Full consultation. We're not trying to take anything. We're trying to *amplify*."

Naila's pencil stopped.

Frik watched her catch it — the word — the way she caught a measurement that was three feet off. *Amplify*. She didn't say anything. She wrote something small in the corner of her circle and started shading again.

The thing about the Octagon, Frik had decided, was that it refused to be impressive in the way that pays. It wasn't tall. Stonehenge had the courtesy to be made of enormous rocks you could photograph against a sunset, and the Pyramids had the decency to be visible from orbit, and both of them said *look how much* in a single glance. The Octagon said nothing. You stood inside it and the walls came up to your shoulder, soft green hills you could walk over without noticing, and the genius of it was completely, perfectly hidden from the body. You could only see it from above, or from inside a number.

Webb wanted the version that paid. Frik didn't blame him for it. The man wasn't a villain; he was a guy who'd correctly understood that the world ignores anything it can't put on a poster, and had decided to help. That was almost the worst kind of person to argue with — the kind who was *trying*.

"It's a generous offer," Tessa said.

Webb's shoulders came down half an inch. He'd been braced. The praise had been the wind-up; the bracing was the tell.

"It is," she said again, and let it sit, and then: "And it's not for sale."

"I — sorry. I don't think I framed it as a sale. There's no purchase here. We'd be *funding* —"

"You'd be funding the right to put your name beside ours on what it means." She unfolded her hands and laid one flat on the page he'd slid over, not picking it up, not pushing it back. Just covering it. "That's the only thing in this building anybody actually wants. The dirt's not for sale either, but nobody's tried to buy the dirt in a while. They've moved on to the meaning. It costs less to ship."

Frik's knee stopped bouncing.

"Dr. Redwing," Webb said, and his voice did the thing where it got reasonable, which is the most expensive thing a voice can do, "the meaning is going to be told whether we tell it or not. Someone's going to make the documentary. Wouldn't you rather it was made by people sitting at this table, in consultation, with the nations credited, than by some content farm that's never set foot in Licking County?"

It was a good argument. Frik hated that it was a good argument. He felt his mouth opening — *the reach*, he thought, *find the reach* — and Naila got there first, quietly, not even looking up.

"The cycle's eighteen point six years," she said.

Webb blinked. "I — yes."

"You can't see it." She set her pencil down. "I want to make sure that's clear before anyone schedules a documentary. You can stand on that wall for a year and you see the Moon go up and come down. You stand there ten years, you see it wander north and south and you start to think there might be a pattern but you can't prove it,

because the swing's so slow and your eyes are so short. The only way to *know* it — to know it well enough to dig it into the ground at half a degree — is to watch it, and write it down, and hand the writing to your kid, and have your kid hand it to theirs, until somebody two or three generations down the line is holding a record long enough to see the shape close." She looked at Webb now. Directly. It cost her something; Frik knew the cost; she did it anyway. "The first person who started watching died before anyone could've known if it was worth it. They kept the record on faith that someone would finish reading it."

The room was quiet. The ceiling tile dripped its old dry drip, which was just a stain, but Frik heard it anyway.

"So when you say the meaning's going to be told whether you tell it or not," Naila said, "I think you've got the timescale wrong. The meaning's been getting told for two thousand years. By people who didn't rush it. I don't actually know how you'd amplify that. You can't make it louder. It's already the patientest thing on the continent."

Frik wanted to applaud. He didn't, because the room wasn't his, and because Naila would've hated it. He settled for catching the Lens's eye at the back. The Lens lifted one eyebrow a fraction — the closest he came to a grin.

Webb regrouped. He was good; he'd been edged before and lived. "That's exactly the story I'm talking about. The patience. The relay. That's the documentary. That's the —" he caught himself before *poster*, Frik was almost sure of it — "that's the thing the public needs to feel. And with respect, the descendant nations don't have the distribution to —"

"With respect," said a new voice.

The OHC's liaison had been so quiet at the end of the table that Frik had half-forgotten her — a woman named Okafor with a stack of folders and reading glasses pushed up into her hair. She slid one folder forward now, and it landed with the specific weight of a document that has been to a lawyer.

"The interpretive authority for the Hopewell Ceremonial Earthworks," she said, "rests with the consultation framework. That's not a cour-

tesy line in a grant proposal. It's the framework we operate the World Heritage stewardship under. The descendant nations have the last word on interpretation. Not co-equal. Not amplified." She tapped the folder. "Last."

"Of course," Webb said. "We'd honor —"

"You'd honor it by signing onto a media rollout that puts your imprint on the spine and theirs in the acknowledgments." Okafor said it without heat, which made it worse. "We've read the offer, Mr. Webb. It's the same offer in a clean shirt. The Foundation gets the narrative; the nations get a credit and a feeling of having been consulted. We did that for a hundred years here. There was a golf course on it. We're not doing it anymore."

Frik glanced at Naila. She'd gone back to shading her circle, but she was smiling at the paper, the small private smile of someone who'd watched a number come out right.

It should have been clean after that. It nearly was.

Webb tried two more doors. The first was money — a bigger number, said softer. Tessa let it sit and didn't pick it up, same as the page. The second was time — the gentle suggestion that the research window the Foundation held still had weeks left on it, that legally, technically, they could publish first, that wouldn't it be *better* to do this together than to have it come out adversarial.

That one had teeth. Frik felt the table feel it. The Foundation *did* have the window. That was the ugly arithmetic of the whole chase — that a lease and a grant could put a clean-handed outfit inside the gate before the people whose ancestors built the thing got the keys back. Webb wasn't bluffing. He genuinely could be first.

"You could," Tessa agreed.

She said it so easily that Webb's face did something complicated.

"You've got the window. You could put a paper out next month — restate Hively and Horn, lay your imprint over the top, call it the Foundation's reading of the Octagon. You'd be first." She finally moved the page he'd slid her — picked it up, squared its edges against the table, and set it back down in front of him. Returned. "And then for

the rest of however long this place stands, the descendant nations would have to tell people that the version with the loud distribution wasn't the version with the authority. They'd spend the next two thousand years correcting your head start. You'd have your first. We'd have the work of un-firsting you, forever."

She let that breathe.

"Or," she said, "you don't. You let the window close. And the first real telling of what this is for comes from the people it belongs to, in their own time, which — you may have noticed — they're historically very good at keeping. I can't make you choose the second one. I can only tell you which one I'll spend my life on, and it's the second one, and I've got the framework and the History Connection and the World Heritage listing standing behind me, and you've got a budget line and three weeks." She picked up her coffee at last, the gone-cold cup, and drank it like punctuation. "It's a generous offer, Mr. Webb. It's not enough."

Webb closed the folio.

Frik watched him do it and understood that this was the turn — not a slammed door, nothing so loud. The man simply stopped reaching for the handle. He'd come to read the Octagon first and take the meaning to market, and he'd run into the one thing the market can't price, which is a keeper who out-waits you, who has already decided to spend a life she hadn't finished living on un-firsting anyone who tried.

"I'll take it back to the board," Webb said, which is what people say when the answer is no and they need the no to have been someone else's.

"You do that." Tessa stood. So did Okafor, the folder back under her arm.

Outside the window, the parking lot held its cars. Three miles east, the Moon was below the horizon and the gateway pointed at nothing yet, waiting the way it had waited for two thousand years, patient past any single life, the longest single sentence anyone on the continent had ever written and the one nobody had finished reading out loud — not yet, not for sale, not his.

The Lens lifted the camera then. Just for a second. He caught Tessa at the door with the squared page left behind on the table beside Webb's expensive water, and he caught Naila's circle, drawn for the tenth time, the same true thousand feet.

Then he lowered it again, the way the room wanted, and let the moment belong to the people whose moment it was.

Frik counted the ceiling tiles one more time on the way out. Still forty-eight. Still one with the old stain where something patient had gotten in.

Chapter 17 — Who Holds the Record

The thing about the relay is that nobody at the start of it ever held the finish.

Naila kept coming back to this, the way you keep returning your tongue to a chipped tooth. She was sitting on the low rise where the Observatory Circle's wall ran flattest, the morning frost still bluing the grass in the wall's shadow, and she had her notebook open to a page she'd ruled into columns. Year. Standstill. Observer. The third column was empty all the way down because there was nothing she could put in it. Nobody knew the names. Nobody would ever know the names. That was the whole trouble.

"You're doing the face," Frik said. He'd folded himself onto the slope beside her with a gas-station coffee and the wired stillness that came over him only when something had genuinely hooked him. "The face where you've found a hole and the hole is in your own head."

"It's not a hole. It's a column."

"Same thing, for you."

She tapped the empty column with her pen. "Eighteen point six years. Say you're twenty when you start watching the eastern horizon. You mark where the Moon clears it at its furthest north — the major standstill. Then you watch the Moon swing back south, season on season, and you wait."

"For how long."

"For nine years, roughly, before the Moon's even at the *other* extreme.

And the extreme isn't a single sunrise you can circle on a calendar. It drifts. It hangs near the limit for months — that's *why* they call it a standstill, the Moon seems to stop wandering and just sit there. You can't catch the day. You catch a *window*, and you mark the edges of the window, and you do it again the next time around. And the next time around is eighteen years and seven months after the first."

Frik did the arithmetic out loud, badly, the way he did everything fast. "So you start at twenty. You see the cycle close once at — thirty-nine? You're thirty-nine before you've watched a single full turn of the thing."

"If you live. And one turn doesn't *prove* anything. One turn could be luck. You need the cycle to repeat to know it's a cycle and not a story you told yourself about some marks on a stick."

"So you don't finish."

"You don't finish." She looked at the wall, the long patient earth of it, fourteen hundred feet of curve held to under four feet of tolerance the whole way round, a circle so true you could not feel it bending under your feet. "Whoever started the count handed it to somebody who handed it to somebody. The person who *understood* the pattern got it as an inheritance. And the person who built it into the ground — into *this* —" she put her palm flat on the cold grass "— built it from a record older than they were, made by people they never met, about a sky none of them had watched all the way through."

The fish, who had been quiet, surfaced.

"So basically," Hecklefish said from somewhere around her elbow, in that tone of a creature who has read the back of the box, "the Hopewell invented homework you have to leave in your will."

Frik snorted into his coffee.

"It's not funny."

"It's a *little* funny. 'Dearest grandchild, the Moon owes us another forty years of observations, please find attached the family stick.'"

Naila didn't laugh, but the knot in her chest loosened a half-turn, which was what Frik was for, and what the fish was for, in the end. They heckled the awe back down to a size you could hold without it crushing you. Stonehenge got the postcards. Stonehenge got the

bus tours and the druids at solstice and the whole world's idea of *ancient and mysterious*. And here was a thing the Ohio History Connection's own signage called twice as precise as Stonehenge — tracking a target eighteen times harder, in a town most of the country couldn't find on a map, with a *golf course* laid across it for the better part of a century, fairways mowed over a lunar instrument like it was rough that needed managing.

You could see the famous thing from space. You had to *deserve* this one. You had to do the math.

The Lens came down the wall toward them, camera at her hip, not raised. That was its own kind of report. When she was working a thing she shot it; when she'd decided the thing wasn't hers to take, the camera rode low and dark like a holstered hand.

"Tessa's coming," she said. "She's been on the phone with Pawnee Nation's office since six." She lowered herself onto the grass with the care of a body that had learned where it hurt. "And the Foundation's man is parked by the causeway with a laptop and a face."

"What kind of face," Frik said.

"The kind that's drafting a press release in his head and waiting to find out whether he's allowed to send it."

Naila felt the old itch start up, the one that had got her into half her arguments and all of her best work: the itch to *close the thing*. She had the columns. She had the geometry — she and Frik had walked the OCD off the ground themselves, the Observatory Circle's diameter stepping out across the site like a unit somebody had carried in a pocket, the same length showing up again fifty-three miles south at High Bank where the only other Hopewell octagon in the world encoded the same Moon. One ruler. Two sites. Generations apart. A shared standard meant a shared *people*, a network of watchers who taught each other the same length, and that was bigger and stranger than any of the things the bookshop fringe shelf would sell you. She had it laid out in her head, clean as the circle, and she wanted to say it. She wanted to walk up to Tessa Redwing and put the whole solved shape of it on the grass between them like a gift.

"You've got it," the Lens said, watching her. Not a question.

“I’ve got the structure of it. The relay. The unit. I can *show* how it had to work — the record handed down, the —”

“Yeah.” The Lens turned her face to the gate where the far walls opened a notch onto the eastern horizon. “I’ve watched people do that. Walk up to somebody’s grandmother’s life with the structure of it all worked out, ready to hand it back to her explained.”

It landed soft and it landed hard, the way the Lens’s things did. Naila shut her notebook.

Tessa came up the causeway with her coat open and her hands jammed in her pockets, and she stood at the lip of the Observatory Circle the way she always stood — like she was checking on something rather than visiting it. She looked at the four of them sitting in a row on her wall and something in her face eased a fraction.

“You found the ruler,” she said. Not asked. She’d seen the chalk-marks they’d left on the survey print, the stepped-off OCD, the line they’d drawn between Newark and High Bank.

“We found the ruler,” Naila said, and then, because the itch was unbearable and because she trusted this woman more than she’d trusted anyone on the project, she let it go. “Dr. Redwing — it’s a relay. It has to be. The cycle’s invisible to one lifetime, so somebody kept the count across — I don’t know — four, five generations of watchers, before anyone could even know the shape, and then they *built* the shape, and they used the same unit at High Bank, which means there was a — a school of it, a line of people teaching the same measure across the whole network, and that’s the answer, isn’t it? That’s what it’s *for*. It’s an almanac. A handed-down almanac in the ground. We can — I can lay the whole thing out, the math closes, it —”

She stopped because Tessa had crouched down to her level, slow, knees in the frost, and was not looking at the notebook at all. She was looking at Naila.

“You’re not wrong about the math,” Tessa said.

“Then —”

“You’re not wrong about the math,” she said again, gently, and the second saying changed the first, the way a wall changes a circle. “The

cycle's invisible. The count had to be kept. The unit's shared. Hively and Horn published the alignments forty years ago and I've spent half my life walking people through them. You can stand on every one of those facts. Nobody's taking them off you."

Naila waited for the *and*. It didn't come the way she expected.

"What you're calling the answer," Tessa said, "is the *mechanism*. How a record gets kept longer than a life. That's a beautiful piece of work and it's true and you can have it." She tilted her head toward the gate, toward the notch of horizon waiting empty for a Moon that wouldn't come for years. "But you asked what it's *for*. And that's a different word."

"I — isn't it the same word? The function. What it does. It tracks the Moon —"

"It tracks the Moon the way a church tracks Easter," Tessa said. "Yes. And no. You can write down the mechanism of a church — angles, dates, who rings the bell. You'd have all of it correct and you'd have missed the entire thing." She wasn't unkind. That was the part Naila would think about for a long time afterward — there was no edge in it, no *gotcha*, no closing of a door in a face. "Somebody kept this count for two hundred years before it was true enough to build. *Why* did they keep it. Not how. Why does a person hand a record to a grandchild they'll never see grown, about a sky they'll never watch close? What did they think they were *doing*."

The fish, for once, said nothing.

"And that," Tessa said, standing, knees cracking, "is ours."

Naila felt it the way you feel a step that isn't there. The itch to close the thing met something that wasn't a counter-argument — she could have fought a counter-argument, she was *good* at counter-arguments — but a wall. A true one, four feet of tolerance, held the whole way round.

"I'm not telling you that to shut you up," Tessa said, reading her exactly, which was its own small mercy. "I'm telling you because you're the kind of person who'll get it. You found the squared circle by *seeing* it. You didn't need me to point. So I'll trust you with the part

I'm not going to explain, which is *that there's a part I'm not going to explain*, and it's not a hole in your work. Your work's whole. It just stops where mine starts."

Down by the causeway, the Foundation's man had got out of his car. He was looking up at them, and at the survey print, and Naila understood with a cold clarity that he wanted exactly what she had wanted ten minutes ago — to be the one who *said it first*, who laid the solved shape on the table and put his name on the lid. The difference was that he'd say *what it's for* and mean *what it'll sell as*. He'd publish the almanac and own the meaning by being fastest to the microphone.

She thought about her empty third column. Observer. Blank all the way down. She'd been treating it like a failure of the record. It wasn't. The names weren't lost. They were *kept* — somewhere that wasn't a peer-reviewed journal, in a line that hadn't ended just because a golf club fenced the wall for a century. Tessa wasn't standing at the lip of the circle as a visitor. She was the next watcher in a relay that the math could describe and could not enter.

"What do you want us to do," Naila said. It came out smaller than she meant and exactly the right size.

"Shoot the alignment when it comes," Tessa said, looking at the Lens. "Get the math right — get it *exactly* right, because the fast people get it sloppy and the sloppy version is how they steal it. Show everybody the relay, the ruler, the genius of the people who did it with baskets and cord and patience nobody alive has anymore." She put her hands back in her pockets. "Tell the whole true thing right up to the edge. And then leave the last word in the ground where my people can pick it up. The almanac's published. The *meaning* isn't yours to hand out, and it isn't his —" a tip of the chin at the causeway — "and it isn't even mine to give to *you*. It's mine to *keep*. Those aren't the same."

Frik, who punctured everything, including himself, especially the reach, looked at the gate and then at Naila and said nothing at all, which from Frik was a kind of bow.

The Lens raised the camera. Slowly. Pointed it not at the gate, not at the survey print, but down the long true curve of the wall — the work, not the workers — and then, deliberately, lowered it again without pressing the shutter.

Naila opened her notebook. She drew a line under the columns, closing the table, and where the third column ran blank to the bottom of the page she did not write a name. She wrote, in small letters in the margin, the thing she could stand on and the thing she couldn't, side by side, and left the gap between them exactly the width it was.

The frost was lifting off the grass now. Somewhere past the eastern walls the highway hummed, and beyond it the horizon held the notch where, in a year that none of them could circle on a calendar, the Moon would clear the far gateway within half a degree of the dead centre of a figure that nobody who started it had lived to see.

Somebody had kept that count anyway. Somebody was keeping it still.

What for, she did not write down.

Chapter 18 — The Shared Measure

The drive to High Bank took an hour and ten minutes on a Tuesday, with one stop for gas station coffee that tasted like a penny dissolved in hot water, and Naila spent most of it with a length of nylon mason's cord coiled in her lap, running it through her fingers like a rosary she didn't believe in yet.

Fifty-three miles. She'd done the arithmetic three times in the back seat because the number wouldn't sit still. Fifty-three miles, near enough due south, from the Octagon at Newark to a second octagon outside Chillicothe — the only other one the Hopewell ever built, as far as anyone has found, and that *as far as anyone has found* was the kind of phrase that kept her awake. Two octagons. Two circles. One ruler.

That was the thing she'd come to test.

"You're doing the cord thing," Frik said from the front, not turning around. "When you do the cord thing you've decided something and you're not telling us."

"I haven't decided anything. I want to walk it."

"She wants to walk it," Frik told the windshield. "Fifty-three miles of *I want to walk it.*"

Behind the camera, the Lens said nothing, which was how the Lens said most things.

The official story, said plainly, is not in dispute, and Naila held it the way you hold a tool you respect. The Observatory Circle at Newark measures one thousand and fifty-four feet across, give or take a tolerance that surveyors with theodolites have spent careers being impressed by — under four feet of error around a circle a fifth of a mile wide, laid out with cord and stakes and baskets of carried earth, no draft animals, no metal, no writing. That diameter — call it the OCD, the way Hively and Horn did, because somebody had to give the ruler a name — is the unit. The circles at Newark sit six of them apart. The square nested into the geometry is a function of it. The whole figure unfolds from that one length the way a sentence unfolds from a grammar.

And the same length lays out High Bank.

She'd read it. She'd checked the citation, then checked the citation's citation, the way she did with everything, the way that had gotten her called difficult in rooms full of people who'd never been wrong out loud. The number held. A measuring standard, shared across two monuments built generations apart and a long day's walk away from each other.

She wanted to feel it in her feet.

High Bank does not announce itself. There's no signage worth the name, no fence with a story, just a stretch of low green ground off a county road, the walls slumped by two thousand years of rain into long soft swells you could mistake for landscaping if nobody told you. A red-tailed hawk worked the thermals over the river bottom. Somewhere a dog was losing its mind about something.

Tessa Redwing was already there when they arrived, standing where the octagon's geometry opened toward the southeast, hands in the pockets of a canvas jacket, looking at nothing in particular the way people look at a place they've already memorized. She'd driven down separately. She did most things separately, around them, and Naila had stopped reading it as coldness. It was a person keeping her own counsel on her own ground.

"You brought string," Tessa said.

"I brought a hundred meters of mason's line and a long tape and a

notebook.” Naila held up the coil like evidence. “I want to lay out the unit. Here. With my hands. To know it’s the same.”

Tessa looked at the cord for a moment. “You know it’s the same. It’s in three papers.”

“I know it the way you know a thing you read.” Naila heard her own voice do the flat thing it did when she was being honest past the point of comfort. “I don’t know it the way I know my own height. I want to know it that way.”

Something moved across Tessa’s face that wasn’t quite a smile. “All right,” she said. “Then walk it.”

So they walked it.

It is harder than it sounds, and that is the point — that is always the point, the whole engine of the thing. From above, in a drone shot or a survey plate, the octagon at Newark and the octagon at High Bank look like the same hand drew them, clean as type. On the ground they are vast and vague and slumped, and the only way to know the measure is to *make* the measure, stake to stake, sighting along a line you cannot see all of at once.

Frik took one end of the tape and Naila took the other and they paced the long axis of a wall that had been five hundred and fifty feet of deliberate earth before the rain got to it, and by the third measurement Frik was sweating and complaining and gloriously alive.

“Okay but here’s my problem,” he said, jamming a stake. “Coincidence. Right? You take any two big round things built by the same culture and you go looking for a shared number, you’ll *find* a shared number, because numbers are everywhere, that’s what numbers do, they’re sluts—”

“Frik.”

“—they’ll match anything. That’s the debunk. That’s the real one, the one that isn’t dumb. It’s not aliens, it’s confirmation bias. You want the ruler to be shared so you find a length that’s shared.” He stood, wiped his forehead, looked genuinely troubled, which was Frik at his best. “Tell me why that’s wrong. Because I’d buy it. I’d buy it right now.”

It was the right question. It was the only question that mattered, and Naila loved him for asking it when he could have just agreed with her.

She crouched and read the tape and wrote the number down and sat with it.

“Because it’s not a number,” she said finally. “It’s the *governing* number. At Newark the OCD doesn’t just match one wall. It lays out the circle, and the spacing of the circles, and the square, and the square is what generates the octagon’s diameter. Pull that one length out and the whole figure dissolves. It’s not a coincidence you can find lying around — it’s the grammar the sentence is written in.” She tapped the notebook. “And the sentence is written twice. Here and there. In the same grammar.”

Frik chewed on it. “Huh,” he said, which from Frik was a standing ovation.

From a few feet off, the fish — well. There was no fish. There was Frik, doing the voice he did when he wanted to heckle himself without taking the blame, pitching it high and aggrieved: “*Oh sure, ‘the same grammar,’ very poetic, you measured a hill with a tape from the hardware store and now it’s a school of geniuses—*” and then, in his own register, quietly: “but the number held. Three times. It held.”

It had.

Tessa let them work. That was a thing about her Naila had started to notice and then started to revere: she didn’t correct, didn’t hover, didn’t perform expertise. She let the crew find the floor with their own feet and only spoke when they reached the edge of what feet could find.

They reached it around the long axis of the second wall, when Naila straightened up with the number confirmed and the cord slack in her hand and the obvious, enormous, unanswerable thing sitting in the middle of the green field like weather.

“Someone carried this,” she said. “The length. From there to here. Or from here to there. Across two days’ walk. Across — how many years between the two builds?”

“We don’t have it to the year,” Tessa said. “Generations. Plural.”

“Generations.” Naila looked south, the way the octagon opened, toward a horizon two thousand years of someone’s people had watched. “So it’s not one surveyor with a long memory. It can’t be. Whoever set out Newark was dust before High Bank’s walls were knee-high. The ruler outlived the people who used it. Somebody *taught* it. Somebody kept it and handed it forward and somebody else received it and trusted it enough to build a fifty-acre figure on it.” The cord was very still in her hand now. “That’s a school. That’s a — a discipline. Two thousand years ago. In Ohio.”

This was the place she usually got into trouble. She could feel the next sentence forming, the one that reached, the one her whole life had taught her to be embarrassed about an instant too late — *somebody must have taught them, somebody from somewhere, somebody who came with the measure already in their hands—*

She didn’t say it. She’d learned, this season, where that road went and who it robbed when it got there. But she felt it pull, and she was honest enough to know she’d felt it.

Tessa was watching her. She always seemed to be watching the exact thing you were trying not to show.

“You want to know who taught them the ruler,” Tessa said.

“Yes.” No point lying to her. “And I can feel it being the wrong question. I can feel it wanting an outside hand. I’m not going to say it. But I can feel it.”

Tessa nodded slowly, like that admission had bought something the easy version wouldn’t have.

She turned and looked along the wall, north, the long invisible line back toward Newark, fifty-three miles of farmland and strip mall and interstate laid over a sightline somebody once kept in their head.

“You keep saying *somebody taught them*,” Tessa said. “Like the teaching came from above, or outside. Down a ladder.” She let it sit. “That’s not how a measure travels between two of your own places.”

“Then how—”

“It means we taught *each other*.”

The hawk turned overhead. The dog had given up. Somewhere far off

the county road sighed a truck past, and the truck didn't know, and the field didn't care that it didn't.

"Newark and High Bank," Tessa went on, and her voice had changed register, dropped into something that wasn't lecture and wasn't quite story either, "they're not master and copy. They're not the original and the imitation. They're a conversation. Somebody walked the length of it — the ruler, the sky, what the gateway was supposed to hold — and brought it down here, and somebody down here knew enough to receive it and check it against their own horizon and build it true. That's not a gift handed down by a stranger. That's a people keeping faith with itself across distance and time. Across more time than anybody got to live." She looked at Naila. "Your missing teacher is the whole network. It's the relationship. It was never one hand."

Naila stood there with the cord and the number and the slow cold opening of it in her chest.

"That's bigger," she said.

"It's bigger," Tessa agreed.

The Lens had been filming this. Naila knew without looking because she always knew, and she knew too the way the Lens filmed Tessa — from a respectful distance, never close on the face, the frame leaving room around her like you leave room around someone whose ground you're on. Later there'd be the question of what to do with the footage, whether the Foundation's lawyers would want it, what the rollout people would try to make this exact moment say. Naila didn't want to think about that here. Here was for the field.

"The papers stop at the unit," she said. "Hively and Horn, the survey work — they prove the ruler's shared and they stop. They don't say who or how. They can't. The evidence runs out exactly there."

"The evidence runs out exactly there," Tessa said, "because that's where it actually stops. Not because we lost the answer. Because the answer that's left is the kind you can't dig up. You can dig up the unit. You can't dig up the *teaching*." She turned the canvas collar up against a wind that had found them. "People want a tablet. A name. A king with a measure. There isn't one. There's a length that two places share and a cycle nobody living could've seen on their

own, and the only thing both of those facts require is patience and trust and a great many ordinary people who never saw it finished.” She almost smiled again. “That’s the part the Foundation can’t sell. There’s no genius to brand. The genius is everyone.”

Frik, who had been quiet for an unprecedented stretch, said: “So the answer to *who taught them* is — them. They taught them. It’s recursive. It’s a people teaching itself across two hundred years and fifty-three miles.” He sounded faintly stricken, the good kind. “That’s so much weirder than a tablet.”

“Yes,” said Tessa.

“And we can’t close it.”

“No.”

“And you don’t *want* it closed.”

Tessa looked at him for a long moment. “I want it *kept*,” she said. “There’s a difference. Closing it makes it a fact in a book. Keeping it means it’s still being handed forward.” She gestured, just barely, at the green swell of the wall, at the cord in Naila’s hand, at the four of them standing in a two-thousand-year-old measurement they’d just verified with a tape from the hardware store. “You came here to walk the ruler. You walked it. It held. Now you know it the way you know your own height.”

“I do,” Naila said.

“Good. That’s yours. The number’s yours. You can publish the number; it’s already published.” Tessa’s eyes were level and entirely without apology. “What the conversation *was* — who said it to whom, and what it cost them to keep, and what it was *for* down to the marrow — that’s not in the number. That’s not yours to give. It came down the line to us. We’re still on the line.”

She picked up the far end of the mason’s cord, where Frik had staked it, and walked it back, coiling it neat as she came, and handed it to Naila — the whole length of it, gathered, the unit they’d just confirmed together resting light in her two hands.

“Same ruler,” Tessa said. “Two sites. Somebody kept handing it forward.” She let go. “And somebody still is.”

The hawk dropped off the thermal and was gone behind the tree line. Fifty-three miles north, under a golf course they'd stopped pretending was the strangest part, the same length was lying in the ground waiting for the next person who'd walk it and find it held. Naila stood in the open figure with the cord warm where Tessa's hands had been, and let herself not know — let it stay open, the way the field stayed open, the way the line stayed open — who, and how, and across how many lives, somebody had decided this measure was worth carrying further than any one of them would ever get to see.

It was real. It held. And it was not finished being handed.

Chapter 19 — The Overreach

Frik had been awake since four, and by the time the light came up grey over the Observatory Circle he'd filled eleven pages of a steno pad with numbers he was almost certainly going to embarrass himself with.

He knew the shape of it. He'd felt it the way you feel a sneeze building — that lovely, dangerous itch right behind the eyes that meant a pattern was about to arrive whether he'd earned it or not. ADHD gives you a thousand of these a day. The trick, the whole discipline of his life, was knowing which one to feed. Most of them you starved. This one he'd been feeding since midnight, and it had grown teeth.

He found Naila where he knew she'd be: standing dead centre of the Circle, where the morning frost hadn't burned off the grass, holding her arms out a little from her sides the way she did when she was reading a space with her whole body instead of her eyes. The wall ran around them, a thousand feet of it, low and green and impossibly round. From the ground it was just a hump in a field. You had to know to feel the size of it.

"Okay," Frik said. "Don't do your face yet. Let me finish before you do your face."

"I'm not doing a face."

"You're loading a face."

"I'm cold." But she turned, and there was something almost fond in it, which he chose to take as license.

He flipped the steno pad around so she could see the two ratios he'd boxed and underlined three times each.

"Great Circle to Observatory Circle," he said. "Diameter ratio. One point one three one. Right? You measured it, Tessa's people measured it, Hively and Horn measured it. Not in dispute."

"Not in dispute."

"Now." He tapped the second box, and his voice did the thing he hated and couldn't stop, climbed half an octave like a kid showing you a beetle. "The Moon. At perigee — closest — it looks *bigger*, right, fat harvest Moon. At apogee — furthest — smaller. The ratio of how big it looks, near to far. Apparent size. Perigee to apogee."

"Frik."

"One point one two seven."

He let it sit there in the frost. The numbers looked back at him from the page, and they were so *close*, so obscenely close, that he could feel the floor of the field tilting toward the idea like water toward a drain.

"One thirty-one versus one twenty-seven," he said. "That's a zero point three five percent match. The ratio of the two circles is the ratio of the Moon's two faces. They built the Moon's *breathing* into the ground. They put the fat Moon and the thin Moon fifty acres apart and made the dirt remember the difference." He was talking faster now, the sentences crowding each other. "And that's not even the alignment. That's not the standstill. That's a *second channel*. They encoded the cycle in the gateways and they encoded the size-change in the circle diameters, two completely different — Naila, that's two independent astronomical—"

"Where'd you get the perigee-apogee figure?"

It stopped him the way a hand on the chest stops you. Not unkind. Just there.

"Astronomical apparent diameter," he said. "Standard values. I checked three sources."

"And the Great Circle diameter."

"Three sixty-six metres. OHC's own number."

“To what tolerance?”

He looked at the page. The thing about a sneeze is that the moment somebody asks you to describe it, it's gone.

“It's an earthwork,” he said, more quietly. “It's a wall of dirt that's two thousand years old and got plowed and golf-carted and rained on for a century. The diameter's not a *number*, it's a — it's a fuzz. There's a few feet of fuzz on it either way.”

“How many feet of fuzz turns one point one three one into one point one two seven?”

He did the arithmetic he should have done at four in the morning instead of underlining things. It didn't take long. It never takes long once you're honest about the error bars. A couple of feet on a thousand-foot circle was enough to swallow the whole 0.35 percent and ask for seconds.

“Okay,” he said. “Okay. The match is inside the noise.”

“*Frik*,” said the fish, who had appeared on the wall-top with the radiant cruelty of a creature that has slept eight hours, “*you just discovered that if you measure a fat thing and a slightly-less-fat thing, the bigness goes down. Call the journals.*”

“I hate you and you're not real.”

“*The Moon's apparent size also depends on where you're standing, what the air's doing, and whether you've had coffee. You did this on no coffee.*”

“I did this on rage and a granola bar.”

Tessa came up the causeway as the sun finally cleared the eastern trees, and she had two paper cups, and she gave one to Naila first, which Frik noted and respected. She'd done that every morning. The hierarchy of the cups was its own small grammar. He was learning to read it.

“You've got the look,” Tessa said to him. Not a question.

“He found a ratio,” Naila said.

“The size one,” Tessa said. “The one-point-one-three-one.” She blew across the cup. “Everyone finds the size one. It’s the prettiest thing on the whole site that nobody can prove.”

Frik felt the floor steady itself, because here at least was someone who’d walked the same drain. “So you’ve—”

“I had a grad student once spend a summer on it.” Tessa looked out across the Circle, at the wall doing its slow impossible round through the burning-off frost. “Good student. She was sure. She had the perigee-apogee to four decimal places and she had a survey and she ran it past me like she was bringing me a kill.” A small breath that might have been a laugh once. “I sent her to remeasure the Great Circle. Three times, three different baselines. You know what she got.”

“Three different diameters,” Frik said.

“Three different diameters.” Tessa nodded slowly. “All within a few feet of each other. All within a few feet of three sixty-six. And the ratio she’d staked her summer on lived or died in those few feet. So.” She drank. “It’s not nothing. I want to be careful how I say this, because the Foundation will quote me. It is not *nothing*. The Hopewell knew the Moon’s near face from its far face — you can’t watch the eastern horizon for forty years and not see the Moon get fat and thin. They *had* that knowledge. I’d bet my career they had it.” She turned, and the look she gave Frik was the one he’d come to think of as the floor under the floor, the place where she stopped being a host and was only the keeper. “But knowing a thing and *building* a thing are different verbs. I can’t show you they built it. The dirt won’t say. And I won’t put words in my ancestors’ mouths to win an argument with a man who’s well-funded.”

Frik thought about the Foundation rep — Devlin, perpetually pressed, perpetually warm — who that very week had used the phrase *the Moon’s heartbeat encoded in the earth* in a slide deck Frik had seen over a shoulder. It had a beautiful kerning. It was, Frik now understood with a cold drop in the gut, *exactly the thing he’d been about to do at four in the morning*. Same reach. Better suit.

“He’s going to publish the ratio,” Frik said. “Isn’t he.”

“He’s going to publish the ratio,” Tessa agreed, “with a graphic. The

fat Moon and the thin Moon and a little dotted line. And it'll be the thing people remember, because it's *cute*, and the standstill cycle isn't cute, it's just true and it takes eleven minutes to explain." She didn't say it bitterly. She said it like weather she'd lived under a long time. "That's the part that gets to me. The real thing here doesn't need the pretty thing. The real thing is the biggest thing I know."

Naila had crouched to the grass while they talked, and now she said, without looking up, "Walk me through what you'd actually have to do."

"To prove the ratio?"

"To *do* the thing he's claiming they did." Her voice had gone flat and careful, which meant she was building. "Not see the Moon fat and thin. Anyone can see that. To put the difference *in the ground*. You'd have to decide, before you laid a single basket, that this circle is the fat Moon and that circle is the thin Moon. You'd have to fix the ratio first. Then survey it. Then hold that ratio across — what, fifty acres — basket by basket, by hand, no instruments, no writing, for however many years it took. And the whole time you'd be holding two numbers in your head that no one could check until it was done, and that no one watching could ever see were *there*, because from the ground" — she gestured at the wall, the long green nothing of it — "it's just round. You can only see what they made from the sky." She stood up. "That's the potato."

"The what," said Tessa.

"It's a thing we say. The easy version of the story. The version that makes it small. 'The Hopewell saw the Moon change size and copied it into two circles' — that *sounds* easy. Like tracing." She turned to Frik. "But you just walked me through how it isn't easy. It's the opposite of easy. The easy story has more genius hidden in it than the hard story does."

Frik looked at his steno pad. Eleven pages. He thought about tearing them out and decided not to, because they were a record of a man being wrong in a way that taught him something, and that was worth keeping. Patience, basket by basket. He could do that with a steno pad, at least.

"So I don't get to have it," he said. He meant the ratio. He meant the

lovely 0.35 percent. He meant the slide with the dotted line.

“You get to say *maybe*,” Tessa said.

And there it was — the thing Frik had come to think of as the gift she kept handing out and watching to see who’d actually take it. Devlin couldn’t take it; *maybe* didn’t fundraise. The decoy debunkers couldn’t take it either, the ones who needed every wonder reduced to coincidence by Tuesday. *Maybe* was the hardest stance in the whole field, because it didn’t sell tickets in either direction.

“The ratio’s a *maybe*,” Tessa went on. “A real one. Not a ‘we don’t know yet because we’re lazy’ *maybe* — a ‘the evidence genuinely doesn’t reach’ *maybe*. The dirt has a few feet of fuzz and the fuzz is bigger than the claim. So we stand in the fuzz. We say: they could have. It’s there to find, if you measure the right way someday, with the right baseline, and even then it might just be that round things have ratios. *We don’t know*.” She said it like it cost nothing, which Frik now understood meant it had once cost everything. “And the not-knowing isn’t the hole in the wonder. The not-knowing *is* the wonder. The standstill cycle’s not a *maybe* — that one’s nailed, half a degree, all eight points, go and check Hively and Horn, it’s in the journals. But the size-ratio? That’s mine to leave open. And I’m leaving it open.”

The fish, who had been suspiciously quiet, said from the wall-top, “*He wanted to bring you a kill, too*,” and for once there was no heckle in it. Just the observation, landed once.

“Yeah,” Frik said. “I did.”

“*And she sent you to remeasure the circle*.”

“She did.”

The frost was almost gone now. Where it hung on in the wall’s shadow you could still see, faintly, the lay of the land the way the morning had first shown it — the long curve, the gateway off to the east where in eighteen and a half years’ time the major-standstill Moon would heave itself up and clear the far cut within half a degree of the centre of a figure no single person who started it had lived to see close.

Frik stood in the middle of it with his wrong pages and his cold coffee and the most useful failure he’d had in months, and he found he didn’t

want the ratio anymore. The ratio was a thing you could own — publish it, kern it, dot the line. The maybe was a thing you could only stand inside.

Devlin would never stand in it. That was, Frik thought, the whole quiet war of this place: not whether the math was real — it was, the hard part was settled and revered and twice as precise as the famous stones everyone looked at instead — but whether you could love a record enough to leave its softest part *un*-resolved. To let the keeper keep the gap.

“I’m going to write it as a maybe,” he said.

“Good,” said Naila.

“And I’m going to let her say the maybe. Not me.”

Tessa didn’t answer that. She was looking east, at the gateway, at the long slow door cut into the dirt by people who’d handed the watching forward across more years than any of them would get. She had the cup in both hands. She was a scientist and she was a descendant and right now she was only one of those, and it wasn’t Frik’s to name which.

He capped his pen. The highway sound came faint over the wall, the way it always did, the present tense leaning on the deep past, and the Moon was somewhere below the eastern trees getting fat or thin, breathing its 0.35 percent that maybe — *maybe* — somebody two thousand years ago had bet a lifetime of baskets on.

Frik decided he could live in that maybe.

He decided it was the best room in the building.

Chapter 20 — What Is It For

The frost came in overnight and laid itself down across the whole fifty acres like a sheet pulled flat by careful hands, and the Lens was awake to photograph it before anyone said he could.

He'd learned that much in twenty years of standing where people would rather he didn't. You shoot the light while you have it. The light here was the colour of skim milk going blue, and it filled the eastern gateway of the Octagon the way water fills a doorway, level and patient, and the grass inside the walls held the cold in a way the grass outside did not, so the figure showed itself in a faint pale geometry, eight walls breathing white against the green of the fairway beyond.

A century of golf carts had run their wheels over this. He kept that in his head as a kind of penance. The eighteenth hole had been right there, by the embankment the colour of a held breath, and men in soft shoes had three-putted on a lunar instrument and gone home for lunch.

Behind him, the others were coming up the causeway, and Jakobus was with them, which meant the morning had a question in it.

Naila walked the way she always walked a thing she loved, which was slowly and with her chin slightly lifted, reading the air above the walls more than the walls themselves. Frik had a thermos and a grievance about the cold. The Lens kept the camera down for now and let the frame build itself in his eye instead — Tessa Redwing already stand-

ing at the centre, where she had been since before any of them arrived, in a coat that had seen a lot of dawns, her hands in her pockets and her face turned to the gate.

Dr. Tessa Redwing did not perform waiting. She simply stood inside the figure her ancestors had loaded into the ground basket by basket, and let the rest of them arrive into her morning.

“Twice as precise as Stonehenge,” Frik said, to the air, because somebody had to say it and the Ohio History Connection had said it first. “Their words. I checked.”

“It’s their phrase,” Tessa said, not turning. “I’d say it slower. Stonehenge tracks the sun. Anybody can find the sun. You stand in one place for one year and the sun gives you the whole story — longest day, shortest day, the two even ones. A child finds the sun.” She did turn, then. “This tracks the moon.”

“Harder target,” Naila said.

“The hardest one in the sky you can build for.” Tessa’s breath showed. “And the long way round.”

The Lens raised the camera and got her like that, the gate behind her shoulder, and then he lowered it, because there is a thing his trade had taught him about taking a face: the face is a loan. You hold it for a moment. It does not become yours.

He had photographed people in the worst hours of their lives and learned the difference between a witness and a thief too late to undo all of it. So now he counted. One frame of the woman, then the camera down. Let the rest belong to her.

Jakobus had come to a stop a little off from the group, the way he always did, as though there were a chalk line on the floor he’d agreed not to cross. He had been to a thousand made things. He read them the way Naila read pattern and the Lens read light. He never led. He arrived, and asked, and stepped back, and the answer was somebody else’s to give.

“They came here to prove it wasn’t real,” Naila said. She was looking at the far gateway now, the one across the whole figure, the one the moon would clear. “Hively and Horn. Eighty-two. A physicist and a

philosopher. They thought the astronomy was wishful — that people see calendars in old dirt because they want to. So they came to measure it and put it to bed.”

“And?” Frik already knew. He asked the way you ask so the other person gets to say it.

“All eight,” Naila said. “Every standstill point. The moon’s whole wander, north and south, rise and set, the full envelope of where it can come up over eighteen and a half years. Built into the walls. Built into the gates. Half a degree.” She held up a thumb against the sky and the thumb was wider than the error. “The far moonrise clears that gate within half a degree of the dead centre of the figure. They came to bury it and it stood up.”

“That’s the part I like,” Frik said. “Skeptics. Real ones. Not the kind who sneer — the kind who do the trig. They wanted it to be nothing and the dirt told them no.”

The Lens framed the far gate. Empty. The moon was somewhere below the horizon doing the only thing it has ever done, which is keep its appointment.

“Here’s what won’t sit down in my head,” Frik said, and now the fish in him was up and swimming. “I’m going to be annoying. It’s my function. — You can’t see this cycle.”

“No,” Tessa said.

“Like — not ‘it’s subtle.’ Not ‘you’d have to be clever.’ You physically cannot. One person, one life, standing right here, watching that horizon every clear night —”

“Never sees it close.” Tessa’s voice was flat and kind. “The moon’s standstill swings out wide, then comes back, then wider the other way, over eighteen point six years. You’d watch it creep. You’d write down where it came up. You’d get old. You’d be most of the way through the only life you get before the pattern even *suggested* its own shape — and you’d be dead, probably, before it confirmed.”

“So somebody watched,” Naila said quietly. “And wrote it down without writing.”

“Kept it,” Tessa said. “Watched, and kept, and handed it on. Marker by marker. Person to person. Somebody started a record they had no reason to think they’d live to read — and they were right, they didn’t — and they handed it to someone younger, who handed it on again, and *that* person, or that person’s grandchild, finally had enough of the record to know the shape.” She let it sit. “And then they built the shape into the ground at half a degree. With baskets.”

Frik had nothing, which was rare and worth a photograph the Lens did not take.

“No metal,” Tessa said. “No draft animals. No wheel that mattered. No writing. Cord, and baskets, and the patience to keep a number longer than a marriage, longer than a chief, longer than a war.” She looked around the eight walls. “People say *they couldn’t have*. Out loud, sometimes. To my face. As if the impressive thing would be aliens.” Her mouth did something that wasn’t quite a smile. “The impressive thing is that they *did*. That is the whole sentence. They did.”

The light changed. It does that fast at the end, when the sky’s already decided.

The moon came up through the far gateway.

It did not do anything dramatic. That was the part the Lens would remember — that the most precise object he had ever stood inside delivered its proof so quietly. The disc cleared the embankment between the two raised walls of the distant gate, low and fat and the colour of old ivory, and it sat in the gap as though the gap had been cut for it, which it had, two thousand years ago, by people working from a number they could not have finished gathering in a lifetime.

He had the camera up. He shot it. Three frames, the gate, the disc, the frost. Then he lowered the camera, because the moment after the proof belonged to the woman standing in the centre of the proof, and not to him.

Naila had gone completely still. She did that with things that were true. Frik was making a sound that wasn’t a word.

“Half a degree,” Naila said, when she could. “I keep saying it like it’s

a number. It's not a number. It's a person standing exactly *there* —" she didn't point, she just was aimed — "and a person standing exactly *here*, and a length of cord between them they got *right*. And then the same right answer fifty-three miles south."

"High Bank," Tessa said.

"Same octagon. The only other one you ever built. Same alignments. Same unit." Naila's hands had come out of her sleeves. "The Observatory Circle Diameter. The same *ruler*. You used the same ruler at two sites built generations apart and fifty-three miles apart and it's the *same length*." She turned. "Somebody taught the same standard to people who never met. That's a school. That's a — a network. A people keeping one measure across time and distance —"

"Naila," the Lens said, gently, because he could hear where the sentence wanted to go.

She stopped herself. She was good at stopping herself, when she heard it. "—that's bigger than a master race," she finished, smaller. "I wasn't going to say a master race."

"You were going to say *who taught them*," Tessa said. Not unkind. Just ahead of her, the way she had been ahead of all of them all week. "Everybody wants to say *who taught them*. Nobody wants to say *they taught each other*."

This was the part where Frik usually reached, and he reached.

"The size thing, though," he said. "The proportion. Great Circle to Observatory Circle, the ratio's eleven thirty-one, and the moon's own apparent size, perigee to apogee, that's eleven twenty-seven — they built the moon's *zoom* into the diameters, they —"

"That's a pretty one," Tessa said.

"Point three-five percent off! That's —"

"That's a pretty one," she said again, and the way she said it the second time closed a small door very softly, "and I won't stand on it, and neither should your book. The alignment is steel. The shared ruler is steel. Hold those cold. The size ratio is a lovely thing somebody

noticed and might be a coincidence and might not, and the difference between a fact and a wish is whether you'd still believe it if you wanted to be wrong." She looked at him almost fondly. "You came here to puncture reaches. Don't grow your own."

Frik laughed, caught clean. "Yeah," he said. "Fair. Fair." And he wrote *PRETTY REACH — do not stand on* in his notebook, and underlined it, and the Lens loved him a little for it.

The Foundation's car was at the bottom of the causeway. It had been there a while, engine off, the rep declining to walk up uninvited, which the Lens gave him real credit for — he was a clean-handed man with a clean-handed grant and a media rollout that wanted the moonrise photographed and captioned and copyrighted before the descendant nations had the last word on what the caption should say. He wanted to be first. Being first was the whole of his ask. He had not, this morning, been first; he had been at the bottom of the hill while the keeper stood at the centre, and that was the chase, and the chase was decided not by cleverness but by who was allowed to stand where.

The OHC people were up here, at least. Public stewardship, restored. The lease wound down. The carts gone.

And Jakobus, who had said nothing the whole morning, chose the moment the moon cleared the trees, when it was just a moon again and the proof had finished proving, to ask the only question he ever brought.

He didn't ask it loudly. He asked it the way you'd ask the road.

"What is it for?"

He was looking at Tessa when he asked. Everyone was. Naila's recorder was running. The Lens had the camera down at his hip, lens cap on now, which was a thing he did on purpose and only sometimes, and this was one of the times.

Tessa Redwing stood inside the figure her people had handed forward across more lifetimes than anyone in the circle could count, and she took her time, because the answer was hers, and a thing that is yours you may give slowly.

"You all keep asking me what it *does*," she said. "And I've told you.

It tracks the moon. It keeps the standstill. It holds the cycle a single person can't hold. That's what it *does*." She looked at the gate, where the ivory disc had been. "What it's *for* is a different word."

She did not lower her voice for the next part. She didn't have to.

"It is a record," she said. "And it is ours."

Naila did not ask the follow-up. The Lens watched her not ask it, watched her hold the question she most wanted answered — *whose hand, whose school, who carried the ruler the fifty-three miles* — watched her decide that seeing it clearly was the whole of her job and the answer was not hers to take. She let the recorder run on the silence. The silence was the most honest thing they'd record all week.

Frik closed his notebook.

The Foundation's car, eventually, started, and went.

The Lens lifted the camera one last time and did not photograph Tessa. He photographed the empty gate, where the moon had been and would be again in eighteen and a half years, when most of them would be older and some of them gone, the gap doing the only thing it has done since the baskets stopped: waiting, exact, for an appointment it would keep with or without anyone there to see.

He left the centre of the frame empty.

That was for the reader to stand in.

Chapter 21 — The Almanac Remains

The frost came in over the Octagon the way it had every morning that week, laying a grey skin on the fairway grass that had not yet learned it was no longer a fairway. Naila stood at the foot of the parallel walls with her hands in her sleeves and watched the eastern gateway hold its piece of sky. The flags were gone from the greens. The little white markers, the rakes, the bunkers — all of it staying for now, because a thing fenced inside a lease for a hundred years does not become public the morning the paperwork clears. But the cups were filled in. Somebody had filled in the cups.

She found that detail unreasonably moving and could not have said why.

“It’s smaller than it looks in the survey,” Friik said, coming up beside her with two coffees and handing her the one that was hers because he’d remembered, again, that she didn’t take it touching his. “Or bigger. I can’t decide. From the air it’s a diagram. From down here it’s just — a long hill that won’t stop.”

“Five hundred and fifty feet a wall,” Naila said. “Eight of them. You can’t see the figure from inside the figure. That’s the whole point of it.”

“Is it the point?”

“It’s a point.” She drank. The coffee was too hot and she held it anyway. “They built a shape you can only read from a place no one standing here could go.”

Behind them the highway made its low continuous sound, the sound

that had been under everything for the entire visit, the sound she'd stopped hearing on the second day and now heard again because they were leaving and her ears had gone greedy. A truck downshifted somewhere out past the trees. The frost held.

The official story is not in dispute, and Naila had learned, over eleven days, to stop reaching past it.

This is what is known. Two thousand years ago, give or take, people the archaeology calls Hopewell — a name borrowed from the farmer whose land held the first dug mounds, not a name those people ever called themselves — raised the largest set of geometric earth-works on the planet across this stretch of central Ohio. Three thousand acres of it once. Walls of basket-carried earth, packed by hand, hauled load by load by people with no draft animals, no metal, no wheel, no writing to leave a foreman's note in. The Great Circle a few miles south, three hundred and sixty-six metres across its ditch, the ditch sitting *inside* the wall where no fort would ever put it. And here, the Octagon, fifty acres of it, locked to a circle eleven hundred feet wide by a causeway, the two figures set six of their own diameters apart as if someone had measured the distance against the thing itself and found it good.

And the gates point at the Moon.

Not the Sun. The Sun is easy; any year gives you the Sun. The gates of the Octagon point at the Moon's standstills — the slow swing of moonrise along the horizon that takes nearly nineteen years to complete one breath out and one breath back. All eight of the turning points are built into these walls. The far gateway frames the major-standstill moonrise to within half a degree of the figure's exact centre.

Twice as precise as Stonehenge, the Ohio History Connection likes to say, and they have earned the right to say it, because two physicists came here in 1982 to prove the whole alignment was wishful pattern-finding and went home having proved the opposite and spent the rest of their working lives unable to make it go away.

Naila had read their papers on the plane in. She had read them again the first night in the motel, and a third time after she'd walked the causeway, because there is a difference between a number on a page

and the same number standing in your boots, and she had wanted to feel the seam between them close.

It had closed. That was the trouble. It had closed completely and left her holding the harder thing.

“Tessa’s here,” Frik said.

Dr. Redwing came across the grass without hurry, in the canvas coat she’d worn every day, a travel mug in one hand and a roll of something — site maps, Naila guessed — under the other arm. She stopped a polite distance off and looked at the gateway the way Naila had been looking at it, and for a moment none of them said anything, which Naila appreciated. Tessa Redwing had a gift for the silence that wasn’t waiting to be filled.

“You’re driving south today,” Tessa said. Not a question.

“High Bank, then home,” Naila said. “I wanted to see the second one. Before we go.”

Something moved at the corner of Tessa’s mouth. “The ruler.”

“The ruler.”

Because this was the part Naila could not stop turning over, the part that had kept her awake more than the half-degree. The standard length that lays out this site — call it the Observatory Circle Diameter, the OCD, a touch over a thousand feet — is the same length that lays out High Bank, fifty-three miles down the country. The only other octagon those people built. Same unit. Same Moon. A ruler carried across two generations and two days’ walk and laid down again true.

“You want me to tell you who carried it,” Tessa said.

“I want to know if it’s knowable.”

“That’s a better question.” Tessa shifted the roll of maps under her arm. “Most of your colleagues don’t ask that one. They ask the first one and then they answer it for me.”

Naila said nothing. The LENS, twenty feet off, had the camera up and then — Naila watched it happen — lowered it, deliberately, let it hang

against his chest. He'd been doing that more. The lowering had become a thing he did on purpose, like a man setting down something heavy to remember his hands were his own.

Here is where the easy story goes to die, and Naila had watched it die a dozen ways in eleven days, and every death had been clean and final and had only made the wonder bigger.

It's a fort. The ditch is inside the wall. No one digs their moat behind their rampart. Dead.

It's a coincidence — eight gates, eight directions, of course one of them points somewhere. But it isn't one alignment, it's all eight standstill points, and the major moonrise lands inside half a degree of centre, and half a degree is the width of the Moon itself held at arm's length, and the people doing this had cord and stakes and packed dirt, not a transit. Dead.

It's one clever person, a Hopewell genius, a single brilliant sky-watcher. And here the death is the slowest and the most beautiful, because the eighteen-point-six-year cycle does not close inside a human life's attention. You can watch the eastern horizon every clear morning from the day you are old enough to be sent to watch until the day your eyes fail, and you will see the moonrise creep north and creep south, and you will never, alone, see the pattern come back around enough times to *know* it for a pattern and not a drift. You'd need to watch it, and write it — except they had no writing — so you'd need to watch it, and say it, and have it heard, and have it kept, and have it handed to a watcher not yet born who keeps it and hands it on again. The record outlives the recorder. It has to. There is no other way to get the number into the dirt.

So the genius is not in one head. The genius is in the keeping.

Frik had put it best, on the fourth night, in the diner, with a fish-shaped salt shaker he'd been narrating in a heckling falsetto all week: *"So somebody starts a calendar he knows he'll die before it finishes. He tells his kid, 'keep watching the Moon, write it down — oh wait, we don't have writing — okay, remember it perfectly and tell your kid, who will tell their kid, and in four generations someone you'll never meet will build a door out of the answer.' That's not a monument.*

That's a relay where nobody sees the finish line. Including the guy who fires the gun."

And the salt shaker, in falsetto: "*Sounds fake. Where's the spaceship.*"

"There's no spaceship," Frik had told the salt shaker, with real tenderness. *"That's what's wrong with you. You can't handle that there's no spaceship."*

"I'll tell you what I told the Foundation," Tessa said now, looking at the gateway and not at them.

The Foundation rep had flown back to wherever Foundations keep their reps, two days ago, after the consultation had landed where it landed: access through the tribal nations, interpretation through the tribal nations, the Ohio History Connection holding the public trust and the gate, and the glossy first-publication ephemeris reading that the rep had wanted to drop on a Tuesday with a press embargo — that, quietly, not happening on anyone's schedule but the descendants'. The rep had been gracious about it. They are always gracious about it. Graciousness is the clean hand's specialty.

"I told them they were asking *what does it mean* like the meaning was lying on the ground waiting for the fastest reader," Tessa said. "It isn't a code. You don't crack it. The alignment is real — go measure it, it'll hold, Hively and Horn did the math and the math is good, I'd stake my career on the math and I have." She turned, finally, and looked at Naila directly, and Naila held the look the way she'd learned to hold the coffee. "But the math is the *least* of it. The math is the part that's yours to take home. The relay isn't. The relay is what my ancestors *did with their whole lives* so that a person they'd never meet could stand where you're standing and know which way to face. You don't get to make that mean something for me. You get to know that it's true. Those aren't the same thing, and your whole field keeps walking off the property holding the first one and calling it the second."

Naila felt the gap open under that, the real one, the one she'd come eleven hundred miles partly to find and partly to stand inside without falling.

"I don't think it's mine," she said. "The meaning. I never thought it

was mine.”

“I know,” Tessa said. “That’s why I’m out here at frost o’clock seeing you off and I didn’t do that for the Foundation.” A breath of something almost like warmth. “You read the shape. You’re good at the shape. The shape is real and it’s astonishing and you can say so as loud as you like. Just don’t tell my grandmother’s grandmother’s grandmother what she was thinking when she carried the basket.”

The Moon was not rising; it was the wrong morning in the wrong year for the gateway to do its trick, and Naila had known that coming out here and had come anyway, because the gateway holds its sky whether or not the Moon obliges, and there is something to standing in front of a door that points at an event you cannot see today and will not see for years and that the people who built the door also could not see most mornings of their lives. They built it for the morning it *would* happen. They built it for a sky they had to wait for. The waiting was the instrument.

Frik came and stood at her shoulder. He’d stopped heckling. Even the salt shaker, somewhere in his coat pocket, had gone quiet.

“You want to know if it’s one hand,” he said. Low. Just to her. “The ruler. Newark and High Bank, same length, same Moon. You want there to be one person, or one school, or one teacher who set the standard and sent it out. Don’t you.”

“I want it to be knowable,” Naila said again. “I keep wanting the answer to close like the math closed.”

“And?”

She looked at the gateway. Fifty acres of held breath. A ruler that walked fifty-three miles and laid down true, generations apart, and nobody alive could tell her whose feet carried it or who first paced out the length and said *this, this is the measure, teach it on*.

“And it doesn’t close,” she said. “And I think that’s allowed. I think I’m allowed to stand here and say I don’t know who kept the same ruler across two days’ walk and four generations, and let it be the true thing instead of the failed thing.”

Tessa, behind them, said nothing. But when Naila turned, the archaeologist had rolled out one of her maps against the parallel walls, weighted at the corners with two stones from her coat pocket, and it was not High Bank. It was older than that, and hand-drawn, and Naila did not get to see what was on it, because Tessa rolled it back up the moment she saw Naila looking, unhurried, no rebuke in it, just a record being kept where a record was kept.

“Drive safe,” Tessa said. “The Moon will be over the far gate in March. Come back and watch it clear the centre. Bring your half a degree.”

“Will you tell me then?” Naila asked. She knew the answer. She asked because the asking was honest.

“No,” Tessa said, and smiled, and it was the smile of a person who holds a thing and means to keep holding it. “But I’ll stand next to you while you don’t find out. That’s the better thing anyway.”

They drove south with the frost burning off the fields and the highway humming the same note it had hummed over the walls. Somewhere behind them the gateway kept its angle. Somewhere ahead, fifty-three miles on, a second octagon held the same number to the same sky, laid out with the same ruler by hands that had learned the length from hands that had learned it from hands, back and back, into a keeping older than the figure and longer than any one watcher’s eyes.

The frost was almost gone when Frik finally took the salt shaker out of his pocket, set it on the dash, and looked at it for a long moment.

“Go on,” Naila said.

“*Still no spaceship,*” the fish said, in falsetto, sadly.

“No,” Frik agreed. “Worse than that. A bunch of regular people, watching the Moon, for two hundred years, on purpose, and getting it right.”

The fish considered this.

“*That’s terrifying,*” it admitted.

Naila watched the gate-shaped door fall away in the mirror and let the question stand open in the car, the way Tessa had let it stand open in the grass — *who keeps the same ruler longer than a life* — and did not answer it, and found, fifty-three miles’ worth of road ahead of her,

that not answering it felt exactly like wonder, and nothing at all like defeat.

Chapter 22 — Appendix: Real, Debated, Invented

The last thing Naila did before they left Newark was sit on the cold causeway between the Octagon and Observatory Circle with a notebook on her knees and write down everything she was *sure* of. Frost on the grass. A jogger going by on the path with earbuds in, no idea she was crossing a lunar instrument the size of fifty football fields. The highway breathing somewhere past the wall.

She made three columns. Tessa had suggested it, weeks back, in the OHC archive when the Foundation rep was still circling and everyone was arguing about who got to say things. *Keep your columns clean*, Tessa had said, not unkindly. *Real. Debated. Invented. Most fights are people mixing the three on purpose.*

So here it is. The columns. Check them yourself.

REAL

These are not in dispute. Not by us, not by the archaeologists, not by the descendant nations whose ancestors stacked the baskets. Stand on the floor; it holds.

The earthworks at Newark are the **largest set of geometric earthen enclosures on Earth**. Built by the people we call the **Hopewell** — a name they did not give themselves; it belongs to a nineteenth-century farmer whose land sat over another site — between roughly **100 BCE and 400 CE**. The complex once covered around **3,000 acres**. Less

than **ten percent** survives. The rest is under Newark's streets and lawns and the parking lots of ordinary errands.

What's left, mostly: the **Great Circle** and the **Octagon with its attached Observatory Circle**.

The **Great Circle** is about **366 metres** across, with walls standing **8 feet** and a ditch *inside* them. Write that down twice. The ditch is on the inside. A fort puts its ditch outside, so the enemy has to climb up out of a hole into your spears. This ditch faces the wrong way to kill anyone. It was never a fort.

The **Octagon** is eight walls, each about **550 feet** long, enclosing **50 acres**, joined by a causeway to the **20-acre Observatory Circle**. Every cubic foot of it was carried by hand, in baskets, by people with no draft animals, no metal tools, and no writing. Basket. By basket. By basket.

In **1982**, two scholars from Earlham College — **Ray Hively**, a physicist, and **Robert Horn**, a philosopher — came to the Octagon to *disprove* the claim that it was aligned to anything. They were skeptics. They were doing the right thing. They wanted to put the astronomy nonsense to bed.

They found that the Octagon encodes the **full 18.6-year lunar standstill cycle**. All **eight** of the Moon's standstill rise-and-set points are built into the walls and the gateways, to **sub-degree precision**. At the major standstill, the Moon rises and clears the far gateway within about **half a degree** of the figure's exact centre.

The Ohio History Connection's own phrase for the surveying: **"twice as precise as Stonehenge."** Their words, not ours. We just kept reading them.

And the part Naila underlined three times: **a single human being cannot find this cycle**. Not in a year. Not in ten. The 18.6-year wobble only resolves through observation kept, written into something — earth, cord, memory, relay — and handed forward across more than a lifetime. Somebody watched the eastern horizon and held the record before the shape could even *be* a shape. Then they laid it into the ground at half a degree.

There is a **standard unit** at Newark — call it the **Observatory Circle Diameter**, the **OCD**, about **1,054 feet**. The circles sit six of those

apart. The geometry nests a circle inside a square inside the octagon — the **squared circle** the math people get quiet about.

And the same OCD lays out the **High Bank Works**, 53 miles south near Chillicothe — the **only other octagon the Hopewell built** — which encodes the **same lunar alignment** with the **same ruler**.

The earthworks were inscribed on the **UNESCO World Heritage List in 2023**, as part of the Hopewell Ceremonial Earthworks. The United States' twenty-fifth such property.

For most of a century, the Octagon lay inside a private golf course lease. People played the back nine across a lunar observatory and didn't always know it.

That is the REAL column. Every line of it is checkable, footnoted in the people we cite at the back, and astonishing without a single visitor from anywhere off this planet.

"That's the thing that gets me," Frik said, leaning over her shoulder, reading upside down because he reads everything upside down. "We came in ready to fight someone, and the truth just — *sat* there. Being weirder than anything we'd have made up."

DEBATED

Here's where you keep a cold eye. These are not lies. They are *maybes* that haven't been nailed down, and the honest thing is to label them as maybes and let you stand in the gap.

The big one is the **apparent-size proportion**.

The ratio of the Great Circle to the Observatory Circle comes out around **1.131**. The ratio of the Moon's apparent size at its nearest, perigee, to its farthest, apogee, is about **1.127**. That's a **0.35 percent** difference. Tantalisingly close. It would mean the builders had not only mapped where the Moon rises but *how big it looks* across its orbit, and built that ratio into the ground.

Frik wanted it so badly. You could see it on him at High Bank, the day the geometry lined up — he had the look of a kid who's found the

secret panel. “It’s *right there*,” he’d said. “Point-three-five percent. That’s not luck. That’s somebody clocking that the Moon swells and shrinks—”

And Tessa, walking the inner curve with Naila, hadn’t shut him down. She’d just looked at him the way you look at someone whose enthusiasm you respect too much to flatter. “It’s pretty,” she said. “It might even be true. But you can’t stand on a maybe and call it a floor.”

Naila said it more bluntly, the way only she could, because she’d run the numbers and they made her itch: “Two ratios being close isn’t intention. I can find you a 0.35 percent match to something in any pile of measurements if I’m allowed to choose what I’m matching. The alignment is a fact. This is a *coincidence that might be a fact*. Those are different shelves.”

Frik took it well. He usually does, when he’s the one over-reaching — he’s faster than anyone at catching the reach when it’s his own. “Fine,” he said. “DEBATED column. But I’m allowed to like it.”

“You’re allowed to like it,” Tessa said. “Just don’t sell it.”

So: **apparent-size ratio — debated. Possibly designed. Not a thing to build a book on.** We didn’t.

INVENTED

This column is short, and it’s the one the genre fills first, so we’ll be careful.

There were no ancient astronauts at Newark. Nobody from elsewhere taught the Hopewell their geometry. There is no lost master race, no Atlantean survey crew, no cabal handing down secret cubits.

We want to be exact about *why* this matters, because the polite version of the lie wears a lab coat. The lie is: *they couldn’t have done this themselves*. That somebody — wiser, paler, off-world, whatever — must have shown them. That lie is older than the earthworks’ UNESCO plaque and it has done real damage, and the whole reason this book exists is to refuse it.

Because the actual evidence points the other way, and it points somewhere *better*.

A shared ruler across two octagons, 53 miles apart, generations apart, doesn't mean an outside hand. It means a **people**. A network. A practice of sky-watching with a standard length passed teacher to student, river town to river town, the way you'd pass any craft worth keeping. Somebody learned the OCD and taught the OCD and went and laid the OCD into the ground at Chillicothe, and the only thing stranger than a master race is *that* — that this was schooled, social, ordinary genius, kept by hand for longer than anyone alive could verify.

The wonder doesn't need rescuing from the Hopewell. The Hopewell are the wonder.

The fish, who had been quiet through the math because the math bored him, surfaced for this one. *Oh good. The aliens column is empty. Tell the History Channel. They've got a marathon to fill.*

A STATEMENT ABOUT WHO THIS BELONGS TO

We need to say a thing plainly, and then stop talking.

We came to Newark to read something. We're a crew that reads things — it's what our wiring is *for*, the watching nobody ever let us switch off. Naila reads geometry the way you read a friend's face. Frik reads the seam where a claim's been stretched too thin. The Lens reads the room, and the cost of seeing it. We're good at this. We're proud of it.

And the single most important thing that happened on this book is that we *didn't finish the puzzle*. Because it was never ours to finish.

There's a moment in the field, near the end, that Naila still chews on. The dawn alignment had just happened — the Moon clearing the far gateway, dead centre, exactly as Hively and Horn said it would forty years ago, exactly as somebody intended two thousand years ago. Everyone there had gone silent. And Jakobus, who'd driven in from wherever Jakobus drives in from, asked the question he always asks.

The one that re-opens the wonder instead of closing it.

What is it for?

He didn't ask Naila. He didn't ask us, the clever outsiders with the notebooks and the camera. He turned to **Dr. Tessa Redwing** — archaeologist, Tribal Historic Preservation Officer, descendant of the people who carried the baskets — and he handed her the question and got out of the way.

She let it sit a long time. The frost was burning off.

"It's a record," she said. "And it's ours."

That's the answer. That's the whole answer, and it is complete, and it does not belong to us to expand, footnote, monetise, or improve.

The earthworks are returning to public stewardship under the Ohio History Connection, in consultation with the descendant nations — Eastern Shawnee, Shawnee, Absentee-Shawnee, Miami, and others whose ancestors and relations are tied to this ground. The lease that fenced an observatory inside a golf course for a century is ending. A clean-handed outfit would have liked to read it and frame it and say *what it means* first. They don't get to. The people whose record it is get the last word on what it's for, including the word *no*.

We held the camera. We did not hold the meaning. If anywhere in these pages it reads like we cracked somebody's mystery, we failed, and Tessa would tell you so, and she'd be right.

GAZETTEER — GO STAND IN IT YOURSELF

You should. The scale doesn't survive a photograph; it lives in your shins when you walk a 550-foot wall.

Octagon Earthworks & Observatory Circle — Newark, Ohio, off North 33rd Street. As the golf lease winds down, public access is expanding; **check the Ohio History Connection** for current hours, open days, and the scheduled alignment viewings (the major standstill is rare — worth planning a life around). Go at dawn if you can. Bring a coat. Stand in the Observatory Circle and try to feel the fig-

ure you can't see from the ground. You can't. That's the point — they built a shape only the sky could read whole.

Great Circle — a few miles south, on Cooper Avenue. Open, walkable, free as of writing. Walk into the ditch. Note which side of the wall it's on. Argue with anyone who says "fort."

High Bank Works — near Chillicothe, 53 miles south. Access is limited and partly conservation-managed; **confirm before you go.** This is the second octagon. The same ruler. The shared measure. If you stand here and at Newark in the same week, you will feel the network in your feet.

FURTHER READING: Ray Hively & Robert Horn's papers on the Octagon alignments (the 1982 work and after) — the primary source, and a model of skeptics following the evidence past their own intentions. Ohio History Connection publications and the UNESCO inscription dossier for the Hopewell Ceremonial Earthworks. And, before anything else, the **statements of the descendant nations themselves.** Read them first. Read them last. Let them be the ones to tell you what it's for.

Naila closed the notebook. Three columns, clean.

The Moon had set hours ago; it was just sky now, ordinary blue over an ordinary Ohio town that happened to be holding a calendar somebody started knowing they'd never see it close.

"You okay with not knowing?" Frik asked. "The whole *who taught them the ruler* thing. We never got the bottom of it."

"There's no bottom to get to," Naila said. "There's a shared measure. Somebody keeps leaving the same length in the ground." She looked south, toward Chillicothe, toward fifty-three miles of nothing-in-particular that wasn't nothing at all. "We didn't make that up. We just noticed it."

And here's the part I love, the fish said, almost gently, for once. *You're going to keep noticing it. Aren't you. The same ruler. Site after site.*

Nobody answered him.

That's the door. We're leaving it open.



Klaus

Custos, non Conditor